Polynomial depth in Bayesian neural-network learning: unrestricted Gaussian priors, trained networks, and numerical stability

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Abstract

We continue the investigation of depth dependence in Gaussian Bayesian learning of Boolean functions. Without an additional robustness restriction, a Gaussian-distance argument gives an all-width finite-variance theorem for every fixed exponent $\gamma > 1$. Its target-dependent cost is polynomial in $Q_* = [(K+2)/L]^{L/2}$, where K is the energy of a unit-margin target witness. A zero-padding and scalar-transmission construction makes this cost polynomial for every bounded-weight one-hidden-layer target, uniformly over all learner depths and every width above the actual depth-specific complexity. In particular, parity, symmetric Boolean functions, and polynomial-size DNF/CNF have polynomial sample bounds for every fixed $\gamma > 1$. Any fixed intrinsic target depth is also covered, with polynomial degree depending on that depth. Linear thresholds with inverse-polynomial geometric margin also have joint polynomial bounds. It does not establish a joint polynomial bound for every small circuit or resolve arbitrary-width mean-field learning. We also give an elementary polynomial-width result uniform in all $\gamma \geq 1$. At $\gamma = 1$, a separate construction shows that absolute prior-cost counting has exponential depth dependence. At a common depth $O(d^2)$, every Boolean function even has critical rate bounded by a universal constant. These are not simple-target sample lower bounds. For alternative priors, numerical output stability removes the all-input clean-margin requirement: a margin is required only on the training examples, and the test guarantee holds at a smaller margin. Trusted input regions and soft stability penalties give further variants. Experiments on synthetic Boolean tasks and small handwritten-digit networks separate near-boundary failures from stability of confident predictions. Explicit gradient-descent basins show that the original robustness condition is not a universal consequence of successful training.

Status. This report contains mathematical proofs, independent adversarial proof reviews by other AI agents, and reproducible experiments. These new results have not been formalized in Lean. Pointwise sufficient certificates generally use floating-point arithmetic. Twenty global positive certificates and selected negative witnesses are additionally verified with interval arithmetic and exact rational checks. Bounds are qualitative polynomial bounds with deliberately loose constants, not practical sample estimates. The foundational posterior-compression proof and Gaussian target-cylinder construction from the preceding investigation are reproduced in the appendices, so the new statistical statements can be read independently. No novelty claim is made for standard finite-parameter counting, product-norm Rademacher methods, Gaussian concentration, or the general idea of learning through stability and compression.

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1 What is resolved, and what remains open

Throughout, "unrestricted" means no additional internal robustness restriction. It retains the original all-input output-margin conditioning unless a theorem explicitly changes that convention.

1. **Unrestricted prior.** If the learner width is polynomially bounded in the target parameters, the sample bound is polynomial in depth for all $\gamma \geq 1$. At arbitrary widths, a stronger result holds for each fixed $\gamma \geq 1 + \zeta$, $\zeta > 0$: a unit-margin witness of energy K gives a bound polynomial in d, L, C, K, Q_* and the statistical parameters. The degree may depend on $1/\zeta$. For every bounded-weight shallow target, a new representation makes Q_* polynomial uniformly over learner depth. Consequently XOR, all symmetric Boolean targets, and polynomial-size DNF/CNF have joint polynomial bounds at every learner depth and every admissible width. Linear thresholds with inverse-polynomial geometric margin also have such bounds. For an arbitrary fixed deeper implementation, the bound is polynomial in added depth but can still be exponential in the original implementation energy. The result also extends to any fixed intrinsic target depth, with polynomial degree depending on that depth. A single polynomial jointly in target size and intrinsic depth, and arbitrary-width $\gamma = 1$ learning, remain open. More quantitatively, if a target representation has depth D_0, width C, and weight bound R, the unrestricted $\gamma > 1$ sample bound is $(2 + d + C + R)^{O_\zeta(D_0)}$ times a polynomial in learner depth and the statistical parameters, for every $L \geq D_0$. Logarithmic target depth therefore gives a quasipolynomial target-size bound. At $\gamma = 1$ we additionally prove an obstruction to a common proof strategy: at depth $L = 20(d + 1) + 4$, every Boolean function on d bits has critical prior cost at most $100000L$, with a direct finite-width Gaussian-mass bound. Thus counting all functions below an absolute prior-cost threshold necessarily incurs an exponential-in-depth term. By using small covariance changes instead of mean shifts, we further show that at one common depth $L_d = O(d^2)$, every Boolean function has critical rate at most a universal constant. Thus an absolute-rate counting argument fails even at a fixed rate threshold. These results do not compare competing functions with a simple target and are not sample lower bounds.

2. **Gradient-trained networks.** The original all-cube condition can fail because of small clean outputs away from the data, even when observed predictions are accurate and stable. Positive finite experiments do not establish a universal training law. We prove a stronger negative statement: there are open basins of gradient descent converging to exact, low-energy, internally fragile fits. Parameter flatness alone also does not imply internal activation stability.

3. **Other conditions.** The most useful new condition controls the change in the numerical output rather than requiring its sign to be stable everywhere. With positive training margins, it gives a joint polynomial theorem for every $\gamma \geq 1$ and every width above a layered Boolean circuit's width, without an all-input clean margin. It can be localized to a trusted input region or imposed as a soft prior penalty. The region can be chosen and validated using independent unlabeled samples. Networks may alternatively have their own exceptional regions from a family of controlled input-set complexity, at an explicit error cost. These are specified changes of prior or likelihood, rather than conclusions about unchanged Gaussian weights.

The open unrestricted question is genuinely target-specific. A large class of low-energy functions is not by itself a lower bound for this Bayesian learner: one must compare their Gaussian posterior mass with that of a small target. Conversely, fragility of one implementation does not imply that its Boolean function intrinsically needs many samples. The same function may have a robust

implementation. In fact the fragile trained dictator construction below has energy at most $L+76$, a class already controlled by a polynomial-depth product-norm bound. Internal fragility alone is therefore insufficient evidence for a sample lower bound.

Related representation-cost work studies infinite-depth limits for homogeneous networks [1] and depth separation in norm-controlled ReLU learning [2]. These results provide useful comparisons, but their activation functions and learning rules differ from the tanh Gaussian posterior here.

2 Model, conventions, and reusable statistical ingredients

Let $d \geq 1$, $L \geq 2$, and $N \geq 1$. Inputs lie in $\{-1,1\}^d$ and $h^0 = (1, x)$. There are $J = L-1$ hidden layers of width N, no biases after the augmented input, and a linear readout:

$$h^s = \tanh(W^s h^{s-1}) \quad (1 \leq s < L), \qquad z_w(x) = v^T h^{L-1}.$$

The Gaussian energy and readout energy are

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{L-1} \|W^s\|_F^2 + N\|v\|_2^2, \qquad E_v = N\|v\|_2^2.$$

The raw prior $\mu_{N,t}$ has independent entry variances $t^2/d, t^2/N, t^2/N^2$, respectively. Its density in energy coordinates is proportional to $\exp[-N\kappa_N/(2t^2)]$. Put $\sigma^2 = t^2/N$; exponent γ means $t^2 = N^{1-\gamma}$. Our unified robustness theorems use $0 < t \leq 1$, hence all $\gamma \geq 1$.

The original prior is this Gaussian conditioned on $|z_w(x)| > 1/2$ at *every* Boolean input. Given independent training inputs $x_i \sim D$ with $y_i = f(x_i)$, condition on correct training signs. The resulting posterior is written Q_S. Its Gibbs error is the expected test sign error of a posterior draw. If $p_S(x)$ is the posterior probability of the correct label, then

$$D\{x : p_S(x) \leq q\} \leq \frac{\mathrm{GibbsErr}(Q_S)}{1-q} \quad (0 < q < 1).$$

Thus the upper bounds here control both confidence thresholds $2/3$ and $1/2$, with factors three and two. Unlike certain lower-bound arguments, these upper bounds allow arbitrary posterior mixtures.

For robustness, use additive postactivation and output disturbances

$$h_\eta^s = \tanh(W^s h_\eta^{s-1}) + \eta^s, \quad z_w(x;\eta) = v^T h_\eta^{L-1} + \eta^L, \qquad \|\eta\|_*^2 = \sum_{s<L} \frac{\|\eta^s\|_2^2}{N} + |\eta^L|^2. \tag{1}$$

The previously proposed reference radius is $r_{\mathrm{ref}}(w) = .01/\sqrt{1 + JE_v(w)}$. A second calibration uses $S_N = (d/N)\|W^1\|_F^2 + \sum_{2\leq s<L} \|W^s\|_{\mathrm{op}}^2 + E_v$ and $r_S = .01/\sqrt{1+S_N}$. A network is sign-robust at an input if every disturbance in the stated closed ball retains its clean sign strictly. Clean zeros fail this condition. Set $\mathrm{sign}(0) = +1$ for auxiliary classifiers only. Write $\mathcal{R}_{r,N,f}$ for width-N networks with clean signed margin greater than $1/2$ for f and sign robustness at radius r on the entire cube.

Circuit terminology. For the robustness theorems, a width-C layered Boolean circuit has $L-1$ stages over fan-in-two AND, OR, NOT, COPY, and constant gates, including its constant rail. This is distinct from the minimum width of an arbitrary analog tanh network at exactly the same depth. Whenever the latter is used, the needed margin and weight bound are stated explicitly. On inputs in $\{-1,1\}$, the preactivations $u+v-1$, $u+v+1$, $-u$, and u implement the four nonconstant gates with signed margin at least one. A constant wire supplies the apparent biases. Multiplying these gate preactivations by a common gain and applying tanh gives the saturated target construction in Appendix B.

Posterior compression and target mass. Appendix C proves that a law Q with $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$ can be approximated at each input by a common law on a finite network class with $\log|\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1})$, provided its networks are sign-robust at radius r. The coupling can depend on the input; its marginals do not. Before robustness is used, the candidate output is exactly the original network under a small simultaneous postactivation perturbation except on probability at most α. This stronger construction is what permits the pointwise and numerical variants below. Compression through noise stability and all-layer margins has precedents in [10, 11]; the exact normalizations and Gaussian conditioning conventions here matter.

For every layered digital target, Appendix B supplies a raw Gaussian target cylinder of mass $\exp[-ND_*/t^2]$, with $D_* = \mathrm{poly}(d, L, C)$, uniformly at all $N \ge C$ and $0 < t \le 1$. Its networks have clean target margin greater than one and internal robustness radius inverse polynomial in C. If a posterior is the raw Gaussian conditioned on an event containing this cylinder, its KL is at most ND_*/t^2. Also $E_v = (t^2/N)\chi^2_N$ under the raw prior. For

$$B = \max\{8, 8[D_* + \log(1/\beta)]\},$$

its posterior mass on $E_v > B$ is at most β. The retained conditional posterior has KL at most $N(D_* + 1)/t^2$ for $\beta \le 1/2$. These facts follow by the chi-square exponential moment and division by the target probability bound. They explain why a radius calibrated to readout energy gives uniform all-width learning without imposing a hard energy cutoff.

3 Unrestricted learning at bounded widths

A robustness restriction is unnecessary when the learner's width is itself polynomially bounded. The following elementary argument retains the original all-input clean margin and Gaussian conditioning conventions, and is uniform in the Gaussian scale. It also covers analog target networks at their original depth, without digital simulation.

Let $P_N = N(d+2) + (L-2)N^2$ be the parameter count, and let $\mu_{N,t}$ denote the raw Gaussian with variances $t^2/d, t^2/N, t^2/N^2$, respectively. Thus $\kappa_N = (t^2/N)\chi^2_{P_N}$ under $\mu_{N,t}$. Fix integers $1 \le C \le M$ and suppose, for some $D \ge 1$, that

$$\mu_{N,t}\{w : f(x)z_w(x) > 1/2 \text{ for every } x\} \ge \exp(-DN/t^2). \tag{1}$$

This holds for $C \le N \le M$ and $0 < t \le 1$. Only this target-mass hypothesis is needed; posterior networks may be fragile.

Theorem 3.1. *For $0 < \varepsilon, \delta < 1$, set*

$$B = 4D + 2[d + 2 + (L-2)M]\log 2 + 4\log(2/\varepsilon),$$
$$A = \sqrt{MB}, \qquad T = d + 1 + M(L-1), \qquad q = [4T(MA)^{L-1}]^{-1}.$$

If

$$n \ge \frac{2}{\varepsilon}\left[\log M + P_M \log(5A/q) + \log(1/\delta)\right],$$

then, with probability at least $1 - \delta$ over independent examples from any input distribution, the unrestricted Gaussian posterior has Gibbs error at most ε, simultaneously for every $C \le N \le M$ and $0 < t \le 1$. Here "unrestricted" retains the original conditioning on $|z_w(x)| > 1/2$ for all inputs, but imposes no internal robustness condition. The same sample event works after any number of additional correctly labeled examples. In particular, if D and M are polynomial in d, L, C, the sample bound is polynomial in $d, L, C, \varepsilon^{-1}, \log(1/\delta)$, uniformly for all $\gamma \ge 1$ with $t^2 = N^{1-\gamma}$.

Proof. For every sample, the event used to condition the raw Gaussian contains the target event in (1). The chi-square exponential moment therefore gives

$$Q_{S,N,t}\{\kappa_N > B\} \le \exp\left(\frac{DN}{t^2} + \frac{P_N}{2}\log 2 - \frac{NB}{4t^2}\right) \le \varepsilon/2.$$

The last inequality uses $P_N/N \le d+2+(L-2)M$, $N \ge 1$, and $t \le 1$. This bound holds for every sample, without a probabilistic qualification.

If $\kappa_N \le B$, every physical parameter has magnitude at most A. Round each to the nearest multiple of q, using at most $5A/q$ grid values per coordinate. For the maximum coordinatewise hidden-state discrepancies, boundedness and the Lipschitz property of tanh give

$$e_1 \le (d+1)q, \qquad e_s \le NAe_{s-1} + Nq, \qquad |z - z_{\text{grid}}| \le NAe_{L-1} + Nq.$$

Consequently, uniformly over Boolean inputs,

$$|z - z_{\text{grid}}| \le q[d+1+M(L-1)](MA)^{L-1} = 1/4.$$

Rounding thus preserves the entire Boolean function of every posterior network with energy at most B.

The union of these grid classifiers over $C \le N \le M$ has logarithmic cardinality at most $\log M + P_M \log(5A/q)$. A consistency union bound gives one sample event on which every consistent grid classifier has true error at most $\varepsilon/2$. Every good-energy posterior draw has the same function as such a classifier; the remaining posterior mass is at most $\varepsilon/2$. This proves the Gibbs bound. The finite class is independent of t, and consistency only becomes more restrictive as examples are added. □

The logarithm of the grid size satisfies

$$\log(5A/q) = \log(20AT) + (L-1)\log(MA).$$

The exponentially small parameter precision costs only a polynomial in L in this logarithm. The unresolved unrestricted problem is therefore uniformity at learner widths *larger than every polynomial in the specified target complexity and depth*; this theorem does not show those larger widths actually require more samples.

Corollary 3.2 (Same-depth bounded-weight analog targets)**.** *Suppose a width-C, depth-L tanh network realizes f with $f(x)z(x) \ge 1$ on every Boolean input and weights of magnitude at most $R \ge 1$. Fix $M \ge C$ and set*

$$\begin{aligned} K &= 2[d(d+1) + (L-1)C^2]R^2, \\ a &= [4(d+1+M(L-1))(MR)^{L-1}]^{-1}, \\ D &= \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}. \end{aligned}$$

Then (1) holds with this D. Thus Theorem 3.1 supplies a polynomial sample bound at the original depth for every $C \le N \le M$, with dependence on R only through $\log R$ in the final sample bound. No robustness condition on the target implementation or prior support is needed. A target with the original strict margin $> 1/2$ satisfies the displayed hypothesis after doubling its readout.

Proof. Clone the C neurons into balanced nonempty groups of sizes n_i summing to N. Copy first-layer rows and replace internal and readout weights by W_{ij}/n_j and v_i/n_i. This preserves the exact function. Since $\max_i n_i / \min_i n_i \le 2$, its energy is at most K, and its weights remain bounded by R. A physical-coordinate perturbation of magnitude at most a changes every output by at most $1/4$, by the recurrence above.

Use isotropic coordinates $y = (\sqrt{d}\, W^1, \sqrt{N}\, W^{\text{internal}}, Nv)$, so $y \sim \mathcal{N}(0, t^2 I)$ and $\|y_*\|^2 \le NK$ for the cloned center. The standard-normal-coordinate box $|y_j/t - y_{*,j}/t| \le a$ lies within the physical box, because all coordinate scalings are at least one and $t \le 1$. Its probability is at least

$$\left(\frac{2a}{\sqrt{2\pi}}\right)^{P_N} \exp\left[-\frac{1}{2}\sum_j (|y_{*,j}|/t + a)^2\right]$$
$$\ge \exp[-NK/t^2 - P_N(2 + \log(1/a))] \ge \exp(-DN/t^2).$$

Here $(u+a)^2 \le 2u^2 + 2a^2$ and $a \le 1$ suffice. Every network in this box has target margin at least $3/4$. □

This is an elementary finite-parameter learning result, rather than a new claim about the general VC dimension of neural networks. Its role is to isolate which width regime remains open under the exact prior conventions.

4 A width-independent tanh bound without robustness

A second unrestricted result addresses arbitrary learner width, but with an exponential dependence on the energy needed by a fixed target circuit. Its depth dependence is polynomial. The following Rademacher argument accounts for the fact that tanh is not positively homogeneous.

The normalized-activation and exponential-moment method is closely related to Nakhleh–Nowak's treatment of deep neural variation spaces [9], which explicitly uses $\tanh(st)/s$ and slope discretization, and to the exponential peeling argument of Golowich–Rakhlin–Shamir [8]. We give a self-contained product-Frobenius version for the present Gaussian-energy conventions; we do not claim the underlying proof technique is new.

Theorem 4.1 (Product-Frobenius bound for tanh)**.** *Let $\mathcal{F}_{L,P}$ contain all scalar-output tanh networks with $L-1$ hidden layers, arbitrary finite widths, a linear readout, and product of layer Frobenius norms at most P. Biases are absent except through an augmented input. On every sample $x_1, \dots, x_n$ with $\|x_i\| \le R$,*

$$\widehat{\mathfrak{R}}_n(\mathcal{F}_{L,P}) \le \frac{PR}{\sqrt{n}}\left[2 + \sqrt{2(L-1)\log(3 + n(L-1)^2)}\right].$$

The bound is independent of every hidden width and of the individual layer norms apart from their product.

Proof. If $R = 0$, every sampled input and every network output on the sample is zero, so the bound is immediate. Assume $R > 0$ below. Define $\phi_s(u) = \tanh(su)/s$ for $s > 0$, $\phi_0(u) = u$, and $\phi_\infty(u) = 0$. These functions are odd, 1-Lipschitz, and zero at zero. For finite $s, t \ge 0$ and $r \ge 0$,

$$\phi_s(ru) = r\phi_{sr}(u), \qquad |\phi_s(u) - \phi_t(u)| \le u^2|s-t|, \qquad |\phi_s(u)| \le 1/s \quad (s > 0).$$

The derivative bound follows by differentiating $\phi_s(u) = u\int_0^1 \operatorname{sech}^2(suv)\, dv$.

Let $\mathcal{G}_0$ contain the identity feature map. Inductively allow arbitrary finite-dimensional features

$$g_j(x) = \phi_{s_j}(a_j^T h(x)), \quad h \in \mathcal{G}_{\ell-1}, \quad \sum_j \|a_j\|^2 \le 1, \quad s_j \in [0,\infty].$$

All these feature vectors have norm at most R on the sample. Put $B_\ell(\epsilon) = \sup_{g\in\mathcal{G}_\ell} \|\sum_i \epsilon_i g(x_i)\|$ for independent Rademacher signs. Writing $a_j = r_j u_j$ and using the scaling identity shows

$$B_\ell \le \sup_{h,\, \|u\|\le 1,\, s\in[0,\infty]} \left| \sum_i \epsilon_i \phi_s(u^T h(x_i)) \right|.$$

For any $\eta > 0$, uniformly on $[-R, R]$ the slope family admits an η/n approximation using at most $J = 3 + n^2R^2/\eta^2$ choices: truncate $s > n/\eta$ to the zero function, and round slopes below that threshold down at mesh $\eta/(nR^2)$. The error in the Rademacher sum is at most η.

We use the convex contraction principle

$$\mathbb{E}G\left(\sup_{t\in T} \sum_i \epsilon_i \psi_i(t_i)\right) \le \mathbb{E}G\left(\sup_{t\in T} \sum_i \epsilon_i t_i\right)$$

for convex nondecreasing G and 1-Lipschitz ψ_i vanishing at zero. For completeness, the single-coordinate proof compares $X_\pm = \sup_t(a_t \pm \psi(t_k))$ and $Y_\pm = \sup_t(a_t \pm t_k)$. One has $\max(Y_+, Y_-) \ge \max(X_+, X_-)$ by $|\psi(t)| \le |t|$. An approximately maximizing pair for X_+, X_- also gives $Y_+ + Y_- \ge X_+ + X_-$, by the Lipschitz property. Convex monotonicity therefore gives $G(Y_+) + G(Y_-) \ge G(X_+) + G(X_-)$; iterate over coordinates. Bounded separable classes follow by approximation from finite classes. In our application, u ranges symmetrically and ϕ_s is odd, so the absolute supremum is the same as the one-sided supremum.

Apply this principle with $G(t) = e^{\lambda t}$ to each grid slope, then bound the exponential of a maximum by the sum of the exponentials:

$$\mathbb{E}e^{\lambda B_\ell} \le e^{\lambda\eta} J\, \mathbb{E}e^{\lambda B_{\ell-1}}.$$

Furthermore, $\mathbb{E}B_0 \le R\sqrt{n}$, and bounded differences gives $\mathbb{E}e^{\lambda B_0} \le e^{\lambda R\sqrt{n} + \lambda^2 nR^2/2}$. Writing $H = L - 1$, iteration and Jensen yield

$$\mathbb{E}B_H \le H\eta + R\sqrt{n} + \frac{H \log J}{\lambda} + \frac{\lambda n R^2}{2}.$$

Take $\eta = R\sqrt{n}/H$ and optimize λ to obtain $\mathbb{E}B_H \le R\sqrt{n}[2 + \sqrt{2H\log(3 + nH^2)}]$.

Finally, for a particular original network, write $M_\ell = \|W^\ell\|_F$ and $S_\ell = \prod_{j\le\ell} M_j$. Dividing its ℓth hidden vector by S_ℓ changes its recursion to $\phi_{S_\ell}((W^\ell/M_\ell)g^{\ell-1})$. Thus its normalized hidden features belong to the enlarged classes above, and its scalar output is at most P times a norm-one linear readout of $\mathcal{G}_H$. Zero-norm layers give the zero function and require no normalization. This proves the theorem. All suprema are measurable by countable approximation of finite-dimensional parameter spaces and then union over widths. □

For the Gaussian energy in our model, the product of the L weighted squared layer norms is dP^2. Consequently

$$P \le d^{-1/2}(\kappa_N/L)^{L/2} \le d^{-1/2} e^{B/2} \qquad \text{if } \kappa_N \le L + B.$$

Applying the theorem to the ramp loss $\ell(yz) = \min\{1, \max\{0, 1 - 2yz\}\}$ gives the following explicit bound.

Corollary 4.2 (All widths at bounded excess energy)**.** *Fix $B \geq 0$. With probability at least $1-\delta$, every sample-consistent depth-L network, at any finite width, with training margins at least $1/2$ and energy $\kappa_N \leq L + B$ has sign error at most*

$$\frac{4\sqrt{2}e^{B/2}}{\sqrt{n}}\left[2+\sqrt{2(L-1)\log(3+n(L-1)^2)}\right]+\sqrt{\frac{\log(1/\delta)}{2n}}.$$

No internal robustness assumption is imposed.

Proof. Augmented Boolean inputs have norm $\sqrt{d+1}$, so $PR \leq \sqrt{2}e^{B/2}$. The ramp loss is bounded by one and is 2-Lipschitz. Symmetrization, scalar contraction, and bounded differences give population loss at most empirical loss plus four times the network Rademacher bound and the stated confidence term. Empirical loss is zero and population loss upper-bounds sign error. □

Lemma 4.3 (A scalar transmission construction)**.** *For every real $D > 0$ and integer $m \geq 2$, there are positive scalar weights $\alpha, b_2, \ldots, b_{m-1}, c$ such that*

$$r_1 = \tanh\alpha, \quad r_s = \tanh(b_s r_{s-1}), \quad c r_{m-1} = 1/2,$$

and, writing $D' = \max\{D, 4\}$,

$$D\alpha^2 + \sum_{s=2}^{m-1} b_s^2 + c^2 \leq (m+12)(D'/4)^{1/m}, \qquad c^2 = (D'/4)^{1/m}.$$

Proof. Set $q = (D'/4)^{1/(2m)}$, $k_s = \min\{s, m-s\}$, and $r_s = q^s/(\sqrt{D'}k_s)$. Then $0 < r_s \leq 1/(2qk_s) \leq 1/2$ and $r_{m-1} = 1/(2q)$. Choose $\alpha = \operatorname{atanh} r_1$, $b_s = \operatorname{atanh}(r_s)/r_{s-1}$ and $c = q$. For $0 < u \leq 1/2$ the power series gives $(\operatorname{atanh}(u)/u)^2 \leq 1 + u^2$. Writing $R_s = k_{s-1}/k_s$, the reciprocal pairs $k/(k+1), (k+1)/k$ satisfy

$$\left(\frac{k+1}{k}\right)^2 + \left(\frac{k}{k+1}\right)^2 - 2 \leq 4/k^2.$$

Consequently $\sum_{s=2}^{m-1} R_s^2 \leq m - 2 + 7$, $R_s \leq 2$, and $\sum_{s=1}^{m-1} k_s^{-2} < 4$. Using $q^{2m} = D'/4$ therefore gives

$$D\alpha^2 \leq q^2 + 1/4, \qquad \sum_{s=2}^{m-1} b_s^2 \leq q^2(m-2+7) + 4, \qquad c^2 = q^2.$$

Their sum is at most $(m+12)q^2$. Empty internal sums at $m = 2$ satisfy the same estimates. □

Corollary 4.4 (Polynomial cost of additional depth in the strong-prior limit)**.** *Suppose a width-C, depth-L_0 network implements f with global margin at least $1/2$ and Gaussian energy E. Put*

$$A = \max\{2E, 1\}, \qquad B = 2E + 13 + 7A.$$

For each learner depth $L \geq L_0$, Corollary 4.2, with B replaced by $B+1$, gives a sample bound for the original unrestricted Gaussian strong-prior learner at every width $N \geq C$. Specifically, choose n so that the displayed risk bound is at most $\varepsilon/2$. On one sample event of probability $1-\delta$, for each $N \geq C$ the posterior Gibbs risk is at most ε for all sufficiently small Gaussian scales. The scale threshold may depend on N, L, f, ε; no uniform threshold is asserted. The sample bound is polynomial in L and exponential in E.

Proof. Balanced cloning into width $N \geq C$ preserves the target function and increases energy by at most a factor of two. Write $c^2 = N\|v\|^2 \leq 2E$ for its readout cost and $E_{\text{pre}} \leq 2E$ for the other costs. Apply Lemma 4.3 with $D = 4c^2$ and $m = t + 1$. Replace the old readout by a hidden layer whose every row is $2\alpha v^T$, then by the lemma's scalar channels and readout. Its first new preactivation is $2\alpha z(x)$, whose signed magnitude is at least α. Oddness and monotonicity preserve the sign and final margin. The new first block has energy $4\alpha^2 c^2$, and the remaining costs are exactly the scalar ones. This gives, for every $t \geq 1$, a same-width realization at depth $L_0 + t$ with energy

$$E_{\text{pre}} + (t + 13)\max\{c^2, 1\}^{1/(t+1)}.$$

Since $A^a \leq 1 + a(A - 1)$ for $0 \leq a \leq 1$ and $(t + 13)/(t + 1) \leq 7$, the target energy is at most $L + B$. This also covers $t = 0$ by cloning directly.

For fixed N, L, Gaussian small-variance asymptotics imply that prior mass on functions of infimum energy greater than $L + B + 1$, divided by prior mass on f, tends to zero. There are finitely many Boolean functions, and f has infimum energy at most $L + B$. This ratio bounds the posterior mass of those functions for every consistent sample. Relaxed minimum-energy representatives exist at fixed width, have margin at least $1/2$, and have the same infimum as strict-margin representatives. Therefore every remaining consistent function has a representative covered by Corollary 4.2. The sample event is common to all widths, and adding at most $\varepsilon/2$ exceptional posterior mass proves the claim. □

This removes exponential dependence on *added learner depth* for a fixed target witness, even at arbitrarily large learner widths. It does not give a bound jointly polynomial in circuit size and depth, and it does not yet cover unrestricted arbitrary-width posteriors at every nonzero scale.

4.1 A nontrivial unrestricted family: linear threshold targets

The preceding result is not limited to padding a fixed network. It gives a bound polynomial in depth and input dimension for every linear threshold whose geometric margin on the Boolean cube is inverse polynomial.

Corollary 4.5 (Linear thresholds at every learner width)**.** *Suppose*

$$f(x) = \operatorname{sign}(a^T(1, x)), \qquad \mu = \min_{x \in \{-1,1\}^d} \frac{|a^T(1, x)|}{\|a\|} > 0, \qquad D_* = \max\{d/\mu^2, 4\}.$$

For every depth $L \geq 2$ and width $N \geq 1$, the original unrestricted Gaussian strong-prior learner admits the following common sample event. With probability at least $1 - \delta$, its Gibbs risk is at most ε for all sufficiently small Gaussian scales, provided

$$\frac{4e^{13/2}}{\sqrt{n}}\sqrt{\frac{(d+1)D_*}{4d}}\left[2 + \sqrt{2(L-1)\log(3 + n(L-1)^2)}\right] + \sqrt{\frac{\log(1/\delta)}{2n}} \leq \frac{\varepsilon}{2}.$$

The sample event is simultaneous over all widths; the small-scale threshold may depend on width and depth. In particular the sample bound is polynomial in $d, L, \mu^{-1}, \varepsilon^{-1}, \log(1/\delta)$.

Proof. Use the scalar construction with first-layer energy coefficient $D = d/\mu^2$. If its first scalar weight is α, give every first-layer row the vector $\alpha a/(\mu\|a\|)$. Every first preactivation has the correct sign and magnitude at least α. Put b_s/N in every internal entry and c/N in every readout entry, where (α, b, c) is the positive scalar tuple. Oddness and monotonicity preserve the target sign and

the scalar output lower bound. The energy is exactly $D\alpha^2 + \sum_s b_s^2 + c^2$, at every width. Lemma 4.3 therefore gives the following upper bound on target infimum energy:

$$K_L = (L+12)(D_*/4)^{1/L}.$$

Functions with infimum energy at most $K_L + 1$ have representatives with product-Frobenius norm at most

$$P \le \frac{1}{\sqrt{d}} \left(\frac{K_L + 1}{L} \right)^{L/2} \le e^{13/2} \sqrt{\frac{D_*}{4d}}.$$

Here $(D_*/4)^{1/L} \ge 1$ and $(1 + 13/L)^L \le e^{13}$. Theorem 4.1, the ramp-loss argument, and the small-variance comparison with the target prove the claim exactly as in Corollary 4.4. The elementary inequality $\mu \le 1$ follows by averaging $(a^T(1,x))^2$ over the cube; hence $D_* \le 4d/\mu^2$, making the asserted polynomial dependence explicit. □

For the finite-variance extension, the same construction has a *margin-one* center of energy

$$K_L^{(1)} \le (L+15)(D_*/4)^{1/L}.$$

Indeed the scalar construction has readout cost $(D_*/4)^{1/L}$; doubling that readout adds three times this cost. A posterior cutoff $K_L^{(1)} + 1$ therefore has product bound $P \le e^8 \sqrt{D_*/(4d)}$. For Theorem 5.1,

$$Q_*^2 = \left(\frac{K_L^{(1)} + 2}{L} \right)^L \le e^{17} D_*/4.$$

Thus that theorem gives a bound jointly polynomial in $d, L, \mu^{-1}, \varepsilon^{-1}, \log(1/\delta)$ at *every* learner width and every exponent $\gamma \ge 1 + \zeta$, for each fixed $\zeta > 0$. No robustness restriction or infinite-width limit is used. The polynomial degree may depend on $1/\zeta$; the mean-field exponent $\gamma = 1$ remains outside this argument.

5 Unrestricted finite-variance learning for $\gamma > 1$

There is a stronger bridge than concentration of the full Frobenius energy. A typical Gaussian bulk has small operator norms when $\gamma > 1$, even when its total Frobenius energy is large. Gaussian isoperimetry shows that the posterior lies near such a bulk, with the remaining signal having energy close to a target witness's. Removing the bulk then preserves all Boolean signs under an explicit product-norm condition. No additional restriction of prior support is used.

Write $\sigma^2 = t^2/N$, so the original Gaussian has independent entry variances $N\sigma^2/d, \sigma^2, \sigma^2/N$. Its energy coordinates are

$$A_1 = \sqrt{d/N}\, W^1, \qquad A_s = W^s \ (2 \le s < L), \qquad A_L = \sqrt{N}\, v,$$

and all their entries are independent $\mathcal{N}(0, \sigma^2)$. Their squared Euclidean norm is exactly κ_N.

Theorem 5.1 (Unrestricted learning above exponent one)**.** *Fix integers $d \ge 1$, $L \ge 2$, $C \ge 1$, and a real $K \ge L$. Use tanh hidden activations and a linear readout. Suppose that for every $N \ge C$ there is a width-N, depth-L target witness satisfying*

$$f(x)z(x) \ge 1 \quad \textit{for every Boolean input}, \qquad \kappa_N \le K.$$

Put

$$Q_* = \left(\frac{K+2}{L}\right)^{L/2}.$$

For every fixed $\zeta > 0$, every $0 < \varepsilon, \delta < 1$, and every distribution D on the Boolean cube, the original Gaussian learner, conditioned only on the original global clean margin and noiseless training consistency, has a sample bound polynomial in

$$d, L, C, K, Q_*, \varepsilon^{-1}, \log(1/\delta)$$

which, with probability at least $1-\delta$ over independent training inputs drawn from D and labeled by f, gives Gibbs error at most ε, simultaneously for all $N \geq C$ and all $0 < \sigma^2 \leq N^{-(1+\zeta)}$. Thus it covers all $\gamma \geq 1+\zeta$ at their actual finite Gaussian scales. The polynomial degree may depend on ζ; no uniform claim as $\zeta \downarrow 0$ is made.

We prove the statement with explicit width thresholds and sample inequalities. The hypothesis $K \geq L$ costs nothing: averaging over the cube and using AM–GM shows that a global unit-margin witness has $\kappa_N \geq Ld^{1/L} \geq L$.

A perturbation estimate using operator norms. Normalize hidden states by $\sqrt{N}$ and the augmented input by $\sqrt{d}$. The input norm is at most $\sqrt{2}$, all coordinate activations remain 1-Lipschitz and zero-preserving, and the layer maps are the A_s above. If U has energy at most $K+1$, and an additive weight perturbation B has operator norm at most η in every normalized layer, telescoping one layer at a time gives

$$\sup_x |z_{U+B}(x) - z_U(x)| \leq \sqrt{2}\, L\eta \left[\frac{K+1+2\eta\sqrt{L(K+1)}+L\eta^2}{L-1}\right]^{(L-1)/2}. \tag{2}$$

Indeed, each summand is at most $\sqrt{2}\eta$ times a product of $L-1$ other layer norms. The sum of squares of all perturbed layer-norm bounds is at most the numerator, and AM–GM bounds each such product. Set

$$A = \left(\frac{K+2}{L-1}\right)^{(L-1)/2}, \qquad \eta = \min\left\{1, \frac{1}{4\sqrt{L(K+1)}}, \frac{1}{8\sqrt{2}\,LA}\right\}.$$

The numerator in (2) is at most $K+2$, so the output change is at most $1/8$. Furthermore $A \leq \sqrt{e}\, Q_*$, hence η^{-1} is polynomial in L, K, Q_*.

A symmetric typical Gaussian set. For completely explicit constants, put

$$a_d^2 = 2(d+1)\log 2 + 4\log 8, \qquad b_L^2 = 16\log 9 + 8\log(8L), \qquad b = \max\{a_d, b_L\}.$$

Let T consist of bulk matrices whose first and last energy-coordinate Frobenius norms are at most ta_d, and whose internal operator norms are at most tb_L. Then T is closed, convex and centrally symmetric, and

$$\mu_{N,t}(T) \geq 1/2. \tag{3}$$

For the first block, its squared Frobenius norm is $(t^2/N)\chi^2_{N(d+1)}$; the readout is the same expression with $d+1$ replaced by one. The exponential moment of χ^2 bounds each failure probability by $1/8$.

For an internal matrix $tG/\sqrt{N}$, a 1/4-net of the unit sphere has at most 9^N points, and the two-net scalar Gaussian bound gives

$$\Pr\{\|tG/\sqrt{N}\|_{\mathrm{op}} > tb_L\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{4L}.$$

Union bounding over the $L-2$ internal blocks proves (3). Every normalized block of a member of T has operator norm at most tb.

Target mass by a Gaussian shift. Suppose $tb \le \eta$. For a target witness w_*, the shifted set $w_* + T$ has target margin at least 7/8, by (2). Central symmetry and the Gaussian change-of-mean identity yield

$$\begin{aligned}\mu(w_* + T) &= e^{-\kappa_N(w_*)/(2\sigma^2)} \int_T e^{-\langle w_*, B\rangle_\kappa/\sigma^2}\, d\mu(B) \\ &\ge e^{-K/(2\sigma^2)}\mu(T) \ge \tfrac{1}{2}e^{-K/(2\sigma^2)}.\end{aligned}$$

The inequality follows by pairing B with $-B$ and using $\cosh u \ge 1$. This lower bound holds for the raw probability of the conditioning event for every correctly labeled sample.

The posterior is close to the typical set. Let $d_\kappa(w,T)$ be Euclidean distance in energy coordinates. Gaussian isoperimetry [5] and (3) imply

$$\mu\{d_\kappa(w,T) > r\} \le e^{-r^2/(2\sigma^2)}.$$

Dividing by the preceding target-mass lower bound gives, for every sample,

$$Q_s\{d_\kappa(w,T)^2 > K+1\} \le 2e^{-1/(2\sigma^2)}. \tag{4}$$

On the complement, let B be the metric projection of w onto T and write $U = w - B$. The projection is well defined and continuous, since T is nonempty, closed and convex. Then $\kappa_N(U) \le K+1$ and $\sup_x |z_w(x) - z_U(x)| \le 1/8$. The original posterior has clean margin $> 1/2$ everywhere, so U has *exactly the same Boolean function* as w and training margins $> 3/8$. It is enough to use a margin-1/4 ramp loss below.

A polynomial width split. Take $\beta = \varepsilon/2$ and

$$M = \left\lceil \max\left\{C, 2, (b/\eta)^{2/\zeta}, 2\log(2/\beta)\right\}\right\rceil.$$

For every $N \ge M$ and $\sigma^2 \le N^{-(1+\zeta)}$, one has $t \le N^{-\zeta/2}$, hence $tb \le \eta$, and $2e^{-1/(2\sigma^2)} \le \beta$. The product norm of U is at most $d^{-1/2}[(K+1)/L]^{L/2}$. Write

$$G = \sqrt{(d+1)/d}\left(\frac{K+1}{L}\right)^{L/2}, \qquad H = L-1.$$

The product-norm Rademacher bound, now with the 4-Lipschitz ramp loss $\min\{1, \max\{0, 1-4yz\}\}$, gives one sample event of probability at least $1-\delta/2$ on which every such consistent residual network has error at most

$$F_n = \frac{8G}{\sqrt{n}}\left[2 + \sqrt{2H\log(3+nH^2)}\right] + \sqrt{\frac{\log(2/\delta)}{2n}}. \tag{5}$$

This event already includes all finite widths. When $F_n \le \varepsilon/2$, adding the posterior exceptional mass β proves the required bound for every $N \ge M$ and every stated scale.

For $C \le N \le M$, use Theorem 3.1. Here is a target-mass constant requiring only the present witness assumption. Every witness weight is at most $R = \sqrt{MK}$. Put

$$a = [4(d+1+M(L-1))(MR)^{L-1}]^{-1},$$
$$D_M = \max\{1, K + [d+2+(L-2)M][2+\log(1/a)]\}.$$

The same Gaussian-coordinate box calculation as in the finite-width proof has physical perturbation at most a, retains target margin at least $3/4$, and has mass at least $\exp(-ND_M/t^2)$. Applying that theorem with error ε and failure probability $\delta/2$ controls all these widths and all $t \le 1$ simultaneously. Its finite-grid logarithm is polynomial in $M, d, L, \log K$. The union of the two sample events proves the theorem. Finally, $G \le \sqrt{2}Q_*$, $b^2 = O(d + \log L)$ and $A \le \sqrt{e}Q_*$ show that M and a sample size satisfying all the displayed inequalities are polynomial in the stated arguments for fixed ζ. This proves the asserted quantitative dependence.

Scope. This argument does not apply at $\gamma = 1$: the Gaussian bulk's operator norm does not tend to zero with width there. It also does not make Q_* polynomial for every small circuit. If a fixed base witness can be padded with energy $K = L + B$, where B is independent of the added depth, then $Q_* \le e^{(B+2)/2}$. The theorem therefore removes exponential dependence on *additional depth*, at every width and every fixed exponent above one, while retaining a potentially exponential dependence on base-witness energy. Stronger target-specific energy estimates can make Q_* itself polynomial.

6 Polynomial unrestricted learning of shallow targets at every depth

A shallow target can have a small first-layer Gaussian energy at a polynomial base width simply by adding zero neurons. Its readout energy is then polynomial. Distributing that readout cost over the learner's remaining layers makes the product-norm parameter polynomial uniformly in depth. This gives a family for which neither arbitrary width nor arbitrary added depth requires a robustness restriction.

Theorem 6.1 (Shallow targets, all depths and all admissible widths)**.** *Suppose a Boolean target $f : \{-1,1\}^d \to \{-1,1\}$ has a one-hidden-layer representation*

$$z(x) = \sum_{j=1}^{C} v_j \tanh(a_j^T \bar{x}), \qquad \bar{x} = (1, x), \qquad f(x)z(x) \ge 1$$

with every coordinate of a_j and every v_j bounded in absolute value by $R \ge 1$. Let $C_L(f)$ denote its bounded-weight network complexity at depth L: the least width N admitting a realization with all weights at most N in absolute value and signed output greater than $1/2$ everywhere.

For each fixed $\zeta > 0$, the original unrestricted Gaussian learner has sample complexity polynomial in

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta),$$

for every specified learner depth $L \ge 2$ and every input distribution, simultaneously for every width $N \ge C_L(f)$ and every $\gamma \ge 1 + \zeta$. With probability at least $1 - \delta$ over a sample of that size, posterior

Gibbs error is at most ε simultaneously at all these widths and all prior scales $0 < \sigma^2 \le N^{-(1+\zeta)}$. The polynomial's degree may depend on ζ. The prior is conditioned only on the original global clean margin and the training labels.

Proof. Let A have rows a_j^T, and set

$$M = \left\lceil \max\{4, C, 2R, d\|A\|_F^2\} \right\rceil, \qquad T = \max\{1, M\|v\|_2^2\}.$$

Both are polynomial in d, C, R. Add zero hidden neurons to obtain width M. This preserves the function, and its first-layer and readout energies satisfy

$$E_{\text{pre}} = \frac{d}{M}\|A\|_F^2 \le 1, \qquad c^2 = M\|v\|_2^2 \le T.$$

In fact $M\|v\|_2^2 \ge 1$, since unit margin and $|\tanh| \le 1$ imply $\|v\|_2 \ge 1/\sqrt{C}$, but the maximum makes this unneeded.

For any $N \ge M$, balanced cloning of this shallow network preserves its output exactly and gives

$$E_{\text{pre}} \le 2, \qquad c^2 = N\|v_N\|_2^2 \le 2T.$$

Explicitly, assign each of the M original neurons either $\lfloor N/M \rfloor$ or $\lceil N/M \rceil$ clones, copy its incoming weights, and divide its readout weight by its number of clones. The first cost increases by at most $M\lceil N/M \rceil/N \le 2$, and the readout cost by at most $N/(M\lfloor N/M \rfloor) \le 2$. For learner depth $L = 2$ we already have a unit-margin witness with

$$K_2 = 2 + 2T, \qquad Q_*^2 = \left(\frac{K_2 + 2}{2}\right)^2 = (T+2)^2.$$

For $L \ge 3$, apply Lemma 4.3 with $D = 4c^2$ and $m = L - 1$. Replace the old readout by a layer whose every row is $2\alpha v_N^T$. Follow it with the scalar transmission layers, implemented by equal coordinates with dense matrix entries b_s/N, and final readout entries q/N, where

$$q^2 = \max\{c^2, 1\}^{1/(L-1)} \le (2T)^{1/(L-1)}.$$

The target had margin at least one, hence in particular at least $1/2$, so the scalar construction gives margin at least $1/2$. Double its final readout to restore margin at least one. This adds $3q^2$ to the energy. Thus every width $N \ge M$ has a depth-L unit-margin witness with

$$\kappa_N \le K_L := 2 + (L + 14)(2T)^{1/(L-1)}.$$

Writing $q_*^2 = (2T)^{1/(L-1)} \ge 1$, we have $K_L + 2 \le (L + 18)q_*^2$. Consequently

$$Q_*^2 = \left(\frac{K_L + 2}{L}\right)^L \le e^{18}(2T)^{L/(L-1)} \le e^{18}(2T)^{3/2} \qquad (L \ge 3). \tag{6}$$

The exponent $L/(L-1)$ is at most $3/2$ for $L \ge 3$. Theorem 5.1, with base width M, therefore provides a sample bound polynomial in the stated quantities for all $N \ge M$.

We verify admissible smaller widths rather than silently identifying M with the target's minimum complexity. First, $C_L(f) \le M$ for all $L \ge 2$. At width M, the original incoming weights have magnitude at most $R \le M/2$. In the added scalar channel, $\alpha \le \operatorname{atanh}(1/2) < 0.55$, so its first

weights $2\alpha v_j$ have magnitude at most $1.1R < M$. The scalar construction has $b_s \le 2.2q$ and final weights $2q/M$. Since $T \le MCR^2 \le M^4$ and $L \ge 3$,

$$q \le (2T)^{1/4} \le 2^{1/4}M, \qquad b_s/M < 3 \le M, \qquad 2q/M < 3 \le M.$$

Thus this depth-L, width-M circuit satisfies the bounded-weight convention and has margin at least one. The case $L = 2$ is immediate.

For $C_L(f) \le N < M$, start from a defining width-$C_L(f)$ circuit, clone it to width N, and double its readout. It has margin greater than one, all weights at most $2M$, and Gaussian energy bounded by a polynomial in d, L, M. The Gaussian box estimate and Theorem 3.1 consequently give a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$, simultaneously for all these widths and all $t \le 1$. Use failure probability $\delta/2$ for this event and $\delta/2$ for the preceding event for $N \ge M$. Their intersection proves the claim. □

The complexity parameter controlling this theorem's sample size is the size and weight bound of a *shallow* representation. Its width admissibility is the actual depth-L complexity $C_L(f)$. The theorem does not identify shallow complexity with the minimum complexity of an arbitrary deep circuit, and therefore does not settle the unrestricted problem for all small deep circuits. It also makes no claim at $\gamma = 1$.

Corollary 6.2 (Parity at arbitrary depth and admissible width)**.** *For every fixed $\zeta > 0$, parity of any $k \ge 1$ chosen coordinates in $\{-1, 1\}^d$ has unrestricted Gaussian posterior sample complexity polynomial in $d, k, L, \varepsilon^{-1}, \log(1/\delta)$, for every $L \ge 2$, every $N \ge C_L(f)$, and every $\gamma \ge 1 + \zeta$. The bound holds for every input distribution and for the original finite-variance posterior.*

Proof. Let $q = (k - \sum_{i \in S} x_i)/2 \in \{0, \dots, k\}$ count negative coordinates, and put $a = \log(16k)$. Define

$$\tau_j(x) = \tanh\big(a(q - j + 1/2)\big), \qquad 1 \le j \le k.$$

Every argument has magnitude at least $a/2$, so $|\tau_j - \operatorname{sign}(q - j + 1/2)| \le 2e^{-a} = 1/(8k)$. Writing $c_k = 1$ for even k and $c_k = 0$ for odd k, the exact step identity is

$$(-1)^q = c_k + \sum_{j=1}^{k} (-1)^j \operatorname{sign}(q - j + 1/2).$$

Hence $F = c_k + \sum_j (-1)^j \tau_j$ satisfies $|F - (-1)^q| \le 1/8$. Represent the constant by one extra hidden neuron with augmented-input weight $\operatorname{atanh}(1/2)$ and no other incoming weights. A linear readout equal to $2F$ has signed margin at least $7/4$. Its threshold-neuron readout weights are $2(-1)^j$, and its constant-neuron weight is $4c_k$. Its incoming weights have magnitude at most $a(k + 1)$. Thus $C = k + 1$ and $R = \max\{4, a(k + 1)\}$ are polynomial in k. Apply Theorem 6.1. □

Corollary 6.3 (Symmetric Boolean targets)**.** *The conclusion of Corollary 6.2 holds for any Boolean function of the Hamming weight of k selected coordinates.*

Proof. If its values are $h(0), \dots, h(k) \in \{-1, 1\}$, then

$$h(q) = \frac{h(0) + h(k)}{2} + \sum_{j=1}^{k} \frac{h(j) - h(j-1)}{2} \operatorname{sign}(q - j + 1/2).$$

Each coefficient has magnitude at most one. Replacing the signs by the same τ_j and doubling the output gives margin at least $7/4$, using the same width and weight bounds as in the parity proof. □

Corollary 6.4 (Polynomial-size DNF and CNF targets)**.** *A Boolean function on d bits represented by a DNF or CNF with $s \geq 1$ clauses has unrestricted Gaussian posterior sample complexity polynomial in $d, s, L, \varepsilon^{-1}, \log(1/\delta)$, for every specified $L \geq 2$, every $N \geq C_L(f)$, and every $\gamma \geq 1+\zeta$, with $\zeta > 0$ fixed.*

Proof. For a DNF, let $q_j(x)$ be the number of false literals in clause j, and put $a = \log(16s)$ and $\tau_j(x) = \tanh(a(1/2 - q_j(x)))$. The exact clause indicator is $[1 + \operatorname{sign}(1/2 - q_j)]/2$. Thus

$$H(x) = s - 1 + \sum_{j=1}^{s} \operatorname{sign}(1/2 - q_j(x))$$

is -1 when no clause is satisfied and at least 1 otherwise. Replacing the signs by τ_j changes H by at most $1/8$. Doubling the resulting expression gives global signed margin at least $7/4$. It is a shallow tanh network with $s+1$ features: the extra feature is the constant $1/2$, with readout weight $4(s-1)$. After removing duplicate literals and discarding contradictory clauses, each clause has at most d literals. Each q_j is affine in the bits, so the remaining incoming and readout weights have magnitude at most $R = \max\{4s, a(d+1)\}$. The shallow-target theorem applies. A CNF is the negation of a DNF in negated literals, and negating the readout preserves every bound. Constant functions are immediate. □

6.1 Targets of any fixed intrinsic depth

The shallow-target argument extends to any fixed target depth by using different numbers of clones at its successive hidden layers. The degree of the resulting polynomial may depend on the target depth, but not on the learner's additional depth.

Corollary 6.5 (Fixed target depth, arbitrary learner depth)**.** *Fix an integer $D_0 \geq 2$ and $\zeta > 0$. Suppose f has a width-C, depth-D_0 tanh realization in the present architecture, with signed output at least one on the Boolean cube and all weights bounded by $R \geq 1$. For every specified learner depth $L \geq D_0$, the original unrestricted Gaussian learner has sample complexity polynomial in*

$$d, C, R, L, \varepsilon^{-1}, \log(1/\delta).$$

The degree and constants may depend on D_0 and ζ. For every input distribution, one sample event of probability at least $1-\delta$ gives Gibbs error at most ε simultaneously for every $N \geq C_L(f)$ and every $0 < \sigma^2 \leq N^{-(1+\zeta)}$. Here $C_L(f)$ is the actual bounded-weight network complexity defined in Theorem 6.1. The prior retains the original global clean-margin conditioning and sign-only training likelihood.

Proof. Put $H = D_0 - 1$. Pad the target's hidden layers to width C with zero neurons if necessary, and write $W^1, \ldots, W^H, v$ for its weights. Choose an integer

$$B = \left\lceil \max\left\{4, 2R, \frac{d}{C}\|W^1\|_F^2, \|W^2\|_F^2, \ldots, \|W^H\|_F^2\right\}\right\rceil, \qquad M = CB^H,$$

omitting the internal-layer terms when $H = 1$. Since $\|W^1\|_F^2 \leq C(d+1)R^2$ and $\|W^s\|_F^2 \leq C^2R^2$ internally, B is polynomial in d, C, R, and so is M for fixed D_0.

At hidden layer s, give each original neuron exactly

$$n_s = B^{H-s}$$

active copies, and fill the other positions up to width M with zero neurons. Copy first-layer rows. For $s \geq 2$, give every edge from a copy of source j to a copy of target i weight W^s_{ij}/n_{s-1}. All unused edges are zero. Since $n_H = 1$, retain the original readout on the final active neurons. This preserves the function exactly. The first-layer and internal energy terms are respectively

$$\frac{d}{M} n_1 \|W^1\|_F^2 = \frac{d}{CB} \|W^1\|_F^2 \leq 1, \qquad \frac{n_s}{n_{s-1}} \|W^s\|_F^2 = \frac{1}{B} \|W^s\|_F^2 \leq 1.$$

Thus the energy before the readout is at most H. Put

$$T = \max\{1, M\|v\|_2^2\} \leq MCR^2.$$

It is polynomial in d, C, R for fixed D_0.

Balanced cloning of this width-M network into any $N \geq M$ preserves the function and gives

$$E_{\text{pre}} \leq 2H, \qquad c^2 = N\|v_N\|_2^2 \leq 2T.$$

If $L = D_0$, these are already unit-margin witnesses with $K = 2H + 2T$, and

$$Q_*^2 = \left(\frac{K+2}{D_0}\right)^{D_0} \leq (4T)^{D_0}.$$

If $L > D_0$, set $m = L - H \geq 2$. Replace the old readout by the scalar transmission from Lemma 4.3, using $D = 4c^2$, and double the final scalar readout to obtain unit margin. Its readout cost before doubling is $q^2 = \max\{c^2, 1\}^{1/m} \leq (2T)^{1/m}$. The total energy is at most

$$K_L = 2H + (m + 15)(2T)^{1/m}.$$

This is the desired depth: the H old hidden layers and $m - 1$ new hidden layers total $L - 1$. Since $(2T)^{1/m} \geq 1$,

$$\begin{aligned} Q_*^2 &= \left(\frac{K_L + 2}{L}\right)^L \\ &\leq \left(1 + \frac{H + 17}{L}\right)^L (2T)^{L/(L-H)} \\ &\leq e^{H+17}(2T)^{(D_0+1)/2}. \end{aligned}$$

The last step uses $L/(L - H) = 1 + H/m \leq 1 + H/2$. For $L \geq 2D_0$, the sharper upper bound $Q_*^2 \leq e^{H+17}(2T)^2$ also holds. Both K_L and Q_* are polynomial in the stated parameters when D_0 is fixed, uniformly over the learner depth. Theorem 5.1 now controls every $N \geq M$.

It remains to check admissible widths below M. First $C_L(f) \leq M$ at every $L \geq D_0$. The tapered construction has all original weights at most $R \leq M/2$. In a padded scalar channel, $\alpha \leq \operatorname{atanh}(1/2) < 0.55$, so its first added weights $2\alpha v_j$ have magnitude less than M. The scalar construction has $b_s < 2.2q$, and

$$T \leq MCR^2 \leq M^4, \qquad q \leq (2T)^{1/4} \leq 2^{1/4} M.$$

Its dense internal entries b_s/M and final readout entries $2q/M$ are therefore less than $3 \leq M$. This gives a bounded-weight width-M witness with unit margin at each learner depth; for $L = D_0$ no scalar channel is needed.

For $C_L(f) \leq N < M$, clone a defining width-$C_L(f)$ realization to width N and double its readout. Its margin is greater than one and all weights are at most $2M$. The Gaussian box estimate and Theorem 3.1 give a bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ for all these widths and all $t \leq 1$. Allocate failure probability $\delta/2$ to this event and to the preceding event for $N \geq M$. Their intersection proves the claim. □

Thus polynomial dependence on learner depth holds for targets of any fixed intrinsic depth with polynomial target-size dependence. The width $M = CB^{D_0-1}$ and the exponents above explain why this does not give a single polynomial jointly in the intrinsic target depth and target size. The mean-field exponent $\gamma = 1$ is also still outside this result.

Corollary 6.6 (Quantitative dependence on intrinsic depth)**.** *Fix $\zeta > 0$. There are constants $a_\zeta, b_\zeta > 0$, independent of the intrinsic depth, with the following property. Suppose f has a width-C, depth-D_0 tanh realization with unit signed margin and weights bounded by $R \geq 1$, where $D_0 \geq 2$, and set $S = 2 + d + C + R$. For every specified learner depth $L \geq D_0$, the sample bound*

$$n \geq \left\lceil S^{a_\zeta D_0} \left(L + \varepsilon^{-1} + \log(2/\delta)\right)^{b_\zeta} \right\rceil$$

suffices for posterior Gibbs error at most ε, with probability at least $1-\delta$ for every input distribution, simultaneously for all $N \geq C_L(f)$ and $0 < \sigma^2 \leq N^{-(1+\zeta)}$. The prior and likelihood are those of Corollary 6.5.

Thus the dependence on intrinsic depth is $\exp[O_\zeta(D_0 \log S)]$, while the dependence on learner depth remains polynomial with degree independent of D_0. In particular, if $D_0 \leq c \log S$ for a fixed constant c, the target-parameter factor is at most $\exp[O_{\zeta,c}((\log S)^2)]$. This is a quasipolynomial bound in S, not a single polynomial jointly in intrinsic depth and target size.

Proof. First suppose $L \geq 2D_0$. Use the tapered construction in the preceding proof, with $H = D_0 - 1$, $M = CB^H$, and $T = \max\{1, M\|v\|_2^2\}$. Its explicit definition gives $B \leq S^5$, so $M \leq S^{5D_0}$ and $T \leq MCR^2 \leq S^{7D_0}$. Put $m = L - H$. Since $L \geq 2D_0$, one has $L/m \leq 2$ and

$$K_L = 2H + (m+15)(2T)^{1/m} = O(LT), \qquad Q_*^2 \leq e^{H+17}(2T)^{L/m} \leq e^{H+17}(2T)^2 = S^{O(D_0)}.$$

For the remaining range $D_0 \leq L < 2D_0$, do not taper the target. Instead pad its original realization to $M_0 = \lceil \max\{4, C, 2R\} \rceil \leq S^2$, and put $T_0 = \max\{1, M_0\|v\|_2^2\} \leq S^5$. Balanced cloning to every $N \geq M_0$ gives

$$E_{\mathrm{pre}} \leq 2HS^4, \qquad c^2 \leq 2T_0.$$

At $L = D_0$, retain the readout. For $L > D_0$, append the same scalar transmission with $m = L - H \geq 2$. These constructions give, respectively,

$$K_L \leq 2HS^4 + 2T_0, \qquad K_L \leq 2HS^4 + (m+15)(2T_0)^{1/m}.$$

In either case $K_L \leq 19LS^5$. Therefore

$$Q_*^2 \leq (20S^5)^L \leq S^{7L} \leq S^{14D_0}.$$

The width-M_0 witness has weights bounded by M_0: the original weights satisfy $R \leq M_0/2$, and the scalar-channel check from the preceding proof applies because $T_0 \leq M_0^4$. Thus $C_L(f) \leq M_0$ in this range.

In both depth ranges, the polynomial in Theorem 5.1 has degree and constants depending only on ζ. Substituting the corresponding bounds gives the stated sample form above the respective base width M or M_0. For smaller admissible widths, the defining-witness and Gaussian-box argument in the preceding proof gives a sample bound polynomial in $d, L, M, \varepsilon^{-1}, \log(1/\delta)$ (or with M_0 in place of M), with degree independent of D_0. Substitution gives the same sample form. Split the failure probability between the two width ranges and enlarge a_ζ, b_ζ to finish. □

7 Limits of cloning and absolute-energy counting

These observations delimit two proof strategies. Neither is a lower bound on the sample complexity of the original Gaussian learner.

Cloning preserves an absolute path quantity. For a depth-D_0 realization with $H = D_0 - 1$ hidden layers, define

$$A = |v|^T |W_H| \cdots |W_1| \mathbf{1}_{d+1}, \qquad P_F = \|v\|_2 \prod_{s=1}^{H} \|W_s\|_F.$$

Absolute values are entrywise. By the operator norm bound,

$$A \le \sqrt{d+1}\, P_F. \tag{7}$$

Replacing each neuron by any positive integer number of exact clones, and dividing each outgoing weight equally among its source clones, preserves A. Indeed, the multiplicity of each original path cancels exactly against the product of the source-clone divisors on that path. Adding zero neurons also preserves A.

Suppose the original output attains absolute value one, and append a zero-bias tanh network that takes this scalar output as its input and restores unit signed margin. Write its scalar response as F. Oddness implies $F(1) \ge 1$. Since $|\tanh u| \le |u|$, its own absolute path sum A_F satisfies $1 \le |F(1)| \le A_F$. The appended realization has absolute path sum $AA_F \ge A$. Thus these two operations cannot reduce the lower bound in (7).

For any resulting width-N, depth-L network, its L Gaussian layer energies have product exactly dP_F^2. Consequently

$$\left(\frac{\kappa_N + 2}{L}\right)^L \ge dP_F^2 \ge \frac{d}{d+1} A^2. \tag{8}$$

In particular, cloning followed by scalar output transmission cannot always make the product-norm parameter polynomial in the size of the *given realization*.

A simple example makes the qualification essential. On one bit, let $h_0(x) = x$, $h_s(x) = \tanh(2h_{s-1}(x))$ for $1 \le s \le H$, and use readout $z(x) = h_H(x)/h_H(1)$. This width-one, depth-$H+1$ network represents $f(x) = x$ with exact unit margin. Inductively $h_H(1) > 0.9$, so all weights are at most two, whereas $A = 2^H/h_H(1) \ge 2^H$. The right side of (8) is therefore at least $4^H/2$. The same function has an inexpensive shallow representation. This is an obstruction to the specified representation operations, not to finding a different representation of the function.

For uniform-per-layer clone counts n_s and common padded width M, one has the stronger exact identity

$$a_1 = \frac{dn_1}{M}\|W_1\|_F^2, \quad a_s = \frac{n_s}{n_{s-1}}\|W_s\|_F^2 \ (2 \le s \le H), \quad a_{H+1} = \frac{M}{n_H}\|v\|_2^2, \qquad \prod_{s=1}^{H+1} a_s = dP_F^2.$$

If the hidden-layer energy is at most $H + B$ with $B \ge 0$, AM–GM gives $\prod_{s=1}^{H} a_s \le (1 + B/H)^H \le e^B$, and hence $a_{H+1} \ge dP_F^2 e^{-B}$. This explains why tapering can leave an exponentially expensive readout when intrinsic depth grows.

Proposition 7.1 (An absolute energy sublevel contains every Boolean function)**.** *For every integer* $d \ge 1$, *put*

$$L = 20(d+1) + 4, \qquad M = 64\, 2^d (d+1)^6.$$

For every Boolean function f on d bits and every width $N \geq M$, there is a depth-L, width-N tanh network with unit signed margin on the entire cube and Gaussian energy $\kappa_N \leq 3L$. In particular, the logarithm of the number of Boolean functions in this energy sublevel is $2^d \log 2 = \exp(\Omega(L))$.

Proof. Write $m = 2^d$ and $b = \log(16m)$. For each cube point a, let

$$h_a(x) = \tanh\big(b(a \cdot x - d + 1/2)\big), \qquad s_a(x) = 2\mathbf{1}\{x = a\} - 1.$$

The preactivation has magnitude at least $b/2$, with the correct sign, so $|h_a - s_a| \leq 2e^{-b} = 1/(8m)$. The exact identity

$$f(x) = \frac{1}{2} \sum_a f(a)(s_a(x) + 1)$$

therefore gives a shallow tanh realization

$$z(x) = \sum_a f(a) h_a(x) + \sum_a 2f(a)\,(1/2)$$

with signed margin at least $2 - 1/8 > 1$. Implement the second sum by m constant neurons, each with first-layer bias $\operatorname{atanh}(1/2)$ and all other incoming weights zero. There are $C = 2m$ hidden neurons and $\|v\|_2^2 = 5m$. All weights have magnitude at most $R = 4(d+1)^2$.

Our chosen $M = 32C(d+1)^6$ satisfies $M \geq \max\{4, C, 2R, d\|W_1\|_F^2\}$, since $d\|W_1\|_F^2 \leq dC(d+1)R^2 \leq 16C(d+1)^6$. Thus the zero-padding and balanced-cloning construction in Theorem 6.1 applies at every $N \geq M$ with

$$T = M\|v\|_2^2 = 320m^2(d+1)^6, \qquad \kappa_N \leq 2 + (L+14)(2T)^{1/(L-1)}.$$

Since $d + 1 \leq 2^d$ and $640 < 2^{10}$, $2T \leq 2^{8d+10} \leq 2^{L-1}$. Therefore $\kappa_N \leq 2L + 30 \leq 3L$. □

This proposition concerns representations and energy, independently of the prior variance exponent. It rules out a bound polynomial in d, L, K for the logarithm of the number of all Boolean functions with energy at most $K = 3L$. The exhibited shallow widths are exponential in d; the proposition does not say these are all small-circuit targets. It also does not compare competing posterior masses relative to a particular simple target, and gives no sample-complexity lower bound. The product-norm parameter can still be exponential when $K = 3L$.

8 Two partial results at the critical exponent

The preceding unrestricted learning theorem stops at $\gamma > 1$. At $\gamma = 1$, polynomial-depth target-mass estimates alone are already available from robust neighborhood constructions; they do not settle the posterior's generalization. The next estimate supplies a particularly simple stochastic padding construction whose additional depth costs only a universal constant per layer. It also identifies precisely which part of the argument is still missing.

Proposition 8.1 (Finite-width critical target mass by stochastic padding)**.** *Let f have the shallow unit-margin representation from Theorem 6.1, with C features and all weights bounded by $R \geq 1$. Set*

$$\rho = (8\sqrt{2}CR)^{-1}, \quad B = \frac{1}{2}\left[d \max_j \|a_j\|^2 + (d+1)(\rho^2 - 1 - \log \rho^2)\right], \qquad K = B + 8C^2R^2 + 18L + 2.$$

At the exact critical prior scale $\gamma = 1$, for every $L \geq 2$ and

$$N \geq 512C^2R^2\big[(d+1)\log 2 + \log(8L)\big],$$

the raw Gaussian probability of the target's global clean-margin event is

$$\mu_{N,1}(A_{N,f}) \geq \tfrac{1}{4} e^{-2NK}.$$

No support restriction is imposed on the original prior.

Proof. We construct an auxiliary law $\mathbb{Q}$ on network weights and show that it has relative entropy at most NK from the original Gaussian law, while assigning the target event probability at least $1/2$. Introduce independent uniform labels $J_i \in \{1, \dots, C\}$, also under the reference law; forgetting these labels can only decrease entropy. Under $\mathbb{Q}$, the first-layer row is

$$W_i^1 = a_{J_i} + \rho G_i/\sqrt{d}, \qquad G_i \sim \mathcal{N}(0, I_{d+1}).$$

Its conditional entropy from the reference $\mathcal{N}(0, I_{d+1}/d)$ is at most B, so the entire first layer costs at most NB. Let $h_i(x) = \tanh(W_i^1 \bar{x})$ and

$$u_i = C v_{J_i}/N, \qquad z_N(x) = \sum_i u_i h_i(x).$$

The expected value of z_N differs from the original shallow output by at most $CR\sqrt{2}\rho = 1/8$. Its summands before division by N are bounded in absolute value by CR. Hoeffding's inequality and a union bound over the cube imply, except with probability at most $1/8$,

$$f(x) z_N(x) \geq 3/4 \qquad \text{for every } x.$$

Indeed the probability that the uniform approximation to its expectation exceeds $1/8$ is at most $2^{d+1} e^{-N/(128 C^2 R^2)}$.

For $L \geq 3$, give each row of the next weight matrix the mean $4u$ and retain its reference covariance I/N. The conditional preactivation mean is $4z_N(x)$, and its variance is at most one. If a Gaussian has signed mean at least 3 and variance at most one, its expected signed tanh is greater than $2/3$. For example, its signed value is at least 1 except on an event of probability at most $\Phi(-2) < 1/32$, and $\tanh 1 > 3/4$, giving the lower bound $89/128 > 2/3$. The layer's conditional entropy cost is

$$\frac{N^2}{2} \|4u\|^2 \leq 8 N C^2 R^2.$$

Its empirical mean activation therefore has signed value at least $7/12$ at every input, except on an event of conditional probability at most $2^d e^{-N/288}$.

For every remaining hidden layer, give each incoming weight mean $6/N$ and retain variance $1/N$. Its preactivation mean is six times the preceding empirical mean. Thus the same induction preserves the signed empirical mean bound $7/12$: its next preactivation mean is at least $7/2 > 3$, and the same concentration estimate applies. Each such layer costs exactly $18N$ in conditional relative entropy. The union of the bad events over all hidden layers has probability less than $1/8$ under the displayed width assumption.

Finally, give each readout weight mean $2/N$ and retain variance $1/N^2$. This costs $2N$. Conditional on the hidden layers, its mean has signed value at least $7/6$ and its noise variance is at most $1/N$ at each input. A Gaussian tail bound and the union bound show that its noise is uniformly at most $1/2$, except with probability at most $2^{d+1} e^{-N/8} < 1/8$. Consequently the output has signed margin at least $2/3$, strictly above the required $1/2$. When $L = 2$, omit the extra hidden layers and instead give the readout mean $2u$, at entropy cost at most $2NC^2R^2$. The same final Gaussian noise estimate applies, and its mean has signed margin at least $3/2$.

The relative entropy chain rule gives $D(\mathbb{Q}\|\mu_{N,1}) \le NK$, and the preceding good events have joint probability at least $1/2$. Binary data processing yields, for $p = \mu_{N,1}(A_{N,f})$ and $q = \mathbb{Q}(A_{N,f}) \ge 1/2$,

$$NK \ge q\log(1/p) - \log 2 \ge \tfrac{1}{2}\log(1/p) - \log 2.$$

Rearranging proves the stated target-mass lower bound. □

The estimate is deliberately elementary. In particular, it proves neither polynomial counting of all low critical-rate Boolean functions nor polynomial sample complexity at arbitrary learner width. The existing critical-rate compression argument propagates tilt bounds by $g \mapsto 3g + 2g^2$; a linear-in-depth target rate does not prevent that recursion from growing very rapidly. The unresolved issue is control of *competing posterior functions*, rather than construction of a likely enough target.

Lemma 8.2 (Polynomial attenuation in the unperturbed critical bulk)**.** *Fix $q_0 > 0$, and define*

$$q_{s+1} = \mathbb{E}\tanh^2(\sqrt{q_s}Z), \qquad a_s = \mathbb{E}\,\mathrm{sech}^2(\sqrt{q_s}Z), \qquad Z \sim \mathcal{N}(0,1).$$

Then, as $s \to \infty$,

$$q_s = \frac{1}{2s} + O\Big(\frac{\log s}{s^2}\Big), \qquad \prod_{s=0}^{L-1} a_s = c(q_0)L^{-1/2}(1 + o(1)),$$

where $0 < c(q_0) < \infty$. Moreover

$$\sum_{s=0}^{L-1}(a_s^{-2} - 1) = \log L + O(1).$$

The constants may depend on q_0.

Proof. The strict inequality $\tanh^2 u < u^2$ for $u \ne 0$ shows that q_s decreases to zero; continuity excludes a positive limit. Taylor expansion, with globally bounded derivatives and Gaussian moments controlling the remainder, gives

$$F(q) := \mathbb{E}\tanh^2(\sqrt{q}Z) = q - 2q^2 + O(q^3).$$

For all sufficiently large s this implies $1/q_{s+1} - 1/q_s = 2 + O(q_s)$, first giving $q_s = \Theta(1/s)$ and then, by summation, $1/q_s = 2s + O(\log s)$. This proves the variance estimate. The identity $a_s = 1 - q_{s+1}$ is exact. Hence

$$\log a_s = -q_{s+1} + O(q_{s+1}^2), \qquad a_s^{-2} - 1 = 2q_{s+1} + O(q_{s+1}^2).$$

Since $q_s - 1/(2s)$ and q_s^2 are summable, summing the first identity shows that $\log\prod_{s<L} a_s + (1/2)\log L$ converges to a finite real constant. Exponentiating gives the positive constant $c(q_0)$. Summing the second identity proves the final assertion. □

The factor a_s is the derivative at zero of $m \mapsto \mathbb{E}\tanh(m + \sqrt{q_s}Z)$. Thus a weak coherent signal in the *unperturbed* critical bulk experiences polynomial, rather than exponential, total attenuation. Compensating each infinitesimal attenuation with a gain a_s^{-1} would add only $\log L + O(1)$ to the sum of squared scalar gains. This motivates a possible $L/2 + O(\log L)$ entropy padding cost. It is not a proof of such a posterior result: a finite coherent signal changes the covariance recursion, and the conditioned posterior need not retain the unperturbed Gaussian feature laws.

9 An obstruction to a uniform critical-rate counting theorem

There is a stronger reason to avoid treating the critical exponent as only a technical extension of the preceding proof. Absolute critical-rate sublevel sets cannot have a counting bound jointly polynomial in input dimension, depth, and rate. This is a statement about the actual mean-field rate functional, although it is still not a sample-complexity lower bound for a simple target.

We use the critical finite-domain variational formula proved in Appendix A. For a feature law ν on $[-1,1]^{\mathcal{X}}$, let $Q(\nu)=\mathbb{E}_\nu hh^T$, and let M_Q be the law of coordinatewise $\tanh(G)$ for $G\sim\mathcal{N}(0,Q)$. The critical rate of a global-margin Boolean function is

$$J_L(f)=\inf\left\{D(\nu_1\|\mathsf{M}_{Q_0})+\sum_{s=2}^{L-1}D(\nu_s\|\mathsf{M}_{Q(\nu_{s-1})})+\frac{1}{2}z^TQ(\nu_{L-1})^+z : f(x)z(x)\ge\frac{1}{2}\ \forall x\right\},$$

where the final term is infinite outside the covariance range and $Q_0(x,y)=\bar{x}\cdot\bar{y}/d$.

Theorem 9.1 (Every Boolean function has polynomial critical rate at a polynomial depth)**.** *For each $d\ge 2$, set*

$$r=20(d+1)^5,\qquad L=r+4.$$

Every Boolean function on $\{-1,1\}^d$ satisfies

$$J_L(f)\le 100000L.$$

Consequently no polynomial P can satisfy

$$\log\#\{f : J_L(f)\le K\}\le P(d,L,K)$$

for every d,L,K.

Proof. Write $m=2^d$. First we obtain a crude quantitative lower bound on the smallest eigenvalue of the unperturbed first hidden covariance

$$Q=\mathbb{E}_w h(w)h(w)^T,\qquad h(w)_x=\tanh(w\cdot\bar{x}),\qquad w\sim\mathcal{N}(0,I_{d+1}/d).$$

For each $a\in\{-1,1\}^d$, choose the center

$$w_a=(-b(d-1/2),ba_1,\dots,ba_d),\qquad b=\log(16m).$$

At $x=a$ its preactivation is $b/2$, whereas at every other cube point it is at most $-3b/2$. Thus its feature vector is within $1/(8m)$ coordinatewise of the vector that is $+1$ at a and -1 elsewhere. Let E_a be the coordinate box of radius $\rho=1/[8m(d+1)]$ around w_a. These boxes are disjoint, and changing weights within a box changes every activation by at most $1/(8m)$. The conditional mean vectors $\bar{h}_a=\mathbb{E}[h(w)\mid w\in E_a]$ therefore form a matrix H within operator norm $1/4$ of $2I-\mathbf{1}\mathbf{1}^T$. Since $m\ge 4$, that matrix has smallest singular value at least two; hence $HH^T\succeq I$.

Every box has Gaussian probability at least

$$(2\rho)^{d+1}\left(\frac{d}{2\pi}\right)^{(d+1)/2}\exp\left[-\frac{d}{2}(\|w_a\|+\sqrt{d+1}\rho)^2\right]\ge p:=e^{-20(d+1)^5}.$$

For the last crude estimate, use $b\le 3(d+1)$, $\|w_a\|+\sqrt{d+1}\rho\le 4(d+1)^2$, $(d+1)\log(1/(2\rho))\le 3(d+1)^2$, and $(d/(2\pi))^{(d+1)/2}\ge e^{-(d+1)}$. Conditional Jensen's inequality, together with disjointness, gives

$$Q\succeq\sum_a\Pr(E_a)\bar{h}_a\bar{h}_a^T\succeq pI.$$

Fix any Boolean vector $f \in \{-1,1\}^m$ and put $\eta = \sqrt{p/m} \le 1$. Take the first feature law to be its unperturbed law, at zero entropy cost. For the second feature law, shift its Gaussian preactivation mean to ηf, retaining covariance Q. Gaussian relative entropy and data processing give cost at most

$$\frac{1}{2}\eta^2 f^T Q^{-1} f \le \frac{1}{2}.$$

All diagonal entries of Q coincide, so the new feature mean is exactly $m_0 f$ for some $m_0 > 0$. It satisfies $m_0 \ge \eta/32$: for $0 \le t \le 1$ and variance $q \le 1$,

$$\mathbb{E}\,\mathrm{sech}^2(t + \sqrt{q}Z) \ge \Pr\{|Z| \le 1\}\,\mathrm{sech}^2(2) > 1/32,$$

and integration from zero to η proves the claim.

For any feature law with mean vector u, its second moment obeys $Q \succeq uu^T$, hence u lies in its range and $u^T Q^+ u \le 1$. Therefore shifting the next Gaussian preactivation mean to gu costs at most $g^2/2$ in entropy. All feature variances remain at most one. In our construction all diagonal second moments remain identical, so mean vectors remain of the form $m_s f$. Choose $g = 100$ for r successive layers. If $0 < m_s \le 1/100$, the preceding derivative estimate gives

$$m_{s+1} \ge 100 m_s/32 > 3m_s.$$

If $m_s \ge 1/100$, monotonicity gives $m_{s+1} > 1/32 > 1/100$. Since

$$\log(1/\eta) = 10(d+1)^5 + (d/2)\log 2 < 11(d+1)^5,$$

the choice $r = 20(d+1)^5$ guarantees $m_r \ge 1/100$. Each amplification layer costs at most 5000.

Use one last mean shift with $g = 400$. Its mean preactivation has signed magnitude at least four. With Gaussian variance at most one, the expected signed tanh is greater than $2/3$, by the Gaussian tail estimate used in Proposition 8.1. This layer costs at most 80000. Take the final output z equal to this last feature mean. It has signed margin greater than $2/3$, and its Gaussian readout cost is at most $1/2$, again by $Q \succeq zz^T$. The number of hidden layers is $1 + 1 + r + 1 = r + 3$, so the depth is $L = r + 4$. The total rate is at most

$$\tfrac{1}{2} + 5000r + 80000 + \tfrac{1}{2} \le 100000L.$$

Finally, at these polynomial values of L and $K = 100000L$, the rate sublevel contains every one of the 2^{2^d} Boolean functions. Its logarithmic cardinality is $2^d \log 2$, which exceeds every polynomial in d, L, K for sufficiently large d. □

This does not prove that a simple target is hard for the critical posterior. Such a lower bound would require comparison with the target's *own optimal rate*, sufficiently accurately to control which functions win after conditioning. The theorem instead rules out a specific proposed upper-bound route: a universal counting theorem based only on absolute critical rate, with polynomial dependence on depth. It is compatible with more precise comparison to the target, or with additional structural information about posterior minimizers.

Proposition 9.2 (The obstruction also holds directly at finite width)**.** *With d, L as in Theorem 9.1, for all widths $N \ge \lceil e^{100(d+1)^5} \rceil$ and every Boolean f,*

$$\mu_{N,1}(A_{N,f}) \ge \tfrac{1}{4}\exp\{-200000NL\}.$$

This is a direct statement about the original finite-width Gaussian prior and requires no appeal to a large-deviation limit.

Proof. Retain $m = 2^d$, $p = e^{-20(d+1)^5}$ and $r = 20(d+1)^5$, and now put $\eta = \sqrt{p/(4m)}$. Sample the first layer from its original law. Let H be its $N \times m$ matrix of activations and $\widehat{Q} = H^T H/N$. The preceding covariance bound and entrywise Hoeffding imply

$$\Pr\{\lambda_{\min}(\widehat{Q}) < p/2\} \le 2m^2 e^{-Np^2/(8m^2)} \le 1/8.$$

Indeed entrywise error at most $p/(2m)$ implies operator error at most $p/2$. On the good event, set

$$u = \frac{\eta}{N} H \widehat{Q}^{-1} f;$$

on its complement set $u = 0$. Give every row of the next matrix this mean vector u, preserving reference covariance I/N. On the good event,

$$H^T u = \eta f, \qquad \|u\|^2 = \frac{\eta^2}{N} f^T \widehat{Q}^{-1} f \le \frac{1}{2N}.$$

This changed layer has conditional relative entropy at most $N/4$. Every conditional variance is at most one, so its signed expected activation is at least $\eta/32$ on the good event.

At each of the next r layers give every incoming weight mean $100/N$, and at one last hidden layer use mean $400/N$, always retaining variance $1/N$. Simultaneously over all cube inputs and all these $r+2$ shifted hidden layers, conditional Hoeffding gives absolute empirical-mean error at most $\eta/128$, with total failure probability at most

$$2m(r+2)e^{-N\eta^2/32768} \le 1/8.$$

On this event, the first shifted empirical mean has signed value at least $\eta/64$ everywhere. As long as its minimum signed value is at most $1/100$, a gain-100 layer increases it by at least a factor of two: its expected increase is at least a factor of three, and $\eta/128$ is at most half the current minimum. Once the minimum reaches $1/100$ it stays above that level, by monotonicity and the bound $1/32 - \eta/128 > 1/100$. The choice of r ensures $2^r \eta/64 \ge 1/100$. The final gain-400 layer then has signed expected mean greater than $2/3$ and signed empirical mean greater than $5/8$.

Give the final readout mean $2/N$ per entry and preserve variance $1/N^2$. Its signed mean exceeds $5/4$, and its Gaussian noise is uniformly at most $1/2$ except with probability $2me^{-N/8} \le 1/8$. Its clean margin is consequently greater than $3/4$. The proposed weight law assigns $A_{N,f}$ probability at least $5/8$, hence at least $1/2$.

The relative entropy chain rule bounds this law's entropy from the original Gaussian by

$$N/4 + 5000Nr + 80000N + 2N \le 100000NL.$$

Binary data processing as in Proposition 8.1 proves the mass bound. For completeness, the displayed width assumption dominates each of

$$\frac{8m^2}{p^2}\log(16m^2), \qquad \frac{32768}{\eta^2}\log(16m(r+2)), \qquad 8\log(16m),$$

which are the three concentration thresholds used above. This follows directly by substituting $m = 2^d$, $p = e^{-20(d+1)^5}$ and $d \ge 2$. All adaptive means are measurable functions of preceding weights; the inverse is used only on the event with smallest eigenvalue at least $p/2$. □

9.1 An exponential lower bound for the first tanh kernel eigenvalue

The preceding box argument can be sharpened considerably by exploiting the positive Hermite expansion of the tanh kernel.

Lemma 9.3 (A quantitative first-layer spectrum bound)**.** *Let $d \geq 1$, $D = d + 1$, $m = 2^d$, and let Q be the $m \times m$ matrix*

$$Q(x, y) = \mathbb{E}_w \tanh(w \cdot \bar{x}) \tanh(w \cdot \bar{y}), \qquad w \sim \mathcal{N}(0, I_{d+1}/d).$$

Then

$$\lambda_{\min}(Q) \geq e^{-13D}.$$

Proof. Write $q = D/d \in [1, 2]$, and let h_n be the coefficient of the normalized probabilists' Hermite polynomial $\mathrm{He}_n / \sqrt{n!}$ in $z \mapsto \tanh(\sqrt{q}z)$. The coefficients vanish for even n. For every odd $n \geq 1$, the Fourier sine identity

$$\tanh(\sqrt{q}z) = \frac{1}{\sqrt{q}} \int_0^\infty \frac{\sin(tz)}{\sinh(\pi t/(2\sqrt{q}))}\, dt$$

and $\mathbb{E}[\mathrm{He}_n(Z) e^{itZ}] = (it)^n e^{-t^2/2}$ give

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt.$$

The sine identity follows, for example, by expanding $1/\sinh(\pi t/2) = 2\sum_{j \geq 0} e^{-\pi(j+1/2)t}$ and using the logarithmic derivative of the Euler product for cosh. Fubini is valid in the displayed Hermite integral: near zero use $|\sin(tZ)| \leq t|Z|$, and at infinity the reciprocal hyperbolic sine has exponential decay.

Restrict the positive integral to $[\sqrt{n}, \sqrt{n} + 1]$. Using $n! \leq n^n$, $1/\sinh u \geq 2e^{-u}$, and $1 \leq q \leq 2$ gives

$$|h_n| \geq \sqrt{2} \exp\left[-\frac{n}{2} - \left(1 + \frac{\pi}{2}\right)\sqrt{n} - \frac{1 + \pi}{2} \right] \geq e^{-6n}.$$

The last inequality follows from $\sqrt{n} \leq n$, $n \geq 1$ and $\pi < 4$.

Mehler's identity now yields the nonnegative power series

$$Q(x, y) = \sum_{n \text{ odd}} h_n^2 \left(\frac{1 + \sum_{i=1}^d x_i y_i}{D} \right)^n.$$

The Boolean characters diagonalize this translation-invariant kernel. For a character of degree r, choose the odd integer $n = r$ if r is odd and $n = r + 1$ if r is even. Then $1 \leq n \leq D$. In the multinomial expansion of the nth power, choosing every selected coordinate exactly once, and choosing the constant coordinate once when r is even, contributes $n!/D^n$ to this character's normalized eigenvalue. Every other contribution is nonnegative. Therefore its eigenvalue for Q/m is at least

$$h_n^2 \frac{n!}{D^n} \geq e^{-12n - D} \geq e^{-13D}.$$

Here $n! \geq (n/e)^n$, and $n[1 + \log(D/n)] \leq D$, prove $n!/D^n \geq e^{-D}$. Multiplying by $m \geq 1$ proves the stated lower bound for Q. □

Corollary 9.4 (An exponential-in-depth critical-rate capacity obstruction)**.** *For every $d \geq 1$, set $r = 20(d+1)$ and $L = r + 4$. Then every Boolean function satisfies $J_L(f) \leq 100000L$. Moreover, at every $N \geq \lceil e^{100(d+1)} \rceil$,*

$$\mu_{N,1}(A_{N,f}) \geq \tfrac{1}{4} e^{-200000NL} \qquad \text{for every Boolean } f.$$

Thus at $K = 100000L$ the logarithm of the critical-rate sublevel cardinality is $2^d \log 2 = \exp(\Omega(L))$.

Proof. Repeat the mean-shift constructions from Theorem 9.1 and Proposition 9.2, replacing $p = e^{-20(d+1)^5}$ by $p = e^{-20(d+1)}$ and $r = 20(d+1)^5$ by $r = 20(d+1)$. Lemma 9.3 gives the required covariance lower bound for every $d \geq 1$. The same estimates $\log(1/\eta) < 11(d+1)$ in the variational construction and $\log(1/\eta) < 12(d+1)$ in the finite-width construction ensure that r amplification layers suffice. The three finite-width concentration thresholds remain below $e^{100(d+1)}$ upon the same substitutions. All entropy costs and layer counts are unchanged. □

10 Bounded critical rate for all functions at quadratic depth

Covariance changes are cheaper than coherent mean shifts when the relative change is small. The following sharper capacity obstruction uses that fact. It concerns the critical variational rate, and does not compare the optimal rates of competing functions closely enough to prove a posterior sample-complexity lower bound.

Theorem 10.1 (Universal bounded rate at quadratic depth)**.** *There are universal finite constants $A, K_0 > 0$ such that, for every $d \geq 1$, there is an integer depth $L_d \leq A(d+1)^2$ for which*

$$J_{L_d}(f) \leq K_0 \qquad \text{for every Boolean } f : \{-1,1\}^d \to \{-1,1\}.$$

The depth L_d is the same for all functions on that cube.

We first record exact scalar formulas for a controlled feature law. Put

$$F(q) = \mathbb{E} \tanh^2(\sqrt{q} Z), \qquad a(q) = 1 - F(q), \qquad R(q) = \frac{q[1-F(q)]^2}{F(q)} \quad (q > 0).$$

The value $R(0) = 1$ makes it continuous at zero.

Lemma 10.2 (Rank-one covariance control)**.** *Suppose a feature covariance Q has every diagonal entry equal to q and $Q \succeq v f f^T$ for $v > 0$, where f is Boolean. For any $\epsilon \geq 0$, let the next feature law be coordinatewise tanh of a centered Gaussian with covariance $Q + \epsilon v f f^T$. Its entropy relative to the ordinary next-layer law M_Q is at most $\epsilon^2/4$. Its second moment Q' has constant diagonal q' and obeys $Q' \succeq v' f f^T$, where*

$$u = q + \epsilon v, \qquad q' = F(u), \qquad v' = (1+\epsilon) v [1 - F(u)]^2. \tag{9}$$

Consequently, for $w = v/q$ and $w' = v'/q'$,

$$w' = w \frac{1+\epsilon}{1+\epsilon w} R(u).$$

All these scalar quantities depend only on q, v, ϵ, not on the other entries of Q or on the choice of f.

Proof. The Loewner bound implies $f \in \mathrm{range}(Q)$ and $v f^T Q^+ f \le 1$. The two Gaussian laws have the same support, and the rank-one determinant identity gives their relative entropy as

$$\frac{1}{2}\{t - \log(1+t)\}, \qquad t = \epsilon v f^T Q^+ f \le \epsilon.$$

This is at most $t^2/4 \le \epsilon^2/4$. Applying coordinatewise tanh can only decrease relative entropy. Write the changed Gaussian as $\sqrt{(1+\epsilon)v}\, fZ + G$, where Z is standard Gaussian and G is an independent centered Gaussian of covariance $Q - v f f^T$. Its conditional feature mean given Z is $f\psi(Z)$, since the variances of the coordinates of G all equal $q - v$. Thus $Q' \succeq \mathbb{E}\psi(Z)^2 f f^T$. Gaussian integration by parts and projection onto the first Hermite polynomial give

$$\mathbb{E}\psi(Z)^2 \ge (\mathbb{E} Z\psi(Z))^2 = (1+\epsilon)v\,[\mathbb{E}\,\mathrm{sech}^2(\sqrt{u}Z)]^2.$$

This is the claimed v', and the diagonal formula and ratio identity follow directly. In particular, these formulas automatically retain $0 < v' \le q'$, since they arise from a valid covariance lower bound. □

Proof of Theorem 10.1. Write $D = d+1$ and $m = 2^d$. Begin with the unperturbed first feature law. Its covariance Q has constant diagonal $q_0 = F(D/d) \in [F(1), F(2)]$, and Lemma 9.3 implies

$$Q \succeq e^{-20D} I \succeq v_0 f f^T, \qquad v_0 = e^{-20D}/m.$$

This first layer costs zero entropy. We will use Lemma 10.2 at each subsequent layer, tracking its guaranteed scalar component v, not the largest possible rank-one component of the covariance.

We collect universal scalar estimates. The function F is increasing, $0 < F(q) < q$ for $q > 0$, and Taylor expansion gives

$$F(q) = q - 2q^2 + O(q^3), \qquad R(q) = 1 + O(q^2).$$

There consequently exist universal $u_0 > 0$ and $C \ge 1$ such that, for $0 < u \le u_0$,

$$F(u) \le u - u^2, \qquad R(u) \ge 1 - Cu^2. \tag{10}$$

We may and do take $u_0 \le 1/4$. A useful global lower bound is

$$F(q) \ge \frac{q}{1+3q}. \tag{11}$$

Indeed $\tanh^2 t \ge t^2/(1+t^2)$ follows from $|\sinh t| \ge |t|$. Cauchy–Schwarz applied to $V = qZ^2$ gives $\mathbb{E}[V/(1+V)] \ge (\mathbb{E}V)^2/\mathbb{E}[V(1+V)] = q/(1+3q)$.

Choose a sufficiently small universal $v_* > 0$ satisfying

$$v_* \le \min\{u_0/4, 1/64, 1/(100C)\}.$$

Then choose a positive universal c so small that

$$c \le \min\{v_*/4, u_0/3, 1/3, 1/(144C)\}.$$

Put $\delta = c/D$.

Free cooling. Apply the unmodified next-layer law, that is, $\epsilon = 0$, until $q \le \delta$ for the first time. This costs no entropy. There is a universal $c_F > 0$ with $q - F(q) \ge c_F q^2$ for $0 < q \le F(2)$: the ratio

is positive and continuous, and tends to two at zero. Hence the number of cooling layers is at most a universal constant times $1/\delta$. The final cooled variance q_c lies in $[\delta/2, \delta]$, by monotonicity and (11).

During cooling the ratio $w = v/q$ is multiplied by $R(q)$ at each step. This loses at most a fixed positive factor, uniformly in the number of cooling layers. To see this directly, on $[0, F(2)]$ one has $|\log R(q)| \le C_R q^2$ for a universal C_R, by continuity, positivity and the expansion at zero. Also $\sum q_s^2 \le c_F^{-1} \sum (q_s - q_{s+1}) \le F(2)/c_F$. Thus the cooled ratio satisfies

$$w_c \ge c_0 v_0, \qquad \log(1/w_c) \le A_0 D$$

for universal $c_0 > 0, A_0 < \infty$.

Amplifying the relative component. Now set $\epsilon = \delta$ at each layer until $w \ge 1/2$. Throughout this phase $q \le 2\delta$. Indeed $u = q + \delta v \le (1+\delta)q$, and the increasing function $u - u^2$ on the relevant interval gives

$$q' \le 2\delta(1+\delta) - [2\delta(1+\delta)]^2 \le 2\delta.$$

While $w \le 1/2$, the exact ratio identity and $u \le 3\delta$ imply

$$\log(w'/w) \ge \frac{\delta}{4} - 18C\delta^2 \ge \frac{\delta}{8}.$$

Here we used $\log(1 - Cu^2) \ge -2Cu^2$, valid by the chosen constants. The number τ of these layers is therefore at most $8A_0 D/\delta + 1 = O(D^2)$. Their total entropy is at most

$$\tau\delta^2/4 \le 2A_0 D\delta + \delta^2/4 = O(1).$$

Furthermore, since $u \ge q$, monotonicity and (11) imply $1/q' \le 1/q + 3$ at every layer. Thus, at the end of this phase,

$$q \ge \frac{1}{2/\delta + 3\tau} \ge \frac{c_1}{D^2}, \qquad v \ge \frac{q}{2} \ge \frac{c_1}{2D^2}$$

for some universal $c_1 > 0$. At the same time $v \le q \le 2\delta < v_*$.

Growing the absolute component. Set $\epsilon = 16v$ until $v \ge v_*$. The invariant $w \ge 1/3$, or equivalently $q \le 3v$, is preserved. To check this, while $v < v_*$ one has $u = q + 16v^2 \le 4v$. If $w \ge 1/2$, the ratio identity gives $w' \ge \frac{1}{2}(1 - 16Cv^2) \ge 1/3$. If $1/3 \le w \le 1/2$, it gives

$$\log(w'/w) \ge 4v - 32Cv^2 \ge 0.$$

The initial ratio is at least $1/2$, so induction proves the invariant. Using $F(u) \le u$ also gives

$$v' \ge v(1+16v)(1-2u) \ge v(1+16v)(1-8v) \ge v + 4v^2, \qquad v' \le v + 16v^2.$$

Thus $1/v - 1/v' \ge 2$ until the stopping time, which is $O(D^2)$. The final value is between v_* and $2v_*$, and the total entropy of this phase telescopes:

$$\sum \frac{(16v)^2}{4} = 64 \sum v^2 \le 16 \sum (v' - v) \le 32 v_*.$$

Finally choose output $z = f$. Its signed margin is one, and $Q \succeq v_* f f^T$ bounds its Gaussian readout cost by $1/(2v_*)$. All entropy costs plus this readout cost are bounded by a universal constant K_0. The number of hidden layers, and therefore the depth, is at most a universal constant times D^2. Crucially, q_0, v_0, every subsequent scalar recursion, every control, and every stopping rule depend only on d. Although the full covariance matrices depend on f, the resulting depth L_d is common to all Boolean functions. This proves the theorem. □

This construction sharpens an absolute-rate capacity obstruction. It still does not show that the optimal rate of a complex competitor is below, or sufficiently close to, the optimal rate of a specified simple target. That comparison remains essential for a posterior lower bound.

11 What ordinary training and other stability notions do not guarantee

The following results delimit the empirical question. They do not imply that fragile outcomes are common under standard initialization, or that their Boolean functions are statistically difficult.

Proposition 11.1 (An open gradient-descent basin of fragile exact fits)**.** *For every even $L \geq 256$, there is a width-one, depth-L tanh network on $x \in \{-1, 1\}$ with output $z(x) = x$, energy $\kappa_1 \leq L + 76$, and a postactivation sign-changing disturbance of size at most*

$$\frac{2}{L} e^{-\sqrt{L}/4}.$$

This disturbance is smaller than both calibrated support radii

$$r_{\mathrm{ref}} = \frac{.01}{\sqrt{1 + (L-1)E_v}}, \qquad r_S = \frac{.01}{\sqrt{1+S}}$$

from the preceding report. For sufficiently small fixed step size, full-batch gradient descent on the unregularized squared loss for the two labeled inputs has an open set *of initializations converging to nearby exact fits that still violate both support conditions.*

Proof. Put $k_s = \min(s, L - s)$ and

$$r_s = \frac{\exp[-(k_s - 1)/\sqrt{L}]}{2k_s}, \quad a = \operatorname{atanh}(1/2), \quad b_s = \frac{\operatorname{atanh}(r_s)}{r_{s-1}}, \quad v = 2.$$

Use first-layer input coefficient a and bias zero, followed by scalar weights b_s. Then $h_s(x) = r_s x$ and $z(x) = x$. The reciprocal-ratio estimate from the preceding report's bounded-excess construction gives $\kappa_1 \leq L + 76$; replacing its constant input by x preserves the computation and reduces the first-layer energy charge. Also $b_s \geq e^{-1/2}/2 > 0.30$.

At $x = 1$, the middle-layer disturbance $-2r_{L/2}$ changes the entire remaining computation to its negative and gives final output -1. Its norm is at most $2L^{-1}e^{-\sqrt{L}/4}$. Here $E_v = 4$, so the ratio of this upper bound to r_{ref} is at most $400e^{-\sqrt{L}/4}/\sqrt{L} < 1$ for $L \geq 256$. Since $S = \kappa_1 \leq L + 76$ and $L + 77 \leq 4L - 3$, it is also below r_S.

For the training claim, let $F(\theta) = (z_\theta(1), z_\theta(-1))$ and $\ell(\theta) = \frac{1}{2}\|F(\theta) - (1, -1)\|^2$. At the constructed point θ_*, the columns of DF corresponding to the first input coefficient and bias are $(c, -c)$ and (c, c), respectively, with $c > 0$. Thus DF has full row rank. In a sufficiently small closed ball about θ_*, its least singular value is at least $\mu > 0$, its norm is at most G, and $\|\nabla^2 \ell\| \leq M$. The clean margins remain $> 1/2$, and the *fixed* middle-layer attack remains strictly inside both calibrated balls and gives strictly negative output.

For $0 < \eta \leq 1/M$, the descent lemma gives, as long as the iterates remain in this neighborhood,

$$\ell(\theta_{j+1}) \leq (1 - \eta\mu^2)\ell(\theta_j).$$

Their total movement is at most

$$\frac{\eta G\sqrt{2\ell(\theta_0)}}{1 - \sqrt{1 - \eta\mu^2}} \leq \frac{2G\sqrt{2\ell(\theta_0)}}{\mu^2}.$$

Starting in a sufficiently small open ball makes this displacement smaller than the distance to the boundary, proving by induction that the iterates stay inside. The summable increments give convergence, and the geometric loss decay gives exact interpolation at the limit. Every limit retains the strict fragility witness. □

A nondegenerate Gaussian initialization gives this open basin positive probability. There is no useful lower bound here on that probability; the result rules out a universal guarantee, not a typical-case guarantee. It concerns unregularized full-batch squared-loss GD, rather than arbitrary weight decay, Adam, or a prescribed finite training budget.

The same construction also violates the numerical-stability rule with the radius in (12) and permitted output change $1/16$. For even $L \geq 1024$, the attack-to-radius ratio is at most $4000e^{-\sqrt{L}/4}/\sqrt{L} < 1$, and the attack changes the output by two. The same open-basin argument applies. Thus that alternative condition is also not a universal consequence of ordinary gradient descent.

Proposition 11.2 (Parameter flatness need not give activation robustness)**.** *The networks just constructed preserve their correct signs under every simultaneous Euclidean parameter perturbation of norm at most* $1/4$*. At the exact fit their squared-loss Hessian has operator norm* $O(L)$*. Nevertheless their absolute internal-noise radius is at most* $2L^{-1}e^{-\sqrt{L}/4}$*.*

Proof. Every internal scalar weight and the readout stay positive under the stated parameter perturbation. At both inputs the first signed preactivation is at least $\operatorname{atanh}(1/2) - \sqrt{2}/4 > 0$. Odd monotonic tanh layers therefore retain the signs throughout the parameter ball.

For $t > 0$, $t \tanh'(t)/\tanh(t) \leq 1$, since $\sinh(t)\cosh(t) \geq t$. Each internal log-weight elasticity of the positive output is consequently at most one. All scalar weights are bounded below by a positive numerical constant, so every parameter derivative at either input is bounded by a numerical constant. There are $L+1$ parameters, and at interpolation $\nabla^2 \ell = DF^\top DF$; its norm is therefore $O(L)$. □

This is a separation between stability notions. Parameter flatness could still support a different learning theorem, but it cannot be substituted for absolute activation robustness in the existing proof.

Proposition 11.3 (Eventual membership after readout amplification)**.** *Every fixed network with nonzero outputs on the finite Boolean cube belongs to either calibrated support after sufficiently large positive readout amplification.*

Proof. Continuity and finiteness of the input set give $\rho, m > 0$ such that every hidden disturbance of total normalized norm at most ρ preserves each clean sign and leaves unscaled signed output at least m. Multiplying the readout by A makes either calibrated radius r_A tend to zero. For large enough A, $r_A < \rho$ and $Am > r_A$, so even adding the permitted scalar output noise cannot change a sign. Increasing A also makes the clean output margin exceed $1/2$. □

This does not remove the Gaussian cost: the readout energy grows as A^2, and the required A may be exponentially large. It does mean that empirical membership before and after output rescaling answers different questions. The calibrated condition is a joint restriction on robustness and scale.

Proposition 11.4 (Global and typical margins can differ exponentially)**.** *There is a one-hidden-unit network with bounded input weight norm and Boolean function* x_1 *whose minimum output margin is* $\tanh(2^{-d})$*. Enforcing the original global clean margin* $> 1/2$ *by readout amplification requires a multiplier at least* 2^{d-1}*, while permitting an exceptional fraction of inputs requires only polynomial amplification in the reciprocal exceptional fraction.*

Proof. Take

$$a_d(x) = \sum_{j=1}^{d} 2^{-j} x_j, \qquad z_d(x) = \tanh(a_d(x)).$$

The squared input weight norm is less than $1/3$. The first coefficient exceeds the sum of the others, so the sign is x_1, and the minimum absolute preactivation is 2^{-d}. Thus the global clean-margin multiplier must exceed $[2\tanh(2^{-d})]^{-1} \geq 2^{d-1}$. Under the uniform cube, a_d is uniform on the odd multiples of 2^{-d} in $(-1, 1)$, giving

$$\Pr(|a_d(X)| \leq u) \leq u + 2^{-d} \qquad (0 \leq u \leq 1).$$

Consequently for $A \geq 1$,

$$\Pr(|Az_d(X)| \leq 1/2) \leq \operatorname{atanh}(1/(2A)) + 2^{-d} \leq A^{-1} + 2^{-d}.$$

For example, $A \geq 2/\epsilon$ suffices when $2^{-d} \leq \epsilon/2$. □

The last example concerns the inherited global clean-margin convention, not an additional failure of internal robustness. It motivates distinguishing training, held-out, and worst-input quantities, and investigating support conditions that explicitly tolerate a small exceptional input mass.

Proposition 11.5 (A generic Gaussian halfspace has a tiny cube margin)**.** *Let $d \geq 64$ and $w \sim \mathcal{N}(0, I_d/d)$. Then*

$$\Pr\left\{\min_{x \in \{-1,1\}^d} |w \cdot x| \leq d^2 2^{-d}\right\} \geq 1 - \frac{6}{d}.$$

On this event, the one-hidden-unit output $A\tanh(w \cdot x)$ needs $A > 2^{d-1}/d^2$ to satisfy the global clean margin $> 1/2$. This is a statement about Gaussian coefficients, not the distribution of weights after training and not a sample-complexity lower bound.

Proof. For $0 < \tau \leq d^{-1/2}$ put $Z = 2^{-d} \sum_x \mathbf{1}\{|w \cdot x| \leq \tau\}$ and $p = \mathbb{E}Z = \Pr(|G| \leq \tau)$ for $G \sim \mathcal{N}(0, 1)$. For independent uniform cube points x, x', their Gaussian evaluations have correlation $\rho = d^{-1} \sum_j x_j x'_j$. When $\rho \neq \pm 1$, the bivariate Gaussian density bound gives

$$\frac{\Pr(|w \cdot x| \leq \tau,\ |w \cdot x'| \leq \tau)}{p^2} \leq \frac{e^{\tau^2}}{\sqrt{1 - \rho^2}},$$

using $p \geq \sqrt{2/\pi}\, \tau e^{-\tau^2/2}$. The antipodal or identical pairs have total probability 2^{1-d}. For $|\rho| \leq 1/2$, $(1 - \rho^2)^{-1/2} \leq 1 + \rho^2$. Otherwise, excluding the two degenerate correlations, this factor is at most $\sqrt{d/2}$. Since $\mathbb{E}\rho^2 = 1/d$ and $\Pr(|\rho| > 1/2) \leq 2e^{-d/8}$,

$$\mathbb{E}\left[\frac{\mathbf{1}\{\rho \neq \pm 1\}}{\sqrt{1 - \rho^2}}\right] \leq 1 + \frac{1}{d} + \sqrt{2d}\, e^{-d/8} \leq 1 + \frac{2}{d}.$$

Here the last inequality holds for $d \geq 64$. Using $p \geq \tau/2$ and $e^{\tau^2} \leq 1 + 2/d$, Chebyshev gives

$$\Pr(Z = 0) \leq \frac{\operatorname{Var} Z}{p^2} \leq \frac{2^{2-d}}{\tau} + \frac{5}{d}.$$

Take $\tau = d^2 2^{-d}$; this is at most $d^{-1/2}$ for $d \geq 64$. The right-hand side is at most $6/d$. The output-scaling claim follows from $|\tanh u| \leq |u|$. □

The same observation includes a Gaussian first-layer bias by passing to the $(d+1)$-dimensional cube: absolute affine values on the original cube are exactly absolute homogeneous values on one representative from every antipodal pair of the augmented cube.

12 Expanded experiments on trained networks

The strongest positive observation is that twenty unmodified trained digit networks satisfy the new numerical-stability condition at *every real input*, with interval-verified matrix inequalities and an exact rational final bound. All twenty also fit every training example with signed output greater than 1/2. The condition is nevertheless not universal: a separate well-performing trained network has an interval-verified violation even at an actual held-out image. The old sign condition and the new numerical condition must be kept separate when interpreting these findings.

12.1 Protocol and what is being measured

We trained 168 networks in five retained cohorts: 80 initial synthetic models, 32 synthetic extensions, 32 main digit models, 16 deeper digit models, and eight smaller-step-size controls. Configurations were fixed before training each cohort; later cohorts were chosen after some earlier results were visible. These are exploratory experiments, not a publicly preregistered confirmatory study. Failed fits were retained. No Bayesian posterior was sampled: the experiments test properties of trained parameters, not posterior prediction probabilities or statistical sample complexity. The new numerical condition was proposed after training, and its evaluation is a post-training audit of the resulting networks.

The architecture is $h_0 = (x, 1)$, $h_s = \tanh(h_{s-1}W_s)$, and $z = h_J v$, with no biases beyond the input augmentation. Synthetic widths are 32 or 64, dimensions $8, 10, 12$, and hidden depths $4, 8, 16, 32$. Tasks include majority, parity, random Boolean labels, and labels from a fixed small tanh teacher. Training uses either the whole cube or a fixed subset, with all cube inputs subsequently audited. Image models have width 64 and hidden depths $4, 16, 32, 64$. Initialization uses Gaussian first and final weights and orthogonal internal matrices, with gains 0.7, 1.1, or 1.5 in the synthetic cohorts and 1.1 for images.

Both optimizers run for 6000 steps with 10^{-6} weight decay. The code's label `sgd` means *full-batch vanilla gradient descent*, without momentum or minibatch noise. Its learning rate is 0.05. Adam uses learning rate 0.003, except in the separately declared 0.0003 control; rates are multiplied by 0.3 after 80% of training. Synthetic losses are half the squared error, while image losses are binary logistic. No test-driven rescaling is used for images. Exact saved floating-point weights, data splits, protocols, source, and detailed records accompany the report.

For clarity, write $E_v = N\|v\|_2^2$ and

$$\|\eta\|_*^2 = \sum_{s=1}^{J} \frac{\|\eta_s\|_2^2}{N} + |\eta_{\text{out}}|^2.$$

There are four distinct measurements:

1. correctness relative to the target labels;
2. the inherited global clean-margin requirement $|z(x)| > 1/2$;
3. preservation of the network's own clean sign throughout radius $r_{\text{sign}} = 10^{-2}/\sqrt{1 + JE_v}$;
4. numerical output change at most $1/16$ throughout $r_{\text{num}} = 10^{-3}/\sqrt{1 + JE_v}$, as in Section 14.

The last condition imposes no all-input clean margin. Its likelihood still requires $y_i z(x_i) > 1/2$ for *every training example*; we audit this separately on the raw trained networks. A constant failed fit can be stable without being eligible under this likelihood.

Positive pointwise tests use the finite-radius clean-transport certificate, including nonlinear Taylor remainder bounds; they are not just tests of the clean Jacobian. Unless explicitly marked

interval verified, the sufficient inequalities are evaluated in float64. Failure to certify is reported as unresolved. Negative witnesses use explicit perturbations strictly inside the normalized noise ball. Selected witnesses and the twenty global positive results receive the stronger checks described below.

12.2 Full-cube synthetic audits

Of 112 synthetic runs, 105 fit every training label, and all 105 have correct signed training outputs greater than 1/2. Under the old global sign-support condition, 52 pass a sufficient certificate on their entire cube, 59 fail the clean-margin requirement at an explicit cube point, and one remains unresolved. Seventy networks certify their own sign on the whole cube, independently of the clean-margin requirement; 35 have explicit, strictly budget-checked sign-flipping perturbations.

Cohort	Runs	Fits	Old support in/out/?	All-cube sign certified	Sign attacks found
Full cube, $d = 8$	24	23	23/1/0	24	0
Half cube, $d = 8$	24	23	12/12/0	17	6
Initialization gains, $d = 8$	16	16	7/9/0	10	5
Subset, $d = 10$	16	15	6/10/0	10	5
Subset, $d = 12$	8	8	2/6/0	5	2
Small teacher, $d = 10$	16	13	0/15/1	0	14
Depth 32, $d = 10$	8	7	2/6/0	4	3

Here "out" means a witnessed clean-margin failure; "?" is not a nonrobustness claim. Counts include failed fits.

The apparent fragility is concentrated near the decision boundary. Across all 105 successful fits, *every* cube point whose clean absolute output exceeds 1/2 passes the finite-radius sign certificate: 80,603 input–model pairs in total, with a nonempty qualifying set for every model. Seven networks classify their entire held-out set correctly but fail the global clean margin; all seven certify sign stability on their whole cube. Two depth-16, dimension-12 full-batch-GD models achieve held-out accuracies 99.55% and 99.44% yet admit sign attacks at rare inputs with clean margins about 0.0072 and 0.0021.

Rescaling the readout using the global minimum margin makes 108 of the 112 networks pass the old support certificate, leaving four unresolved. This is a changed-network diagnostic, not a raw training outcome: rescaling raises the margin and simultaneously shrinks the required noise radius, and may incur a large Gaussian energy cost. On a finite cube, eventual membership under sufficiently large rescaling is possible for every network with no zero clean outputs, so this statistic alone provides little evidence of a natural training bias.

At the new numerical radius, 108 of 112 raw networks pass on the entire cube, including *all 105* training-margin-eligible fits. Of 90,112 input–model pairs, 90,108 pass; the four unresolved points belong to failed training runs. These positive counts use floating-point evaluation of sufficient bounds, not interval arithmetic.

12.3 Actual images, depth, and failed optimization

We use the locally bundled `sklearn` handwritten-digit dataset: 1797 images of 8×8 pixels, with tasks "3 versus 8" and "even versus odd". A 70/30 split is stratified by the original digit labels. Boolean features threshold pixels at 8; continuous features map a pixel value p to $p/8 - 1$. These are distinct experiments. The continuous feature results extend the numerical audit, not the Boolean

input premise of the statistical theorem. The main cohort crosses both tasks, both feature maps, depths $4, 16$, both optimizers, and two seeds. All 32 models satisfy the training-margin likelihood.

Task	Features	Test accuracy	Own-sign certificate	Margin and sign
3 versus 8	Boolean	95.37–97.22%	100%	98.15–100%
3 versus 8	Continuous	97.22–100%	100%	99.07–100%
Even/odd	Boolean	93.33–95.00%	99.63–100%	98.89–100%
Even/odd	Continuous	97.04–98.15%	99.81–100%	98.89–100%

Each row contains eight models; ranges are across those models. The last two columns concern *held-out image inputs* and the old radius, not all possible inputs. Every held-out point with $|z| > 1/2$ passes the sign certificate in all 32 models. Thresholding can give identical feature vectors in different images: at most 23 held-out vectors per model occur in the training set. Counts and accuracy restricted to novel feature vectors are recorded separately; these are descriptive robustness experiments, not a benchmark generalization claim.

The separate depth-32/64 cohort uses only the 3-versus-8 task. All eight full-batch-GD models fit the training likelihood and achieve 96.30–100% test accuracy. All eight standard-rate Adam runs fail to fit: depth-32 accuracies range from 50.93% to 75.93%, and all four depth-64 runs collapse to 50.93%. The separately declared smaller-rate Adam cohort is retained alongside these failures: seven of eight fit every training margin and achieve 96.30–100% test accuracy; one depth-64 Boolean-feature run still collapses to 50.93%. Stable constant predictions in failed runs do not support a claim about successful learning. No failed optimization run was replaced by a successful rerun in the reported cohort.

12.4 Off-data sign failures and global numerical stability

For each of the sixteen Boolean-feature models in the main digit cohort, we sampled 10,000 uniform 64-bit inputs. All sixteen have explicit clean-margin violations and sign-flipping perturbations at such inputs; all sixteen attacks still work after deleting scalar output noise. These inputs generally do not resemble digit images. A separate attack on 32 sampled points just above clean margin $1/2$ per model found no flips; a failed attack is not a certificate.

Three hidden-only sign witnesses are checked using exact rational noise budgets and 60-digit interval forward evaluation of the saved binary floating-point weights and perturbations. Their outputs change as follows:

Model ID	clean output	perturbed output
203	$+0.00164981$	-0.0274141
204	$+0.0287959$	-0.364879
205	-0.0192534	$+0.300013$

All three already fail the inherited $1/2$ margin. They therefore do not show that added internal robustness excludes a parameterization which otherwise meets the original global-margin support restriction.

The new numerical condition gives a substantially better positive result. All 10,368 held-out input–model pairs in the main digit cohort pass its sufficient inequality. Twenty of these 32 raw trained models—all sixteen depth-4 models and four depth-16 models—have a *global* positive certificate, covering every real input, not only images or the cube. Ten of these twenty use Boolean features and ten use continuous features. These twenty results are additionally interval verified. For

each hidden-to-hidden matrix, an exact dyadic-integer Gram matrix and a 90-digit interval $LDL^\top$ decomposition prove $W^\top W \prec g^2 I$ for an explicitly stored rational g^2. Cauchy–Schwarz and global Lipschitzness of tanh bound the simultaneous noise gain. The final comparison to $(1/16)^2$ is an exact rational inequality. The resulting output-change upper bounds range from 0.00453 to 0.04950, all below 1/16. This is a numerical proof trusting the interval-arithmetic implementation, not a Lean formalization.

The radius calibration matters. Reusing these same verified operator bounds and rescaling the final inequality with exact rational arithmetic, sixteen, twelve, and four of these twenty models also certify output change at most 1/16 when the radius coefficient 0.001 is increased to 0.002, 0.005, and 0.01, respectively. The last coefficient equals the old sign-radius coefficient, although numerical stability and sign stability remain different properties. A failed sufficient inequality at a larger radius does not establish a violation. The rescaling code and records are supplied separately from the frozen experiment archive.

Cohort	Models	Training-margin eligible	Numerical certificate input–model pairs	Full tested set certified models
Synthetic, full cubes	112	105	90,108/90,112	108
Main digits, held out	32	32	10,368/10,368	32
Deep digits, held out	16	8	1,728/1,728	16
Smaller-rate digits, held out	8	7	863/864	7

The deep numerical certificates in this table concern held-out images, not global input support. Three failed synthetic fits and eight failed standard-rate deep image fits pass their entire tested set; this illustrates why the likelihood eligibility column is essential. The one unresolved smaller-rate image point has the negative witness described next. Pooled input–model counts reuse inputs across models and are not counts of independent test examples for statistical inference.

Model 203 illustrates the distinction particularly clearly. It has the old hidden-only sign flip displayed above, but the new global certificate bounds output changes by 0.004744 throughout the smaller numerical radius, for every real input. Numerical stability intentionally allows small-margin decisions to change sign; it need not rule out otherwise useful trained networks because of a decision boundary away from the data.

There are also genuine counterexamples to automatic numerical stability. The smaller-rate Adam model 400, with depth 32 and Boolean pixels, fits every training margin and achieves 98.15% test accuracy. At held-out image index 25, an exactly budget-checked perturbation changes its output from about $+0.188308$ to -0.006960, an absolute change of 0.195268. A 120-digit interval calculation proves that this exceeds 1/16. The true label is -1, so this particular perturbation *corrects* an existing classification error; it is a counterexample to numerical stability, not an induced label error. Other verified violations occur on uniform-cube inputs. One changes 5.296417 to 5.218443, which violates raw numerical stability while leaving both clipped scores equal to one. This supports considering the bounded-score condition of Section 16; it does not establish that every trained model satisfies that weaker condition. In fact, clipping leaves model 400's small-output held-out violation unchanged, so that trained model also violates the bounded-score condition at the same radius.

12.5 Other diagnostics and limits of the evidence

The product of layer Frobenius norms can be enormous even when the new stability test succeeds. For raw main digit models, the median $\log_{10}(\sqrt{d}\prod_s \|W_s\|_F)$, including the readout as its last factor, is 4.80 at depth 4 and 15.74 at depth 16. In the standard deeper cohort it is 31.34 at depth 32 and

58.96 at depth 64. Thus a small product-Frobenius condition can be much more restrictive on these trained parameterizations than finite-noise output stability. These finite depths do not establish an asymptotic exponential law, and the experiment does not exclude a lower-product implementation of the same function.

A separate, fixed activation-rounding diagnostic rounds every hidden activation and the final output to 8, 12, or 16 binary fractional bits, without rounding weights. Across the main digit models, 8-bit rounding changes two held-out signs; 12- and 16-bit rounding change none. Only 16 bits uniformly place the worst-case rounding budget within the old radius for all 32 models. Success of one deterministic rounding rule is weaker than robustness against every allowed perturbation.

These are modest tanh multilayer perceptrons, not evidence about large contemporary ResNets or Transformers. The strongest positive global certificates occur at depths 4 and 16; favorable on-image behavior at depths 32 and 64 does not prove all-input membership. None of these finite experiments proves a polynomial-depth scaling law or a universal property of gradient descent. They support the naturalness of numerical stability for some successful trained networks, expose its exceptions, and show why training correctness, confident-point stability, global margin, and global numerical stability should be reported separately.

13 A posterior certificate using robustness only where it is needed

The earlier compression proof uses robustness only at its final prediction comparison. This permits a version that does not assume robust support. Fix a radius $0 < r \le 1$. Define

$$b_w(x; r) = \mathbf{1}\{\text{some perturbation of total RMS norm at most } r \text{ makes } \operatorname{sign} z_w(x)\, z_w(x; \eta) \le 0\}.$$

Clean zeros are declared bad. For a parameter law Q, write $b_Q(x; r) = \mathbb{E}_Q b_w(x; r)$, and let $b_D(Q; r)$ and $b_s(Q; r)$ be its population and empirical input averages. The reference measure D here is the input distribution, not a target-mass exponent.

Proposition 13.1 (Compression with a pointwise defect)**.** *For $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$, the same finite class $\mathcal{H}(d, L, K, r, \alpha)$ from the earlier robust entropy-compression theorem admits a probability law Γ_Q with*

$$\left|\mathbb{E}_Q \mathbf{1}\{\operatorname{sign} z_w(x) \ne f(x)\} - \mathbb{E}_{h\sim\Gamma_Q} \mathbf{1}\{h(x) \ne f(x)\}\right| \le \alpha + b_Q(x; r) \qquad (x \in \{-1, 1\}^d).$$

The construction and the class do not depend on the target f.

Proof. Run the earlier construction for the unrestricted law Q. Its entropy budgets, clipping bounds, moment estimates, and common candidate marginal do not use robustness. At each fixed input, its comparison coupling has the original full-network marginal Q and, except with probability less than $\alpha/4$, produces an exact postactivation perturbation of norm less than $r/2$. When $b_w(x; r) = 0$, the candidate has the original sign and margin greater than $r/2$: otherwise an additional output injection within the remaining norm budget could reach zero. The same finite-grid rounding preserves that sign. In the small-width branch, good parameter clipping likewise preserves the original network until rounding; local robustness gives its output margin greater than r. Adding the probability $b_Q(x; r)$ to the original exceptional probability proves the assertion. No conditioning on a robustness event is performed, so no extra entropy cost or inconsistency between pointwise couplings is introduced. □

Theorem 13.2 (Robustness-defect certificate for a consistent posterior)**.** *With probability at least $1 - \delta$ over n iid labeled examples, simultaneously for all N, t, Q with the stated entropy bound,*

$$\mathrm{GibbsErr}_D(Q) \le 2\,\mathrm{GibbsErr}_s(Q) + 3\alpha + b_D(Q; r) + 2b_s(Q; r) + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}.$$

In particular the empirical Gibbs error term vanishes for the original noiseless-consistent Gaussian posterior. This statement imposes no robustness restriction on that prior.

Proof. On the usual finite-class relative-Chernoff event, $\mathrm{err}_D(h) \le 2\,\mathrm{err}_s(h) + 8[\log|\mathcal{H}| + \log(1/\delta)]/n$ for every $h \in \mathcal{H}$. Average under the common law Γ_Q and use the preceding proposition once on the population and once on the empirical inputs. □

For a fixed posterior selected without using an independent unlabeled sample U of size m, Hoeffding also gives

$$b_D(Q;r) \le b_U(Q;r) + \sqrt{\log(1/\delta')/(2m)}$$

with probability at least $1-\delta'$. Sound upper certificates for $b_w(x;r)$ may replace its exact value. This is a mathematical posterior certificate; one trained network is not automatically a draw from the specified Bayesian posterior. The training defect cannot simply be omitted: the posterior depends on the training inputs, so small population defect alone does not establish a small empirical defect.

13.1 A distributional prior with robust training likelihood

There is also a genuine learning theorem under a weaker prior restriction. Suppose the input distribution D is known before the labels, set $J = L-1$, $E_v = N\|v\|^2$, and put

$$r_{\mathrm{ref}}(w) = \frac{1}{100\sqrt{1+JE_v}}.$$

For fixed $0 \le \rho < 1$, condition the raw Gaussian only on

$$D\{x : b_w(x; r_{\mathrm{ref}}(w)) = 1\} \le \rho.$$

Unlike the earlier prior, this condition imposes no all-input clean margin. For every observed example (x_i, y_i), use the likelihood requiring $y_i z_w(x_i;\eta) > 0$ for every perturbation of norm at most $r_{\mathrm{ref}}(w)$. Thus the likelihood enforces a robust answer on training inputs. This is a modified learning rule, not the original sign-only likelihood with a silently omitted empirical defect.

Corollary 13.3 (Polynomial-depth learning up to a distributional defect)**.** *For targets with a width-C, depth-$(L-1)$ layered Boolean circuit, this posterior has Gibbs error at most $\rho+\varepsilon$ after* $\mathrm{poly}(d, L, C, \varepsilon^{-1}, \log(1/\delta))$ *examples, simultaneously for all $N \ge C$ and all $0 < t \le 1$. The polynomial has degree independent of depth, width, and Gaussian scale.*

Proof. The earlier robust target cylinder belongs to this prior support and satisfies the robust training likelihood for every sample. If its raw mass is at least $\exp(-ND_*/t^2)$, posterior KL is at most ND_*/t^2. With $B = \max\{8, 8[D_* + \log(1/\beta)]\}$, the same Gaussian readout tail bound gives posterior mass at most β to $E_v > B$. On the complementary event the required radius is at least $r_0 = 1/[100\sqrt{1+JB}]$. The conditional posterior has entropy bound $N(D_*+1)/t^2$, population robustness defect at most ρ, and empirical defect zero. Apply the preceding theorem and add the discarded mass:

$$\mathrm{GibbsErr}_D(Q_s) \le \beta + 3\alpha + \rho + 8[\log|\mathcal{H}| + \log(1/\delta)]/n.$$

Choosing $\beta = \varepsilon/4$, $\alpha = \varepsilon/12$ and the earlier polynomial sample size proves the claim. The finite class is independent of N, t, which gives one simultaneous sample event. □

If a fixed reference input distribution ν is used instead, and $D \le M\nu$, replace the population defect ρ by $M\rho$. This makes explicit what distributional information enters the prior. The result does not assert that the support can be estimated from arbitrary unlabeled data without an additional uniform-convergence argument.

14 Numerical output stability without an all-input margin

The preceding sign-robustness conditions can be replaced by a condition on *numerical changes in the output*. This permits the clean output to be arbitrarily close to zero on unobserved inputs. Only the observed examples are required to have the original positive training margin.

Let $J = L - 1$, $E_v(w) = N\|v\|_2^2$, and set

$$r_{\mathrm{num}}(w) = \frac{1}{1000\sqrt{1 + JE_v(w)}}. \tag{12}$$

Use the same additive postactivation and scalar output disturbances as before. Say that w is *numerically stable* if

$$|z_w(x;\eta) - z_w(x)| \le \frac{1}{16} \quad \text{for all } x \in \{-1,1\}^d \text{ and all } \|\eta\|_* \le r_{\mathrm{num}}(w). \tag{13}$$

The prior is the raw Gaussian conditioned on this event. There is *no all-input clean-margin condition.* For an observed labeled sample, condition further on

$$y_i z_w(x_i) > \frac{1}{2} \qquad (1 \le i \le n). \tag{14}$$

Thus the original output-margin convention is imposed only at training inputs. With no global margin restriction, this likelihood is stronger than sign-only consistency; that distinction is part of the theorem.

The condition has a simple sufficient example. If every hidden-to-hidden matrix has operator norm at most one, then

$$|z_w(x;\eta) - z_w(x)| \le \sqrt{1 + JE_v(w)}\,\|\eta\|_*.$$

Consequently every such network satisfies (13), regardless of how small its clean outputs are. The first layer is unrestricted because there is no input perturbation, and the final readout may have arbitrary gain. Numerical stability can also hold through saturation and cancellation without nonexpansiveness.

Lemma 14.1 (Numerical posterior compression)**.** *Let $0 < r \le 1/6$, and suppose $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$. Suppose every network in the support of Q obeys*

$$|z_w(x;\eta) - z_w(x)| \le 1/16 \qquad (\|\eta\|_* \le r)$$

at every input x in a specified set G. There is a finite class $\mathcal{H}$ of real-valued *depth-L network outputs with*

$$\log|\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}),$$

independent of N, t, Q, and a law Γ_Q on $\mathcal{H}$, such that at each $x \in G$ a coupling with marginals Q, Γ_Q satisfies

$$\mathbb{P}\{|h(x) - z_w(x)| > 1/12\} \le \alpha.$$

The candidate marginal is the same at every input.

Proof. Run the earlier entropy compressor before using its final sign-robustness step. Its entropy bounds and construction do not require robustness. Except with probability less than $\alpha/4$, its unrounded candidate output at a fixed input is exactly the original network's output under a simultaneous postactivation disturbance of norm less than $r/2$. The numerical stability assumption

bounds the change by $1/16$. The same fixed parameter-grid rounding changes outputs by at most $r/8$, uniformly on the cube. Since $r \le 1/6$, $1/16 + r/8 \le 1/12$. In the small-width branch the unrounded candidate is the original network except on its bounded-weight failure event, and the same rounding bound suffices. Keep the rounded real outputs, rather than only their signs. The finite grid and the candidate marginal are otherwise unchanged. No clean output margin has been used. □

Theorem 14.2 (Polynomial-depth learning without a global clean margin)**.** *For the prior* (13) *and training likelihood* (14)*, a target computed by a width-C layered Boolean circuit has a sample bound*

$$n \ge \mathrm{poly}(d, L, C, \varepsilon^{-1}, \log(1/\delta))$$

such that, with probability at least $1 - \delta$,

$$\mathbb{E}_{w \sim Q_S,\, X \sim D} \mathbf{1}\{f(X) z_w(X) \le 1/4\} \le \varepsilon. \tag{15}$$

One sample event gives this guarantee simultaneously for every $N \ge C$ and $0 < t \le 1$, hence for every $\gamma \ge 1$. The polynomial's degree is independent of depth, width, and Gaussian scale. In particular, it bounds Gibbs sign error, and gives the usual factors three and two for posterior confidence thresholds $2/3$ and $1/2$, respectively.

Statistical proof, given the target cylinder below. The target cylinder established below has raw Gaussian mass at least $\exp(-ND_*/t^2)$ with $D_* = \mathrm{poly}(d, L, C)$, is numerically stable, and has correct signed clean output greater than one on the full cube. It therefore lies in the posterior conditioning event for every sample. As before, posterior KL is at most ND_*/t^2. The Gaussian readout tail and this partition lower bound allow discarding posterior mass at most β while retaining $E_v \le B$, where $B = \mathrm{poly}(D_*, \log(1/\beta))$. The retained law Q' has KL at most $N(D_* + 1)/t^2$ and numerical stability at the common radius $r_0 = [1000\sqrt{1 + JB}]^{-1} < 1/6$.

Apply Lemma 14.1, and define the bounded ramp

$$\psi(u) = \begin{cases} 1, & u \le 1/3, \\ 5 - 12u, & 1/3 < u < 5/12, \\ 0, & u \ge 5/12. \end{cases}$$

On a successful comparison at a training point, the candidate has $y_i h(x_i) > 1/2 - 1/12 = 5/12$, so its ramp loss is zero. Thus the empirical ramp risk averaged under $\Gamma_{Q'}$ is at most α. On a successful comparison at a test point, if $f(x) z_w(x) \le 1/4$, then $f(x)h(x) \le 1/4 + 1/12 = 1/3$, so the candidate's ramp loss is one. Therefore the original test margin-failure probability is at most the candidate's population ramp risk plus α.

The finite-class relative Chernoff inequality applies to these $[0, 1]$ losses, and gives simultaneously for all $h \in \mathcal{H}$

$$R_\psi(h) \le 2\widehat{R}_\psi(h) + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}.$$

For completeness, the bounded-loss version follows from the same exponential argument as for indicators, using $e^{-su} \le 1 - (1 - e^{-s})u$ for $0 \le u \le 1$. Averaging under the candidate law and adding the discarded posterior mass gives

$$\mathbb{E}_{Q_S, D} \mathbf{1}\{f(X) z_w(X) \le 1/4\} \le \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}.$$

Choose $\beta = \varepsilon/4$, $\alpha = \varepsilon/12$, and n large enough that the last term is at most $\varepsilon/2$. The class is independent of N, t, proving simultaneity. Finally, if the posterior gives probability at most $2/3$ to a correct signed output greater than $1/4$, its conditional margin-failure probability is at least $1/3$; Markov gives the factor three. The sign event is larger, and the corresponding factor at $1/2$ is two. □

14.1 A numerically stable target cylinder

The robust digital-cylinder proof from the preceding report can be made numerically stable by tightening its hidden approximation constants. Use balanced clone groups, write P for the group-constant projection, and write $b^s(x) \in \{-1,1\}^N$ for the ideal repeated Boolean wires. Replace that construction's parameters by

$$\begin{aligned} e_* &= \frac{1}{128\sqrt{2C}}, & \tau_0 &= e_*/4, \\ b_L &= \sqrt{16\log 9 + 8\log(4L)}, & A &= \max\{32b_L,\, 2\log(8/e_*)\}, \\ \zeta &= \min\{1/4,\, A/[4(d+1)],\, Ae_*/(64\sqrt{C})\}. \end{aligned}$$

Here ζ is the cylinder tolerance, unrelated to an input-distribution exception mass. Constrain the standardized first-layer coordinates, active internal group coordinates, and readout coordinates within ζt of the cloned saturated circuit, exactly as in that construction. On its free-bulk event,

$$W^s = W^s_* + R^s + E^s, \quad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \zeta t\sqrt{C}, \quad E^s P = 0, \quad \|E^s\|_{\mathrm{op}} \le b_L t.$$

The free-bulk event has conditional probability at least $1/2$.

The old induction, with these tightened constants, proves

$$\frac{\|h^s_\eta(x) - b^s(x)\|_2}{\sqrt{N}} \le e_* \tag{16}$$

for every input, at every hidden layer, whenever each hidden injection has RMS at most τ_0 and output noise has magnitude at most τ_0. To check the constants directly, an error of RMS e_* makes every clone group's mean error at most $\sqrt{2C}e_* = 1/128$. Thus every ideal gate's signed preactivation exceeds $A/2$. The global secant estimate

$$|\tanh(a+u) - b| \le 2e^{-A/2} + (8/A)|u| \quad \text{when } b \in \{-1,1\},\ ba \ge A/2$$

gives the recurrence

$$e_s \le 2e^{-A/2} + (8/A)\zeta\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each of the four terms is at most $e_*/4$. The first-layer perturbation changes each preactivation by at most $\zeta(d+1) \le A/4$, so its signed preactivation is at least $3A/4$; its activation error plus injection is at most $e_*/2$. This initializes the recurrence. In particular the clean computation also satisfies (16).

The cloned readout obeys $\sqrt{N}\|v_*\| \le \sqrt{8C}$, while its parameter perturbation satisfies $\sqrt{N}\|v - v_*\| \le \zeta$. Comparing the noisy and clean computations for the *same* weights therefore yields

$$\begin{aligned} |z_w(x;\eta) - z_w(x)| &\le 2e_*(\sqrt{8C} + \zeta) + \tau_0 \\ &\le \frac{1}{32} + \frac{3}{4}e_* < \frac{1}{16}. \end{aligned}$$

Also $\sqrt{N}\|v_*\| \geq \sqrt{2C}$, so

$$E_v(w) \geq (\sqrt{2C} - 1/4)^2 \geq C.$$

Consequently

$$r_{\mathrm{num}}(w) \leq \frac{1}{1000\sqrt{C}} < \frac{1}{512\sqrt{2C}} = \tau_0.$$

Thus every draw in the cylinder and free-bulk event satisfies the numerical support rule. Its correct signed clean output is at least $2 - \sqrt{8C}e_* - \zeta > 1$, which also establishes the training likelihood.

There are still only $N[d + 2 + (L-2)C]$ constrained Gaussian coordinates, and the standardized center energy is at most $N[9dA^2 + 18(L-2)CA^2 + 8C]$. The preceding Gaussian box calculation therefore gives mass at least $\exp(-ND_*/t^2)$ for $D_* = \mathrm{poly}(d, L, C)$. Tightening e_* by a constant factor changes only constants and logarithmic factors; it introduces no exponential dependence on depth. This proves the target-cylinder assertion used above.

14.2 A trusted-region numerical version

If the numerical support condition is imposed only on a fixed trusted region G with $D(G^c) \leq \rho$, while the training-margin likelihood is still imposed on *all* training points, the same proof uses masked ramp losses $\mathbf{1}_G(x)\psi(f(x)h(x))$. It gives margin-failure risk at most $\rho + \varepsilon$ with the same polynomial bound. A finite catalog of trusted regions can be handled by the same mixture construction as in Corollary 18.2, at an additional $\log M$ cost. The global target cylinder belongs to every component.

14.3 A soft numerical-stability prior with full support

The hard numerical condition can also be replaced by a soft penalty. Define

$$\rho_{\mathrm{num}}(w) = \min_x \inf\{\|\eta\|_* : |z_w(x;\eta) - z_w(x)| \geq 1/16\}.$$

For every finite parameter array, continuity on the finite input cube gives $\rho_{\mathrm{num}}(w) > 0$, including when some clean outputs vanish. Output-only noise gives $\rho_{\mathrm{num}}(w) \leq 1/16$. For fixed $p, \lambda > 0$, use the prior density

$$d\nu_{N,t}(w) \;\propto\; \exp\!\left[-(N/t^2)\lambda\rho_{\mathrm{num}}(w)^{-p}\right] d\mu_{N,t}(w),$$

and retain the training-margin likelihood (14). This density is positive at every finite parameter array; it retains the support of the raw Gaussian and needs no clean-margin exclusion anywhere.

The soft-prior calculation from Theorem 17.1 applies with ρ_{num} in place of the sign margin. The target cylinder has $\rho_{\mathrm{num}} \geq \tau_0$ and raw mass e^{-ND_*/t^2}. Set $B_* = D_* + \lambda\tau_0^{-p}$. The training partition is at least e^{-NB_*/t^2}; posterior KL is at most NB_*/t^2; and

$$Q_S\{\rho_{\mathrm{num}}^{-p} > u\} \leq \exp\{(N/t^2)(B_* - \lambda u)\}.$$

Choosing $u = [B_* + \log(1/\beta)]/\lambda$ discards mass at most β. The retained law has scaled KL parameter at most $B_* + 1$ and numerical stability at radius $r = \frac{1}{2}u^{-1/p} \leq 1/32$. Lemma 14.1 and the same ramp-loss argument prove the conclusion of Theorem 14.2, with a polynomial bound for fixed p, λ. The prior probabilities have changed; this is not a theorem for unchanged independent Gaussian weights. One may also minimize only over a trusted region G, obtaining the corresponding error bound $D(G^c) + \varepsilon$.

This theorem does not identify sign-only conditioning with positive-margin conditioning. It shows that a global clean-output lower bound is unnecessary when numerical internal stability is imposed and the observed labels retain a positive output margin. Numerical stability is not a

universal consequence of ordinary gradient descent: the earlier fragile scalar-chain construction also violates this condition for sufficiently large depth. For example, at even $L \geq 1024$, its displayed attack bound divided by the present radius is at most $4000e^{-\sqrt{L}/4}/\sqrt{L} < 1$, while the attack changes the output from $+1$ to -1. The same strict inequalities persist in its open attraction neighborhood for gradient descent.

15 Finite-noise certificates that permit expansion and contraction

Numerical stability need not be enforced by making every layer nonexpansive in the ordinary Euclidean norm. Diagonal quadratic metrics provide a global, finite-noise certificate that allows intermediate expansion and subsequent contraction. They also give a concrete optimization objective.

Write $H = L - 1$ for the number of hidden layers. Let M_s be diagonal positive semidefinite matrices, and suppose

$$M_H \succeq vv^\top, \qquad (W^s)^\top M_s W^s \preceq M_{s-1} \quad (2 \leq s \leq H). \tag{17}$$

Define

$$Q_M = N \sum_{s=1}^{H} \lambda_{\max}(M_s).$$

Proposition 15.1 (A diagonal-metric finite-noise certificate)**.** *Under* (17)*, for every input and every finite simultaneous postactivation and output disturbance,*

$$|z_w(x;\eta) - z_w(x)| \leq \sqrt{1 + Q_M}\, \|\eta\|_*.$$

In particular this is a global finite-radius statement, not a bound only on the clean Jacobian.

Proof. Coordinatewise 1-Lipschitzness of tanh makes it nonexpansive in every diagonal seminorm $\|u\|_{M_s} = \sqrt{u^\top M_s u}$. If Δh_s denotes the difference of the noisy and clean states, (17) gives

$$\|\Delta h_s\|_{M_s} \leq \|\Delta h_{s-1}\|_{M_{s-1}} + \|\eta_s\|_{M_s}.$$

The first layer has $\Delta h_1 = \eta_1$, since there is no input noise. Therefore

$$|\Delta z| \leq \sum_{s=1}^{H} \sqrt{\lambda_{\max}(M_s)}\, \|\eta_s\|_2 + |\eta_{\text{out}}|.$$

Cauchy–Schwarz in the RMS-normalized disturbance coordinates gives the claim. No assumption on activation signs, margins, or smallness of the perturbations has been used. □

If all internal operator norms are at most one, taking $M_s = \|v\|^2 I$ recovers $Q_M = HE_v$. But the metric condition is weaker. For example, consider a scalar hidden stack with alternating internal weights $2, 1/2, 2, 1/2, \ldots$. Let $p_s = \prod_{j=s+1}^{H} |W^j|$ and take $M_s = v^2 p_s^2$. Every matrix inequality in (17) is an equality, every $p_s \leq 2$, and $Q_M \leq 4HE_v$. The noise gain is therefore only $O(\sqrt{H})$ despite expansive layers at a positive fraction of depths. The product of the factors $\max\{1, |W^j|\}$ would instead be exponential in depth. The certificate accounts for the compensating contractions.

The certificate is also output-specific. For instance, with two hidden coordinates, readout $v = (1, 0)^\top$, and every internal matrix $W^s = \operatorname{diag}(1, 10)$, take $M_s = \operatorname{diag}(1, 0)$. The second coordinate may have a very large local gain but never affects the output; the certificate gives $Q_M = HE_v$ nonetheless. Thus irrelevant expansive directions need not be suppressed.

Computing and training the certificate. For fixed weights, minimizing $\sum_s b_s$ subject to (17) and $0 \preceq M_s \preceq b_s I$, with diagonal M_s, is a semidefinite program: every displayed constraint is affine in the diagonal metric entries and b_s. Its value certifies a finite-noise gain. One can penalize this certified gain during training, or use the parameterization

$$W^s = D_s^{-1} A_s D_{s-1}, \qquad \|A_s\|_{\mathrm{op}} \le 1, \qquad v = D_H u, \quad \|u\|_2 \le 1,$$

with positive diagonal D_s and $M_s = D_s^2$. Spectral normalization or projection enforces the constraints on A_s; the diagonal scales allow expansion and contraction in the original coordinates. This is a proposed training constraint or regularizer, not an assertion that unconstrained gradient descent automatically satisfies it.

For the numerical radius (12), a certificate satisfying

$$\frac{1 + Q_M}{1 + HE_v} \le \left(\frac{1000}{16}\right)^2$$

proves the prior's 1/16 output-stability condition. More generally, let $R(L) \ge 1$ be any fixed polynomial and replace that radius by

$$\frac{1}{1000 R(L) \sqrt{1 + HE_v}}. \tag{18}$$

A certificate $Q_M \le R(L)^2 HE_v$ then guarantees output change at most 1/1000 throughout the ball. The polynomial-depth learning theorem remains valid: the target cylinder still belongs to the support because the radius is smaller, and the common compression radius after the readout-tail truncation is still inverse polynomial in depth. This permits polynomial intermediate amplification, not just contraction. For example, internal maps with operator norms at most $1 + B/L$ have suffix gain at most e^B; if $B = O(\log L)$, this is polynomial and fits such a calibration. Applying this observation to a different residual-network architecture would still require matching that architecture's prior and target-mass calculation.

The metric inequalities are sufficient certificates for membership in the numerical prior. If one instead restricts the prior to these particular certificates, the existing digital target cylinder is not automatically known to remain inside that stricter support; that would be an additional expressivity and probability question.

15.1 A finite-tube version that uses saturation

The global diagonal-metric test can be conservative for saturated units. There is a rigorous finite-tube variant. Fix an input, a proposed radius r, and positive definite diagonal M_s. Let $a_s = W^s h_{s-1}$ be the clean preactivations. Put

$$B_s = N \sum_{j=1}^{s} \lambda_{\max}(M_j), \qquad R_{s,i} = r\sqrt{B_{s-1}} \sqrt{(W^s M_{s-1}^{-1} (W^s)^\top)_{ii}} \quad (s \ge 2),$$

and define diagonal matrices U_s by

$$(U_s)_{ii} = \sup_{|q| \le R_{s,i}} \operatorname{sech}^2(a_{s,i} + q) = \operatorname{sech}^2((|a_{s,i}| - R_{s,i})_+).$$

Suppose

$$M_H \succeq v v^\top, \qquad (W^s)^\top U_s M_s U_s W^s \preceq M_{s-1} \quad (2 \le s \le H). \tag{19}$$

Then every disturbance with $\|\eta\|_* \le r$ changes the output at this input by at most $r\sqrt{1+B_H}$.

Indeed, induction gives $\|\Delta h_s\|_{M_s} \le \sum_{j\le s} \|\eta_j\|_{M_j} \le r\sqrt{B_s}$. The preceding-layer bound places each preactivation displacement in its specified interval $[-R_{s,i}, R_{s,i}]$. The coordinatewise mean-value slopes are consequently between zero and the corresponding entries of U_s. Since M_s is diagonal, (19) gives the same contraction inequality as before. This closes the induction and proves the finite-noise bound, including scalar output noise by Cauchy–Schwarz. The construction is self-consistent: the interval widths are computed before accepting the matrix inequalities, rather than assumed from the clean Jacobian alone.

To certify the prior, this test must hold at every required input, with the resulting output bound at most 1/16. Metrics may be chosen separately at different inputs. Checking only a finite test set supplies evidence on those inputs, not a global certificate. Unlike the global test, optimizing the finite-tube metrics is not in general a single convex semidefinite program, because the slope bounds depend on the candidate metrics.

15.2 Large output amplification does not automatically repair this condition

Let w be fixed with $E_v > 0$, and let $w^{(A)}$ be obtained by multiplying only its readout by $A > 0$. For an input x, let

$$G_{\mathrm{hid}}(w,x)^2 = N \sum_{s=1}^{H} \left\| \left. \frac{\partial z_w(x;\eta)}{\partial \eta_s} \right|_{\eta=0} \right\|_2^2 .$$

This is the squared clean gain from RMS-normalized hidden disturbances; it does not include the scalar output-noise coordinate. With $c = 1/1000$ and $r_A = c/\sqrt{1+HA^2E_v}$,

$$\lim_{A\to\infty} \sup_{\|\eta\|_*\le r_A} |z_{w^{(A)}}(x;\eta) - z_{w^{(A)}}(x)| = \frac{c\, G_{\mathrm{hid}}(w,x)}{\sqrt{HE_v}}. \tag{20}$$

The convergence is uniform over the finite Boolean cube. To prove it, write $\eta = r_A u$. Smoothness gives

$$z_{w^{(A)}}(x; r_A u) - z_{w^{(A)}}(x) = A r_A\, \ell_x(u_{\mathrm{hid}}) + r_A u_{\mathrm{out}} + O(A r_A^2),$$

uniformly on the unit disturbance ball, where ℓ_x is the clean hidden linear response. Since $Ar_A \to c/\sqrt{HE_v}$, $r_A \to 0$, and $Ar_A^2 \to 0$, taking the supremum gives (20). Finiteness of the cube makes the remainder uniform over inputs.

Consequently, if

$$\max_x \frac{G_{\mathrm{hid}}(w,x)}{\sqrt{HE_v}} > 62.5,$$

then no sufficiently large readout amplification makes the network satisfy the numerical support rule. For a nonexpansive hidden stack the ratio is at most one. This distinguishes numerical stability from the earlier calibrated sign condition, whose required radius could always be met by sufficiently large amplification of a nonvanishing finite-cube network. The limiting clean gain is used here only for an asymptotic scaling obstruction; the positive certificates above control finite disturbances.

A limitation of the global metric certificate on the deeper trained models. The global diagonal-metric certificate cannot certify the depth-32 and -depth-64 gradient-descent digit models in this experiment, even after optimizing its metrics. This failure has an elementary exact certificate; an SDP solver is unnecessary. In column-vector notation, define the *unactivated suffix readouts*

$$g_H = v, \qquad g_{s-1} = (W^s)^\top g_s \quad (2 \le s \le H).$$

For any positive semidefinite metric family satisfying $M_H \succeq vv^\top$ and $M_{s-1} \succeq (W^s)^\top M_s W^s$, induction gives $M_s \succeq g_s g_s^\top$. Consequently,

$$Q_M = N \sum_{s=1}^{H} \lambda_{\max}(M_s) \geq N \sum_{s=1}^{H} \|g_s\|_2^2 =: G_{\text{lin}}^2.$$

Thus if

$$\frac{1 + G_{\text{lin}}^2}{1 + HE_v} > 3906.25,$$

no such metric family can meet the proposed numerical certificate. The argument even permits arbitrary positive semidefinite metrics in this necessary inequality; restricting to diagonal metrics cannot help.

We preselected the two successful Boolean-input GD models with seed zero, 302 ($H = 32$) and 306 ($H = 64$). Their respective ratios are approximately 93 243.76 and 63 136 897.94, both strictly above 3906.25. The implied lower bounds on the output-change *certificate* are 0.30536 and 7.94587, versus the required $1/16 = 0.0625$. These inequalities were checked with exact integer dyadic matrix–vector products and rational arithmetic on the saved weights. An exploratory check of the other six successful deep GD models gives the same impossibility, again with exact rational verification: at $H = 32$ the necessary output-change certificate lies between 0.24417 and 0.32754; at $H = 64$ it lies between 6.31046 and 13.65547.

This is an impossibility result for the *global sufficient certificate*, not a violation of numerical stability itself. The unactivated suffixes need not be derivatives at any actual Boolean input. The successful pointwise certificates for these models use saturation, which the global metric inequalities above omit. A positive global certificate for these particular networks therefore needs additional information about the reachable states, rather than merely a better choice of diagonal scales. The exploratory source and exact rational records are kept separately from the frozen experiment archive in `diagonal-metric-exploration/`.

16 Stability of a bounded prediction score

Raw output stability may penalize changes between two very large outputs that represent the same confident prediction. A bounded score avoids this. Write $T(u) = \max\{-1, \min\{1, u\}\}$, fix $0 < c \leq 1/100$, and put

$$r_{\text{clip}}(w) = \frac{c}{\sqrt{1 + (L-1)E_v(w)}}.$$

Restrict the Gaussian to networks satisfying

$$|T(z_w(x;\eta)) - T(z_w(x))| \leq 1/16 \quad \text{for all } x \text{ and } \|\eta\|_* \leq r_{\text{clip}}(w). \tag{21}$$

Impose no all-input clean margin, and retain the training likelihood $y_i z_w(x_i) > 1/2$.

Corollary 16.1 (Bounded-score stability)**.** *This prior and likelihood satisfy the polynomial-depth conclusion of Theorem 14.2, including its test signed-output margin* $1/4$*, every width above Boolean circuit width, and every* $\gamma \geq 1$*. The condition can also be imposed only on a fixed trusted region* G*, at an added test-error cost* $D(G^c)$*.*

Proof. Before rounding, the entropy compressor evaluates the original network at a disturbance of norm less than the retained common radius divided by two. Condition (21) bounds the resulting change of $T(z)$ by $1/16$. Since T is 1-Lipschitz, the grid-rounding error adds at most $r/8$, so the total

bounded-score error is at most 1/12. Apply the ramp proof to $T(h(x))$ instead of $h(x)$. Oddness and monotonicity of T ensure

$$yz > 1/2 \Longrightarrow yT(z) > 1/2, \qquad yz \le 1/4 \Longrightarrow yT(z) \le 1/4.$$

Thus the same training and test implications, and the same finite-class bound, hold.

For target mass, even the original robust digital cylinder from the preceding report suffices: it has correct noisy signed output greater than one at every input throughout per-layer RMS radius $\tau_0 = 1/(64\sqrt{2C})$, and $E_v \ge C$. Since $r_{\text{clip}} \le (100\sqrt{C})^{-1} < \tau_0$, its clean and noisy scores are both exactly $f(x)$. Hence it satisfies the new support condition with zero score change and retains the same Gaussian mass bound. The entropy and readout-tail steps are unchanged. Masked ramp losses give the trusted-region version. □

The raw numerical condition implies the clipped condition at the same radius, but not conversely. Any finite-noise raw output enclosure $z_w(x;\eta) \in [z - B, z + B]$ gives the sound clipped-score certificate

$$\max\{T(z + B) - T(z),\ T(z) - T(z - B)\} \le 1/16.$$

This is often much smaller than B for large clean outputs. An attack that changes only the raw magnitude while leaving both scores saturated does not refute (21).

Soft version. Let $\rho_{\text{clip}}(w)$ be the smallest disturbance norm, minimized over inputs, at which the clipped score changes by at least 1/16. Continuity makes it positive for every finite weight array; output noise makes it finite. A density proportional to

$$\exp\{-(N/t^2)\lambda \min(1, \rho_{\text{clip}}(w))^{-p}\}$$

relative to the raw Gaussian therefore has full support. For fixed $p, \lambda > 0$, the soft-prior tail calculation and the bounded-score compression argument give the same polynomial-depth theorem. After truncating the penalty at u, use common radius $\min\{1/6, \frac{1}{2}u^{-1/p}\}$; this both lies strictly below the first score-change radius and satisfies the rounding constants above.

Readout amplification. The bounded-score condition restores the scaling effect of calibrated sign robustness. If a fixed finite-cube network has no clean zero outputs, then sufficiently large positive readout amplification makes it satisfy (21). Indeed there are a hidden-noise radius $\rho > 0$ and signed unscaled output lower bound $m > 0$ valid on the whole cube. The calibrated radius tends to zero as the readout scale A grows, while $Am - r_{\text{clip}} > 1$ eventually. Both clean and noisy scores then equal the clean sign. The required scale can be very large, and its Gaussian readout-energy cost is $A^2 E_v$; this observation supplies no uniform sample bound by itself. The large-readout obstruction for raw numerical stability in (20) concerns a different condition.

17 Alternative priors and operational stability conditions

This section gives two changes with complete statistical bridges. A soft inverse-margin penalty retains the support of the raw Gaussian. A bounded activation-fault condition weakens the previous hard support rule and has a direct interpretation in terms of numerical errors and neuron failures. Neither is a theorem for the *unchanged* Gaussian prior.

17.1 A soft prior with the same support as the raw Gaussian

Let $\mu_{N,t}$ be the raw Gaussian of the preceding report, $0 < t \le 1$, and put $a = N/t^2 \ge 1$. Retain the postactivation disturbance convention

$$\|\eta\|_*^2 = \sum_{s=1}^{L-1} \frac{\|\eta_s\|_2^2}{N} + |\eta_{\text{out}}|^2.$$

Define the exact all-input distance to a zero output by

$$\rho(w) = \min_{x\in\{-1,1\}^d} \inf\{\|\eta\|_* : z_w(x;\eta) = 0\}, \qquad r(w) = \min\{1, \rho(w)\}.$$

An output-only disturbance makes ρ finite. If all clean outputs are nonzero, continuity and finiteness of the input cube make $\rho > 0$. Every disturbance ball with radius strictly below ρ preserves the clean signs: a sign change would yield a zero along the segment from zero noise. The function ρ is measurable, since the noise balls are compact and the input set is finite. Under the raw Gaussian, all clean outputs are nonzero almost surely. To see this, condition successively on the preceding layers: each nonzero state gives nondegenerate Gaussian preactivations, and finally a nondegenerate Gaussian scalar readout. A finite union over inputs completes the argument.

Fix constants $p > 0$ and $\lambda > 0$, and define

$$\Phi(w) = r(w)^{-p}, \qquad d\nu_{N,t}(w) = Z^{-1} \exp[-a\lambda\Phi(w)]\, d\mu_{N,t}(w). \tag{22}$$

On the null set $\rho = 0$, set the density to zero. The normalizer is positive and finite, and $\nu_{N,t}$ and $\mu_{N,t}$ are mutually absolutely continuous. Thus no set of positive Gaussian probability is excluded. In energy notation, the new density is proportional to $\exp\{-a[\kappa_N(w) + 2\lambda\Phi(w)]/2\}$. One can optionally retain the original clean-margin restriction $\min_x |z_w(x)| > 1/2$; without that optional restriction, even the original hard output-margin condition is unnecessary for the following result. In both versions, the likelihood is noiseless *sign consistency* with the observed training labels.

Theorem 17.1 (Soft robustness prior)**.** *Suppose that, for a Boolean target f, there is a set A_f of exact sign representations such that*

$$\mu_{N,t}(A_f) \ge e^{-aD}, \qquad \rho(w) \ge r_* \quad (w \in A_f),$$

where $D \ge 1$ and $0 < r_* \le 1$ are independent of N, t. For fixed p, λ, the consistency posterior under (22) has a Gibbs-risk sample bound polynomial in

$$d, L, D, r_*^{-1}, \varepsilon^{-1}, \log(1/\delta).$$

If the cylinder hypotheses hold at every $N \ge C$ and $0 < t \le 1$, the guarantee holds on one sample event simultaneously for all those widths and scales. The optional clean-margin version has the same conclusion when A_f also satisfies that clean margin.

Proof. Set $B = D + \lambda r_*^{-p}$. For every sample S from f, its unnormalized partition function satisfies

$$Z_S = \int \mathbf{1}_{\text{consistent with } S} e^{-a\lambda\Phi}\, d\mu_{N,t} \ge e^{-aB}.$$

The global prior normalizer cancels after conditioning. The posterior Q_S therefore obeys

$$D_{\text{KL}}(Q_S \| \mu_{N,t}) = -a\lambda \mathbb{E}_{Q_S}\Phi - \log Z_S \le aB, \qquad Q_S\{\Phi > u\} \le e^{a(B-\lambda u)}.$$

Choose $0 < \beta \le 1/2$ and $u = [B + \log(1/\beta)]/\lambda \ge 1$. Since $a \ge 1$, the displayed tail is at most β. Let Q'_S be Q_S conditioned on $\Phi \le u$. Its explicit density gives

$$D_{\mathrm{KL}}(Q'_S \| \mu_{N,t}) \le aB + \log \frac{1}{1-\beta} \le a(B+1).$$

Its support is strictly sign-robust at radius $r_0 = \frac{1}{2}u^{-1/p}$. Apply the uniform robust posterior-compression theorem from the preceding report with $K = B + 1$ and $r = r_0$. It gives one finite class $\mathcal{H}$, with polynomial log cardinality, and a common candidate marginal whose prediction can be coupled to Q'_S at each input with disagreement at most α. Since Q'_S is training-consistent, the finite-class relative Chernoff bound from that report gives

$$R(Q_S) \le \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}$$

with probability at least $1 - \delta$. Choose β, α proportional to ε. All class parameters are independent of N, t, which also proves simultaneity. The same calculation with the extra clean-margin indicator proves the optional version. □

The robust digital cylinder from the earlier report has $D = \mathrm{poly}(d, L, C)$ and $r_*^{-1} = \mathrm{poly}(C)$. Consequently Theorem 17.1 gives a polynomial-depth bound for every $N \ge C$ and every $\gamma \ge 1$ while retaining the support of the raw Gaussian. The prior probabilities have changed. The result therefore does not settle whether ordinary independent Gaussian weights alone give the same guarantee. An inverse-squared margin ($p = 2$) is a particularly simple choice. A penalty merely logarithmic in $1/\rho$ would only produce an exponentially small retained radius through this proof, so would not give the same polynomial conclusion.

17.2 Robustness only within the activation range

Call a disturbance *bounded-state* if every perturbed hidden activation belongs to $[-1, 1]^N$. This allows replacing an activation by another legal activation value, and leaves scalar output noise unrestricted. Require sign stability only against bounded-state disturbances in the specified RMS ball.

Proposition 17.2 (Bounded-state compression)**.** *The deterministic and posterior compression theorems from the preceding report remain valid under bounded-state robustness in place of unrestricted postactivation robustness. The previously established target Gaussian-mass estimates remain valid.*

Proof. On every successful compression event, every virtual hidden state is a tanh output. First-layer clipping occurs before the activation; active-coordinate clipping, Gaussian replacement, and source reconstruction also feed the activation. The exact postactivation errors comparing the virtual computation to the original weights therefore produce states in $[-1, 1]^N$, and have the same total RMS bound as before. The margin argument adds only output noise. Subsequent finite-grid rounding uses that positive output margin and does not invoke robustness again. The small-width case also uses only output noise. These are all uses of robustness in the two proofs. Finally, the previous target cylinder satisfies the stronger, unrestricted condition, so its probability lower bound is unchanged. □

This relaxation can be substantial at a fixed radius. With one hidden layer, let $n = N-1$, $c = 0.9$, and take a constant hidden state $h = (c, \dots, c, -c)$ and readout $v = (2/(nc), \dots, 2/(nc), 1/c)$. Its output is 1. Unrestricted hidden and output noise can erase it at norm $[N\|v\|^2 + 1]^{-1/2} = O(N^{-1/2})$.

For bounded-state noise, the last coordinate can decrease by at most $1-c$. The remaining coordinates and scalar output noise can decrease the output by at most $\sqrt{4N/(nc^2)+1}\,\|\eta\|_*$. Thus a zero requires

$$\|\eta\|_* \geq \frac{2c-1}{\sqrt{4N/n+c^2}} > 0.26 \qquad (N \geq 2).$$

This separates the notions of robustness; its readout energy grows with N, so it is not a sample lower bound or a separation of the calibrated prior supports.

17.3 Small numerical errors and rare neuron faults

Say a network tolerates (a, p, b) if it preserves its own clean signs at every input under all simultaneous bounded-state disturbances such that, in each hidden layer, at most pN coordinates have $|\eta_i| > a$, and $|\eta_{\text{out}}| \leq b$. Exceptional coordinates may be replaced arbitrarily within $[-1, 1]$. This combines numerical errors at all neurons with an adversarial fraction of neuron faults. Dropout and pruning are particular faults; average performance under random dropout is a weaker property than this adversarial condition.

Lemma 17.3 (Fault and RMS bridges)**.** *Tolerance of (a, p, b) implies bounded-state RMS robustness at $r = \min\{a\sqrt{p}, b\}$. Conversely, unrestricted RMS robustness at τ implies tolerance of (a, p, b) whenever*

$$(L-1)(a+2\sqrt{p})^2 + b^2 \leq \tau^2.$$

Proof. If the total RMS error is at most r, each layer has at most $Nr^2/a^2 \leq pN$ coordinates with error larger than a; output noise is at most $r \leq b$. Conversely, the ordinary coordinates contribute at most $a\sqrt{N}$ to a layer's Euclidean error. Each faulty coordinate differs by at most 2, so their contribution is at most $2\sqrt{pN}$. Sum the squared layer RMS bounds and the scalar output bound. □

For a single rule independent of target and data, retain the original clean margin and define

$$E_v = N\|v\|^2, \qquad \tau(w) = \frac{0.01}{\sqrt{1+(L-1)E_v}},$$
$$a(w) = \frac{\tau(w)}{4\sqrt{L}}, \qquad p(w) = \frac{\tau(w)^2}{64L}, \qquad b(w) = \frac{\tau(w)}{2}.$$

Restrict the Gaussian to networks tolerating these faults, then condition on noiseless training sign consistency. Every network in the previous radius-$\tau(w)$ support belongs to this support: the second bound in Lemma 17.3 is at most $\tau(w)^2/2$. In particular, the previous target cylinder still lies in the support.

Corollary 17.4 (Polynomial-depth learning with activation faults)**.** *For this fixed fault-tolerance support rule, the same qualitative polynomial-depth sample theorem holds for all $\gamma \geq 1$ and every width above the layered Boolean target circuit's width.*

Proof. The same target partition bound and Gaussian readout tail as before allow conditioning on $E_v \leq B$ while discarding posterior mass at most β, where $B = \text{poly}(d, L, C, \log(1/\beta))$. Every retained network has bounded-state robustness radius at least

$$r_B = \frac{10^{-4}}{32L[1+(L-1)B]},$$

because $a(w)\sqrt{p(w)} = \tau(w)^2/(32L) \le b(w)$. The conditional posterior has the same polynomial scaled-KL bound as in the previous argument. Apply Proposition 17.2 and the finite-class generalization step. All parameters are independent of width and Gaussian scale, giving the simultaneous conclusion. □

This condition has an operational interpretation, but it is not established as a universal outcome of ordinary gradient descent. Likewise, random-noise stability alone cannot simply replace it. A network that reads a single hidden coordinate can be stable under small independent per-coordinate Gaussian noise while that coordinate can be erased with only $O(N^{-1/2})$ RMS adversarial noise. Testing particular dropout masks or one quantization grid similarly does not control the disturbances used by the compression coupling. The conservative constants may give $p(w)N < 1$ at small widths; then the allowed number of arbitrary faults is zero, and the condition concerns bounded errors at all coordinates. No claim of tolerance to a practically large dropout fraction follows from these constants.

18 A trusted input region retains the original sign likelihood

The distributional robustness restriction can be combined with the original sign-only likelihood if the exceptional region is shared across hypotheses. This is relevant when non-data-like inputs, rather than ordinary test inputs, are responsible for a failed all-cube certificate.

Fix an input region $G \subseteq \{-1,1\}^d$ before the labeled sample is drawn. Suppose the test input distribution satisfies $D(G^c) \le \rho$. The region may be selected using independent unlabeled information; its coverage is an assumption to be justified separately. It is independent of the unknown labels. Set

$$E_v(w) = N\|v\|^2, \qquad r_{\mathrm{ref}}(w) = \frac{1}{100\sqrt{1+(L-1)E_v(w)}}.$$

Restrict the raw Gaussian only to networks whose own clean sign is preserved under every disturbance of total RMS norm at most $r_{\mathrm{ref}}(w)$ at every input in G. Impose no margin or robustness condition outside G, and no additional global clean margin. Then condition on *ordinary sign consistency at every observed training example*, including examples outside G.

Theorem 18.1 (Trusted-region prior, original likelihood)**.** *For a target represented by a width-C layered Boolean circuit, this posterior has Gibbs error at most $\rho + \varepsilon$ after*

$$n \ge \mathrm{poly}(d, L, C, \varepsilon^{-1}, \log(1/\delta))$$

examples, with probability at least $1-\delta$, simultaneously for every $N \ge C$ and $0 < t \le 1$. The polynomial's degree is independent of width, depth, and Gaussian scale. The likelihood is the original sign-only likelihood, rather than a robust training likelihood.

Proof. The earlier global robust target cylinder belongs to the new support and satisfies ordinary training consistency for every sample. Its raw Gaussian mass is at least $\exp(-ND_*/t^2)$, where $D_* = \mathrm{poly}(d, L, C)$. Therefore the posterior has scaled KL at most ND_*/t^2. The same Gaussian readout tail bound permits discarding posterior mass at most β while retaining $E_v \le B$, where $B = \mathrm{poly}(D_*, \log(1/\beta))$. The retained law Q' has KL at most $N(D_*+1)/t^2$ and is robust on G at the common radius $r_0 = [100\sqrt{1+(L-1)B}]^{-1}$.

The pointwise-defect compression proposition gives a common law Γ on a finite class $\mathcal{H}$ and a prediction coupling with disagreement at most α at each $x \in G$. No prediction approximation is needed outside G. Apply the finite-class relative Chernoff bound to the *masked losses*

$$\ell_G(h, x) = \mathbf{1}_G(x)\mathbf{1}\{h(x) \neq f(x)\}.$$

With probability at least $1 - \delta$, simultaneously for every $h \in \mathcal{H}$,

$$\mathbb{E}_D \ell_G(h, X) \leq \frac{2}{n} \sum_{i=1}^{n} \ell_G(h, x_i) + \frac{8[\log |\mathcal{H}| + \log(1/\delta)]}{n}.$$

The law Q' is training-consistent, so averaging the empirical loss under Γ gives at most α. The population comparison adds at most another α. Hence

$$\mathbb{E}_{w \sim Q', X \sim D} \mathbf{1}_G(X)\mathbf{1}\{\operatorname{sign} z_w(X) \neq f(X)\} \leq 3\alpha + \frac{8[\log |\mathcal{H}| + \log(1/\delta)]}{n}.$$

Outside G the error costs at most ρ, and the discarded posterior mass costs at most β. Choosing α, β proportional to ε proves the theorem. The finite class is independent of N, t, so its one sample event proves simultaneity. □

The fixed exceptional region is crucial. This theorem does not allow each hypothesis to select its own exceptional inputs after seeing the sample. It also does not justify choosing G from that same labeled sample without additional complexity control. A finite collection of independently obtained unlabeled inputs need not cover most of D merely because all its points are certified. Coverage and certification over G are separate requirements.

The soft-prior construction also localizes to G. Replace the global margin in Theorem 17.1 by

$$\rho_G(w) = \min_{x \in G} \inf\{\|\eta\|_* : z_w(x; \eta) = 0\}.$$

For nonempty G, this is positive and finite Gaussian-almost surely. The soft density proportional to $\exp\{-(N/t^2)\lambda \min(1, \rho_G(w))^{-p}\}$ therefore retains the support of the raw Gaussian and does not penalize instability outside G. The same target cylinder and posterior truncation as in Theorem 17.1, followed by the masked-loss proof above, give error at most $D(G^c) + \varepsilon$ with the original sign likelihood and polynomial sample complexity. This combines a soft prior with restriction of the stability requirement to a trusted data region.

Corollary 18.2 (A finite catalog of trusted regions)**.** *Let $G_1, \ldots, G_M$ be fixed before the labeled sample, with $D(G_j^c) \leq \rho$ for every j. Choose any prior distribution on the index j, and conditional on j use the Gaussian restricted to robustness on G_j as above. Condition the resulting joint prior on the original training signs. Then the marginal network posterior satisfies the same error guarantee with polynomial sample complexity also in $\log M$.*

Proof. Conditional on any index j and the observed sample, the network law is the raw Gaussian conditioned on robustness on G_j and sign consistency. The global robust target cylinder belongs to every such conditioning event, so its same raw mass lower bound applies separately to every component. Consequently every component has the same entropy bound, readout tail bound, and common finite compression class $\mathcal{H}$. Apply the preceding masked-loss proof simultaneously to the $M|\mathcal{H}|$ loss functions $\mathbf{1}_{G_j}(x)\mathbf{1}\{h(x) \neq f(x)\}$. Only $\log M$ is added to the finite-class bound. Every component posterior then has error at most $\rho + \varepsilon$ on the same sample event, so averaging under the posterior law of j preserves the guarantee. □

Thus the exceptional region may adapt to the labels through posterior selection from a predeclared catalog, with an explicit complexity cost. The prior here is a mixture of region-conditioned Gaussians; it is not silently identified with the Gaussian conditioned on their union.

18.1 Why a per-hypothesis distributional exception is insufficient

The following abstract example shows why the compression-with-defect and target-mass facts alone cannot remove the empirical defect term under ordinary sign conditioning. It is *not* a lower bound for the exact Gaussian neural prior.

Let the input domain have M points, uniformly distributed, and let the target be $f \equiv +1$. Fix an integer q with $\rho = q/M < 1/2$. For each size-q set S, define the patch predictor

$$h_S(x) = \begin{cases} +1, & x \in S, \\ -1, & x \notin S, \end{cases} \qquad b_{h_S}(x) = \mathbf{1}_S(x).$$

Include the target predictor $h_+ \equiv +1$ with bad-set indicator zero. Every predictor has population bad-set mass at most ρ. The two-element class $\{h_+, h_-\}$, where $h_- \equiv -1$, compresses these predictors exactly outside their bad sets: map h_S to h_- and h_+ to itself. Thus the pointwise prediction-compression conclusion holds with zero approximation error and $\log|\mathcal{H}| = \log 2$.

Give h_+ prior mass $\eta = e^{-aD_*}$ and distribute the remaining mass uniformly over the patch predictors. Suppose $\eta \le 1/2$, $q \ge 2n$, and

$$n \le \frac{aD_*}{2\log(2/\rho)}.$$

If the sample has $k \le n$ distinct inputs, the patch component's probability of consistency is

$$L_S = \frac{(q)_k}{(M)_k} \ge (\rho/2)^n \ge \sqrt{\eta}.$$

Here $(u)_k = u(u-1)\cdots(u-k+1)$. Consequently the posterior mass on the target is at most $\eta/[(1-\eta)L_S] \le 2\sqrt{\eta}$, and its Gibbs error is at least

$$(1-\rho)(1-2e^{-aD_*/2}).$$

The posterior still has population bad-set mass at most ρ, but its empirical bad-set mass is at least $1 - 2\sqrt{\eta}$: every training point belongs to every consistent patch's bad set. Its KL to this abstract prior is at most $\log(1/\eta) = aD_*$, because conditioning has a partition function at least η.

Thus a target-mass exponent of size aD_*, bounded population defect, scaled KL control, and a tiny compressed class do not by themselves yield a bound independent of a. In our neural setting $a = N/t^2$, exactly the quantity that the robust compression argument must eliminate. The example isolates a logical obstruction; showing that the specified Gaussian prior does or does not exhibit it would require additional analysis.

19 Choosing and validating a trusted region from unlabeled data

A trusted region need not be specified before seeing any data. Independent unlabeled sample splitting supplies a simple rigorous version of this idea. This is a standard statistical construction, not a new sample-splitting principle. Its role here is to make the coverage assumption in the preceding robust-prior results independently checkable.

Let D be the test input distribution on $\{-1,1\}^d$. Draw three mutually independent samples: an unlabeled pilot P, an unlabeled validation sample $V = (X'_1, \ldots, X'_m)$, and the labeled training sample $S = ((X_i, f(X_i)))_{i=1}^n$. Using only P, choose any measurable region $G = G(P)$. The selection rule can be arbitrarily complicated. For example, P may determine a density level set, a union of

neighborhoods of observed inputs, or a support estimate in a fixed representation. The theorem does not assert that such regions admit an easy all-input stability certificate. Define

$$\widehat{\rho}_G = \frac{1}{m}\sum_{j=1}^{m} \mathbf{1}\{X'_j \notin G\}, \qquad \overline{\rho}_G = \min\left\{1, \widehat{\rho}_G + \sqrt{\frac{\log(1/\delta_u)}{2m}}\right\}.$$

Theorem 19.1 (An independently learned trusted region)**.** *Fix $0 < \delta_u, \delta_l < 1$ and $0 < \varepsilon < 1$. Suppose the target has a width-C layered Boolean circuit. For the selected $G(P)$, form any one of the following priors by conditioning the raw Gaussian on the indicated condition at every $x \in G(P)$:*

1. *The calibrated sign stability condition of Theorem 18.1, followed by ordinary training sign consistency. Let $R(Q)$ denote Gibbs sign error.*
2. *The numerical stability condition of Theorem 14.2, with signed training outputs greater than $1/2$. Let $R(Q)$ denote $\Pr_{w\sim Q, X\sim D}\{f(X)z_w(X) \le 1/4\}$.*
3. *The clipped-score stability condition of Corollary 16.1, with the same training-margin likelihood and the same test loss as in item 2.*

No stability condition is imposed outside $G(P)$, and no additional all-input clean-margin restriction is imposed. For the clipped condition, fix $0 < c \le 1/100$ once and for all; constants in the following bound may depend on c. The numerical condition retains its stated fixed calibration. There is a polynomial sample bound

$$n \ge \mathrm{poly}(d, L, C, \varepsilon^{-1}, \log(1/\delta_l))$$

such that, with probability at least $1 - \delta_u - \delta_l$ over P, V, S,

$$R(Q) \le \overline{\rho}_G + \varepsilon,$$

simultaneously for all widths $N \ge C$ and scales $0 < t \le 1$. The complexity of the pilot selection rule does not appear in this bound. The Gaussian is conditioned exactly on the selected support condition; no robust training likelihood or soft reweighting is introduced.

Proof. Condition on the pilot. The set G is now fixed and independent of both V and S. Hoeffding's inequality implies $D(G^c) \le \overline{\rho}_G$ with probability at least $1-\delta_u$ over V. The corresponding fixed-region theorem gives $R(Q) \le D(G^c) + \varepsilon$ with probability at least $1 - \delta_l$ over S. Its constants and common finite compression class are uniform over G: in particular, the globally stable target cylinder belongs to the conditioned support for every possible G. A union bound, followed by averaging over the pilot, proves the result. The sample event supplied by the fixed-region theorem is simultaneous in width and Gaussian scale. □

For any of these losses, a pointwise posterior correct-answer probability of at most $2/3$ entails pointwise Gibbs error at least $1/3$. Consequently the measure of such inputs is at most $3(\overline{\rho}_G + \varepsilon)$. For items 2 and 3, "correct answer" can mean the stronger event $f(x)z_w(x) > 1/4$. An exact one-sided binomial upper confidence limit can replace Hoeffding. For example, when the validation sample has no misses, the upper limit is $1 - \delta_u^{1/m} \le \log(1/\delta_u)/m$.

Validation candidates. The pilot may produce a catalog of M regions, denoted $G_j(P)$. Using

$$\overline{\rho}_j = \min\left\{1, \widehat{\rho}_j + \sqrt{\frac{\log(M/\delta_u)}{2m}}\right\}$$

certifies all M coverages simultaneously. One may then choose any single index $J = J(P, V)$ before seeing S, for example the index with the best coverage among regions admitting a convenient certificate. The theorem holds with G_J and $\overline{\rho}_J$. There is no extra $\log M$ cost in the labeled sample bound: conditional on P, V, just one region has been chosen independently of S. The $\log M$ cost is paid in the unlabeled validation confidence bound. If the region index is instead chosen using S, the additional complexity control of the finite-catalog theorem is needed. Reusing validation to generate an unbounded number of new regions also requires further control; the single-region guarantee cannot simply be applied after that adaptation.

What is still required. Coverage validation checks only how often an input lies in G. The prior must still impose stability throughout G, simultaneously for every admissible disturbance. Certifying stability merely on the finite pilot or validation inputs is not sufficient. In particular, high average certified robustness of one trained network does not by itself prove a theorem about the Bayesian posterior. A network-dependent exceptional region can adapt to the training labels, which is precisely the difficulty exhibited by the sparse-patch example. The present construction avoids that difficulty by freezing a shared region before the labeled sample. It modifies the support of the Gaussian prior; it does not establish the same conclusion for the unrestricted Gaussian prior.

20 Network-dependent exceptions with controlled input-set complexity

The exceptional inputs need not be shared by all networks if the family of permitted exceptional sets has controlled complexity. This provides a second way to retain the ordinary training likelihood. The restriction is on where instability may occur, rather than on an altered likelihood.

Fix a family $\mathcal{B}$ of subsets of the Boolean cube, including the empty set. Write M for its number of distinct sets and suppose $\log M \le H_B$. A family learned from an independent unlabeled pilot is also allowed: all definitions below are then conditional on that pilot, and H_B must be a uniform bound. Draw an independent unlabeled sample V of size m and a labeled sample S of size n. Fix $0 \le \rho < 1$. For a network w, permit any exceptional set $B \in \mathcal{B}$ satisfying

$$\widehat{D}_V(B) \le \rho.$$

Condition the raw Gaussian on the existence of such a B for which the calibrated sign-stability condition holds at every $x \notin B$. Here the radius is $[100\sqrt{1 + (L-1)N\|v\|^2}]^{-1}$, as in the trusted-region theorem. Then condition on ordinary training sign consistency at every example, including examples in the exceptional set. The exceptional set may depend on the network and hence on the labels through the posterior. This prior is the Gaussian conditioned on a union of admissible support events; it is not a mixture of separately normalized conditioned Gaussians.

Theorem 20.1 (Low-complexity distributional exceptions)**.** *For every $0 < \varepsilon, \delta < 1$ there are sample bounds*

$$m \ge O\Big(\frac{H_B + \log(1/\delta)}{\varepsilon^2}\Big), \qquad n \ge \mathrm{poly}(d, L, C, H_B, \varepsilon^{-1}, \log(1/\delta))$$

such that a width-C layered Boolean target circuit has posterior Gibbs sign error at most $3\rho+\varepsilon$, with probability at least $1-\delta$. The guarantee holds simultaneously for every $N \geq C$ and $0 < t \leq 1$. There is no all-input clean-margin assumption.

The same result holds when sign stability is replaced by numerical or clipped-score stability, the training likelihood is $f(X_i)z_w(X_i) > 1/2$, and the test loss is $\mathbf{1}\{f(X)z_w(X) \leq 1/4\}$. The numerical radius is unchanged. In the clipped case fix $0 < c \leq 1/100$ once and for all, with bound constants allowed to depend on c.

Proof. Set $u = \varepsilon/20$. Uniform one-sided Hoeffding bounds, one for each sample, imply that, except on events of probability $\delta/3$ each,

$$\begin{aligned} D(B) &\leq \widehat{D}_V(B) + u && \text{for every } B \in \mathcal{B}, \\ \widehat{D}_S(B) &\leq D(B) + u && \text{for every } B \in \mathcal{B}, \end{aligned}$$

provided $m, n \geq [H_B + \log(3/\delta)]/(2u^2)$. Every admissible exceptional set therefore has population mass at most $\rho + u$ and empirical training mass at most $\rho + 2u$. These statements hold uniformly and hence also for sets chosen by a posterior network.

The global robust target cylinder qualifies using $B = \varnothing$, for every realization of V. It has the same raw Gaussian mass lower bound $\exp(-ND_*/t^2)$ as before and is consistent with all labels. Consequently the full posterior—including the union in its support event—has scaled KL at most ND_*/t^2. The usual readout-energy truncation discards mass at most β and leaves a law Q' with a common inverse-polynomial stability radius and the same polynomial scaled entropy bound.

For each retained network choose one witnessing exceptional set; the selection can be measurable because there are only finitely many distinct sets. Let b_D and b_S denote its population and empirical masses, averaged under Q'. The preceding uniform events give $b_D \leq \rho + u$ and $b_S \leq \rho + 2u$. The pointwise-defect compression argument and its finite-class relative Chernoff bound give, on one further event of probability at least $1 - \delta/3$,

$$R(Q) \leq \beta + 3\alpha + b_D + 2b_S + \frac{8[\log|\mathcal{H}| + \log(3/\delta)]}{n} \leq 3\rho + 5u + \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(3/\delta)]}{n}.$$

Take $\beta = \varepsilon/8$, $\alpha = \varepsilon/24$, and choose n so the last term is at most $\varepsilon/2$. The numerical and clipped cases use the same bounded ramp loss as before; an exceptional input contributes at most one to its population or empirical comparison, giving the same inequality. The common finite class is independent of N, t, and the uniform set-mass events do not involve either parameter. This proves the simultaneous statement. □

A VC-dimension formulation. If $\mathcal{B}$ has VC dimension at most $h \geq 1$, its restriction to the 2^d Boolean inputs has at most

$$\sum_{j=0}^{\min(h,2^d)} \binom{2^d}{j} \leq (2^d+1)^h$$

distinct sets. Thus $H_B = h\log(2^d+1) \leq h(d+1)\log 2$ suffices. No finiteness of a continuous parameterization is required. The theorem therefore gives polynomial depth bounds whenever the exceptional-set VC dimension is polynomial in the stated parameters and independent of network width. This use of the finite Boolean domain is intentionally simple; ordinary VC uniform-convergence estimates can also be used.

For example, one can permit exceptional sets that are halfspaces or balls in a fixed p-dimensional input representation, or unions of a bounded number of such sets. The corresponding VC dimensions

are polynomial in p and the number of components. The representation may be learned from the independent pilot. These are candidate restrictions, not a claim that the actual failure sets of unconstrained trained networks have small VC dimension. Whole-region stability outside a witnessing exceptional set must still hold. A certificate proving that all failures lie inside such a region is enough; high empirical robustness alone is not.

The theorem gives a genuine distributional relaxation with network-dependent exceptions and the original sign likelihood. It still changes the Gaussian prior's support. It neither proves an unrestricted Gaussian result nor evades the sparse-patch obstruction without an added assumption: the family of arbitrary sparse patches has large input-set complexity, precisely what is controlled here.

21 The remaining research questions

The general unrestricted theorem still carries $Q_* = [(K+2)/L]^{L/2}$. The shallow-target construction bounds this quantity polynomially for several substantial target classes. Eliminating its exponential dependence for general small deep circuits, or proving that some such targets require it under the specified Gaussian posterior, remains the central mathematical problem. At $\gamma = 1$, the typical Gaussian bulk has order-one operator norm, so the small-bulk proof does not apply. The finite-width result and the robustness-based results cover this exponent under their stated hypotheses. No new sample lower bound for the exact unrestricted prior is established.

The empirical question is also sharper than universal robustness of every trained network. One should test finite-radius numerical stability on relevant data and on a justified region around that data, retaining optimization failures and boundary cases. Testing one trained network does not establish a property of the Bayesian posterior. Testing finitely many inputs does not establish a worst-case condition on an exponentially large cube. The trusted-region and posterior-defect statements show which additional facts would turn such evidence into a learning guarantee.

For designing a prior, stability of the numerical output or of a clipped prediction score, with positive training margins, is a direct candidate: it expresses resistance to computation errors without requiring every unobserved input to have a confident label. The clipped-score version additionally ignores changes that leave the prediction confidently saturated. Diagonal-metric certificates permit compensating expansion and contraction between layers, rather than requiring every layer to be nonexpansive in Euclidean norm. A soft version retains the Gaussian support, while a trusted-region version avoids demanding stability on arbitrary non-data-like inputs. Both change the learning rule in explicit, measurable ways. Whether ordinary gradient descent selects networks satisfying an appropriate version with high probability is not proved by the present finite experiments or by the counterexamples to a universal assertion.

A An exact variational principle at the critical exponent

For completeness we reproduce the finite-domain variational foundation. Here $\phi = \tanh$. For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation. It identifies the functional used in the critical-rate capacity constructions with the actual Gaussian probability rate. Covariance-process large deviations for Gaussian networks have been studied in a more general functional setting by Andreis, Bassetti, and Hirsch [3]. No novelty claim is made for the general large-deviation framework; the finite-domain proof below fixes the precise normalizations and strict-margin convention used here.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \dots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1, 1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E hh^{\mathsf{T}} \, d\nu(h), \qquad Q_0(a, b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = \left(\phi((Q^{1/2}u)_a)\right)_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho} \, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^+ is the Moore–Penrose inverse.

Theorem A.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1, \dots, \nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D(\nu_s \| \mathsf{M}_{K(\nu_{s-1})}) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{23}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z \in O} I(z) \le \liminf_{N \to \infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N \to \infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z \in F} I(z).$$

The infimum in (23) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [4].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on

the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N}\sum_{i=1}^{N}\left(h_i^{s-1}(x^a)h_i^{s-1}(x^b)\right)_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2} V_N. \tag{24}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta\|\Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s\|\Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N},\dots,\eta_{r,N},V_N)$.

The recursive map from these variables to $(\nu_1,\dots,\nu_r,z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j}\, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q\, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (24).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta : T_\#\eta = \nu} D(\eta\|\rho) = D(\nu\|T_\#\rho). \tag{25}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_\#\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the identity in the extended real sense. Apply (25) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v : Q^{1/2}v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (23).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^{\mathsf{T}} z - \frac{1}{2} t^{\mathsf{T}} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (23) is attained. Moreover $K(\nu) \preceq m I_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary A.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1,1\}^m$, set*

$$J(f) = \inf_{z:\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N\to\infty} -\frac{1}{N}\log \mu_{N,1}(A_{N,f}) = J(f). \tag{26}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g\in\mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N\to\infty} -\frac{1}{N}\log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (23) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (26).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g\in\mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

B Robust Gaussian target mass

This appendix supplies the target probability bound used by the robustness-based theorems, for every $N \geq C$ and every $0 < t \leq 1$. Take balanced clone groups of sizes k_a, so $N/k_a \leq 2C$, and let P project onto group-constant vectors. Every unused circuit slot is filled by a Boolean constant, so the ideal wire vector $b^s(x)$ lies in the range of P and has norm $\sqrt{N}$. Put

$$e_* = (16\sqrt{2C})^{-1}, \quad \tau_0 = e_*/4, \quad b_L = \sqrt{16\log 9 + 8\log(4L)},$$

$$A = \max\{32 b_L, 2\log(8/e_*)\}, \qquad \rho = \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}.$$

Use tanh gate gain A and readout coefficient 2. The gain need not be bounded by logical circuit width; it can be accommodated by polynomial padding if invoking the "weights bounded by width" complexity convention.

Allow any $0 < t \leq 1$. The standardized coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv have common standard deviation t; the exponent-γ case is $t = N^{(1-\gamma)/2}$. Restrict every first-layer coordinate, every

internal row's projection onto each normalized clone-group direction, and every readout coordinate to an interval of radius ρt around its clone value. Then

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \quad E^s P = 0.$$

The free Gaussian bulk E^s is independent of the cylinder. A 1/4-net Gaussian operator bound makes $\|E^s\|_{\mathrm{op}} \le b_L t$ simultaneously with conditional probability at least 1/2.

We check robustness to a separate per-layer RMS budget τ_0, which is stronger than the total-budget requirement for $r \le \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \le e_*$, each clone-group mean has error at most 1/16, so the teacher part of every intended signed preactivation is at least $A/2$. For every real u and $\epsilon \in \{-1, 1\}$,

$$\epsilon a \ge A/2 \quad \Longrightarrow \quad |\tanh(a+u) - \epsilon| \le 2e^{-A/2} + 8|u|/A.$$

This is a global secant bound: for $|u| < A/4$ use saturation, and otherwise use the trivial bound 2. It handles concentrated adversarial noises. Since $E^s b^{s-1} = 0$ and noisy states have norm at most $(1+\tau_0)\sqrt{N}$,

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

The first layer starts the invariant because its coordinatewise preactivation error is at most $\rho(d+1) \le A/4$. The readout has noisy signed value at least $2 - 1/8 - \rho(1+\tau_0) - \tau_0 > 1$.

There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. Their squared centers sum to at most NB'_0, with

$$B'_0 = 9dA^2 + 18(L-2)CA^2 + 8C.$$

The elementary Gaussian interval bound and the independent operator event give

$$-\log \mu_{N,t}(\mathcal{R}_{r,N,f}) \le \frac{N}{t^2}\left[B'_0 + q\log\frac{e\sqrt{2\pi}}{2\rho} + \log 2\right] = DN/t^2.$$

Here D is jointly polynomially bounded in d, C, L. Exponentially small coordinate precision would in any case cost only its logarithm in this negative log-mass estimate; it should not be replaced unnecessarily by its reciprocal.

C Robust posterior compression uniformly for $\gamma \ge 1$

This section gives a different statistical bridge. It uses robustness of the original network while constructing a finite candidate, and tracks entropy only in the original weights that have not yet been processed. Consequently the intermediate compressed prefix need not have a density under a Gaussian prior. The result applies to a posterior containing many different functions.

Let $\mu_{N,t}$, $0 < t \le 1$, have independent weight-entry variances t^2/d, t^2/N, and t^2/N^2 in the first, internal, and readout layers. The exponent-γ prior is obtained by $t = N^{(1-\gamma)/2}$. All-layer sign robustness refers to the total noise norm in (1); the clean 1/2 margin may additionally be imposed.

Theorem C.1 (Uniform robust posterior compression)**.** *Fix a scalar 1-Lipschitz activation $|\phi| \le 1$, and parameters $d \ge 1$, $L \ge 2$, $K \ge 1$, $0 < r \le 1$, and $0 < \alpha \le 1/8$. There is a finite class $\mathcal{H} = \mathcal{H}(d, L, K, r, \alpha)$, consisting of signs of bounded-weight depth-L networks, such that*

$$\log|\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

The polynomial is universal and does not depend on N or t. If $Q \ll \mu_{N,t}$ is supported on all-layer sign-r robust networks and

$$D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2,$$

then there is a distribution Γ_Q on $\mathcal{H}$ such that, for every $x \in \{-1,1\}^d$, a coupling of $w \sim Q$ and $h \sim \Gamma_Q$ satisfies

$$\mathbb{P}\{h(x) \ne \operatorname{sign} z_w(x)\} \le \alpha.$$

The coupling may depend on x; its two marginals do not. Neither the support of Q nor its samples need represent a single Boolean function.

We give the proof, including deliberately loose polynomial choices. Two elementary entropy facts organize the construction. First, for a standard D-dimensional Gaussian G, the entropy variational inequality gives

$$\mathbb{E}_P\|Z\|^2 \le 4D_{\mathrm{KL}}(P\|G) + 2D\log 2. \tag{27}$$

Second, Gaussian Talagrand [12] gives $W_2^2(P,G) \le 2D_{\mathrm{KL}}(P\|G)$.

Lemma C.2 (Entropy of the unprocessed weights)**.** *Let A, X, Y be random variables, with fixed reference laws P_X, P_Y, and put*

$$B = \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y).$$

Let J be a finite message, possibly depending on (A,X,Y), and let $H = f(A,J,X)$. If A' is a randomized function of (A,J,H) whose fresh randomness is conditionally independent of Y, then

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'}\|P_Y) \le B + H(J \mid A).$$

Proof. The left side equals $D_{\mathrm{KL}}(Q_Y\|P_Y) + I(Y;A')$. Data processing and

$$I(Y;J,H \mid A) \le I(Y;X,J \mid A) \le I(Y;X \mid A) + H(J \mid A)$$

bound it by $\mathbb{E}_{A,X} D_{\mathrm{KL}}(Q_{Y|A,X}\|P_Y) + H(J \mid A)$, which is at most the claimed bound by the KL chain rule. These expressions can equivalently be justified by finite partitions and lower semicontinuity. □

Here A will be a constructed prefix, X the current original weight block, and Y all later original blocks. The lemma imposes no reference law on A.

Lemma C.3 (Whole-layer active directions)**.** *Condition on (A,Y) and let X be a standardized Gaussian-reference matrix, with conditional relative entropy $R(A,Y)$. For vectors $b_x(A)$ of norm at most one, one can select a list J of inputs and its span $U = \operatorname{span}\{b_x : x \in J\}$ with*

$$\dim U \le \frac{t^2 R(A,Y)}{N\delta}$$

such that preserving $H = tXU$ and replacing the orthogonal coordinates by fresh Gaussian coordinates changes the physical evaluation vectors tXb_x, under a suitable coupling, by expected squared RMS at most 2δ at each x. The candidate matrix law preserves (A,Y,H) and does not depend on x.

Proof. Use a fixed ordering of the finite input set. Whenever the averaged conditional KL of the entire residual evaluation vector Xv exceeds $N\delta/t^2$, add the unit residual direction v of that input. Each query is selected from the conditional law given (A,Y), averaging over active-coordinate values; thus J is a function of (A,Y). Orthogonal Gaussian coordinates are independent, so the active-coordinate entropy increases by precisely this conditional KL. The displayed rank bound follows. At termination, Gaussian Talagrand on the entire evaluation vector, multiplication by the squared residual-vector length, and division by N/t^2 give the error bound. Scalar-vector couplings extend to full matrices by disintegration. The basis of U is a measurable function of A, J under fixed Gram–Schmidt; it has no further dependence on Y once those variables are supplied. □

Parameter choices. Write

$$\begin{aligned} e &= \alpha r^2/(1000L), & \eta &= \alpha/(100L), & q &= \lceil 2K/(e\eta)\rceil, \\ D_0 &= [(d+1)q+2]\log 2, & V &= 8K+2q, & B_H^2 &= 4V/e, \\ T &= 128(q+1)(B_H^2+1)/e, & a &= e/[128(B_H^2+1)], & & \\ K_0 &= (4K+2\log 2)/\eta, & B_v &= 16K_0/r. & & \end{aligned}$$

Choose m to be the least positive integer no smaller than both

$$\begin{aligned} &10000T^4(q+1)^2a^{-2}[d+\log(10(q+1)/\eta)+1], \\ &10000B_v^2r^{-2}[d+\log(10/\eta)+1], \end{aligned}$$

and put

$$N_0 = \max\{m, \lceil LD_0/K\rceil, 1\}, \quad B_G = \max\{2, 2\sqrt{\log(3m/e)}\}, \quad B_1^2 = 4[4K+2(d+1)\log 2]/e.$$

All quantities have a fixed polynomial upper bound in the theorem parameters.

The small-width case. If $N \le N_0$, let $P_0 = N_0(d+1)+(L-2)N_0^2+N_0$. Applying (27) to all standardized weights, and then multiplying by t^2, gives

$$\mathbb{E}_Q\|tZ_{\text{all}}\|^2 \le 4N_0K + 2P_0\log 2.$$

With probability at least $1-\alpha/4$, all actual weights are bounded by

$$B_{\text{small}} = [4(4N_0K+2P_0\log 2)/\alpha]^{1/2}.$$

Keep the original network on this event and use a default classifier otherwise. Padding with nodes having zero outgoing weights preserves the original output. The successful candidate has output margin greater than r. It remains to treat $N > N_0$.

The virtual prefix and its entropy budget. The constructed prefix A has N virtual nodes in its current layer. Each virtual row acts on m stored nodes of the preceding layer. The auxiliary description may keep all N rows; the final network will keep just the rows selected at the following stage. Track the conditional KL of the unprocessed *original* weights. By Lemma C.2, it is at most

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2. \qquad (*)$$

This is not a KL bound for the singular compressed prefix itself.

For the first layer, zero every physical standardized row $tZ_{1,i}$ of norm greater than B_1, and retain all other rows. Bounded activations and (27) bound the expected squared RMS activation change at each input by e. This deterministic prefix channel does not increase the future entropy budget.

An internal layer: retain active data. Condition on the current prefix A and the future original weights Y. Apply Lemma C.3 to $b_x = h_A(x)/\sqrt{N}$ and $\delta = e$. Abort if more than q inputs are needed. By (*) and Markov, this failure has probability at most η. The successful query list, together with an abort flag, has a code of at most $[(d+1)q+2]$ bits. Its entropy is at most D_0.

Let $H = tXU$ be the physical active matrix. Averaging the Gaussian entropy moment bound, with the success indicator rather than conditioning on success, gives

$$\mathbb{E}\left[\frac{\|H\|_F^2}{N};\ \text{success}\right] \le 4t^2 B_s/N + 2t^2 q \log 2 \le V.$$

Zero every row H_i whose norm exceeds B_H. The expected squared RMS activation change from this operation is at most $4V/B_H^2 = e$. All rows are clipped before any later selection, including selections that may depend on future weights.

An internal layer: reconstruct using m source nodes. For the orthonormal active basis let $u(j) = \sqrt{N}\, U_{j,:}$; uniform source sampling gives $\mathbb{E}_j u(j)u(j)^\top = I$. Clip the *whole vector* at radius T: $c(j) = u(j)\mathbf{1}_{\{\|u(j)\| \le T\}}$. Then

$$\mathbb{E}cc^\top \preceq I, \qquad \mathbb{E}\|u - c\| \le q/T.$$

Sample m uniform source indices S_j. Entrywise Hoeffding and a union bound give, with probability at least $1 - \eta$,

$$\|\widehat{G} - \mathbb{E}cc^\top\|_{\mathrm{op}} \le a, \quad \sup_x \|\widehat{v}(x) - \mathbb{E}ch_A(x)\| \le a,$$
$$\sup_{x,y} |\widehat{K}(x,y) - K(x,y)| \le a,$$

where $\widehat{G} = m^{-1}\sum_j c(S_j)c(S_j)^\top$, $\widehat{v}(x) = m^{-1}\sum_j c(S_j)h_{A,S_j}(x)$, and $\widehat{K}$ is the empirical feature covariance; its population counterpart is $K(x,y) = N^{-1}\sum_i h_{A,i}(x)h_{A,i}(y)$. Abort on failure; this event depends only on A, J, S, so it does not reveal further future information.

Set $\widetilde{c}_j = c(S_j)/\sqrt{1+a}$, and let D have rows $\widetilde{c}_j^\top/\sqrt{m}$. Then $D^\top D \preceq I$ and $I - DD^\top \succeq 0$. Independently for every target virtual row draw $\xi_i \sim N(0, I - DD^\top)$ and use coefficients

$$\widehat{w}_{ij} = m^{-1} H_i^{\mathrm{clip}} \cdot \widetilde{c}_j + t\xi_{ij}/\sqrt{m}. \tag{28}$$

These are actual neural-network rows on the m sampled nodes. Their active feature vector is $\widehat{\psi}(x) = m^{-1}\sum_j \widetilde{c}_j h_{A,S_j}(x)$, and the lazy covariance is exactly $t^2(\widehat{K} - \widehat{\psi}^\top\widehat{\psi})$. For the ideal field, write $\psi(x) = U^\top b_x$ and $C(x,y) = K(x,y) - \psi(x)^\top\psi(y)$. Direct calculation gives

$$\sup_x \|\widehat{\psi}(x) - \psi(x)\| \le q/T + 2a =: \zeta, \qquad \sup_{x,y} |\widehat{C}(x,y) - C(x,y)| \le a + 2\zeta.$$

Here $\widehat{C} = \widehat{K} - \widehat{\psi}^\top\widehat{\psi}$. Bessel's inequality and $D^\top D \preceq I$ give $\|\psi(x)\|, \|\widehat{\psi}(x)\| \le 1$, which justify the covariance bound. At a fixed input, coupling scalar Gaussian evaluations with a common standard normal bounds the squared preactivation transport error, per row, by

$$B_H^2\zeta^2 + t^2(a + 2\zeta) \le e/8.$$

Extend these scalar couplings to full row fields. The candidate weight law is the same at every input; there is no claim of a single Gaussian-field coupling that is close at all inputs simultaneously.

Finally clip *all* coordinates ξ_{ij} to $[-B_G, B_G]$ before later row selection. Since each has variance at most one,

$$\mathbb{E}\left|\frac{t}{\sqrt{m}}\sum_j (\xi_{ij} - \mathrm{clip}_{B_G}\xi_{ij})h_{A,S_j}(x)\right|^2 \le 3me^{-B_G^2/4} \le e.$$

Thus every virtual coefficient is bounded by $B_H T/m + B_G/\sqrt{m}$, without a union bound over N.

All newly constructed data are a randomized function of A, J, H. Lemma C.2 therefore proves (*) for the next step. Combining entropy replacement, active clipping, finite reconstruction, and Gaussian clipping by $(a_1 + \cdots + a_4)^2 \le 4\sum_i a_i^2$, the expected squared RMS activation injection at this layer is less than $20e$.

The readout and final network. The original readout marginal has never changed. Its physical moment satisfies $\mathbb{E}_Q[N\|v\|^2] \le 4K + 2\log 2$. Abort if $N\|v\|^2 > K_0$, an event of probability at most η. Zero coefficients with $|Nv_i| > B_v$. This changes the output uniformly on bounded virtual feature vectors by at most $K_0/B_v = r/16$. Sampling m uniform last-layer sources and using readout coefficients $(Nv_i \mathbf{1}_{|Nv_i| \le B_v})/m$ adds at most $r/16$ uniformly on $\{-1, 1\}^d$, except with probability η, by Hoeffding and the chosen m.

At each stage, retain in the final network just the m source rows selected from the preceding virtual layer. Each such row already acts on the previously stored m nodes. The result is an ordinary width-m network; repeated indices are duplicate nodes. Its weights are bounded by

$$B_{\text{large}} = \max\{B_1, B_H T/m + B_G/\sqrt{m}, B_v/m, 1\}.$$

The larger virtual construction is not part of the final predictor.

The original network receives all perturbations. Fix an input x and glue the conditional couplings, retaining the original weight history. The current and future original blocks are untouched until their own stages. Define

$$\eta^s = h^s_{\text{virtual}}(x) - \phi(W^s_{\text{original}} h^{s-1}_{\text{virtual}}(x)).$$

The analogous readout difference is η^L. Later stages leave earlier computational prefixes unchanged, so the final candidate output is exactly the *original* network's output under these postactivation injections. No intermediate robustness assumption is made.

For precision, the canonical construction draws its new Gaussian coefficients independently of the discarded current block given (A, J, H). The comparison coupling may correlate them with that discarded block. It preserves the same joint marginal of the new prefix and the future original blocks, as well as the original full-network marginal, but need not preserve the canonical joint law with all discarded weights. The next entropy step uses the first marginal; original-network robustness uses the second. There is no conflicting demand for independence in the comparison coupling itself.

More explicitly, the induction invariant is that the comparison law has full original-weight marginal Q and the canonical marginal of (A, X, Y). The conditional transports supply a joint law of (A, X, Y, A') with this incoming marginal and the prescribed outgoing (A', Y) marginal. Extend it by the previous conditional law of the past original weights given (A, X, Y). The extension keeps the complete original weight law unchanged, preserves all earlier computational prefixes, and establishes the same invariant for the next layer. This is ordinary disintegration on finite-dimensional standard Borel spaces.

Abort probability is at most $2L\eta$. The unconditional moment bound gives

$$\mathbb{P}\left\{\text{no abort and } \sum_{s<L} \|\eta^s\|^2/N > r^2/8\right\} \le 160Le/r^2 = 0.16\alpha.$$

Otherwise the readout injection is at most $r/8$, and total squared noise is at most $9r^2/64 < r^2/4$. Robustness then gives candidate margin greater than $r/2$ relative to the original network's label: a smaller margin could be cancelled by an additional output shift within the radius-r noise ball. The exceptional probability is less than $\alpha/4$.

A fixed finite class. Pad candidates to width $M = N_0$ and take $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$. Round every parameter to a fixed grid on $[-B, B]$ of mesh at most

$$h = \frac{r}{16(d+L+1)(2M(B+1))^L}.$$

The bounded-activation, Lipschitz layer recursion gives uniform output change at most $r/8$. Thus rounding preserves every successful coupled prediction. Writing $P = M(d+1) + (L-2)M^2 + M$, the grid class satisfies

$$\log|\mathcal{H}| \le P \log\left[3 + \frac{32B(d+L+1)(2M(B+1))^L}{r}\right].$$

Although grid precision is exponentially small in L, its number of bits is polynomial in L. All other bounds above are polynomial too. Rounding is a fixed map on candidate networks, giving the common marginal Γ_Q and proving Theorem C.1.

Corollary C.4 (Polynomial-depth learning at every width)**.** *Suppose the raw Gaussian target event, consisting of networks with all-layer sign margin r and clean margin $1/2$ for f, has mass at least $\exp(-NK/t^2)$. Restrict the prior to all sign-r robust networks having global clean margin $1/2$, regardless of their Boolean function, then condition on n noiseless iid training examples. With probability at least $1-\beta$, the resulting posterior satisfies*

$$\mathrm{GibbsErr}(Q) \le 2\alpha + \sqrt{\frac{\log(2|\mathcal{H}|/\beta)}{2n}}.$$

Consequently the sample bound is jointly polynomial in d, L, K, r^{-1} and inverse desired error, uniformly in N and $0 < t \le 1$. For posterior-confidence error at threshold $2/3$, multiply Gibbs error by three; for threshold $1/2$, multiply it by two.

Proof. The posterior is the raw Gaussian conditioned on an event containing the target event, so its KL is at most NK/t^2. Its training error is zero. The compression couplings imply that Γ_Q has expected empirical error at most α. Finite-class uniform convergence holds for every member of $\mathcal{H}$, hence also for this sample-dependent mixture. Its true error is at most α plus the displayed square-root term. The pointwise coupling adds at most one more α to recover original posterior Gibbs error. This argument does not require a global coupling over inputs or a common function for all draws. The confidence-error statements follow from Markov on posterior error probability at a random test input. □

D A finite-radius certificate using clean Jacobian products

Products of individual spectral norms discard cancellations between layers. The following sufficient test retains the exact clean transition products and bounds the nonlinear remainder separately. It is a finite-radius statement, not an inference from the infinitesimal Jacobian alone.

Write the $H = L-1$ hidden layers in column-vector convention as $h_s = \tanh(W_s h_{s-1})$, with output $z = v^\top h_H$ and width N. Fix an input and its clean preactivations $a_s = W_s h_{s-1}$. Define

$$A_s = \mathrm{diag}(\mathrm{sech}^2 a_s) W_s, \qquad P_{s,t} = A_s \cdots A_{t+1} \ (t < s), \quad P_{s,s} = I.$$

The Gramian for normalized additive hidden disturbances is

$$G_1 = NI, \qquad G_s = A_s G_{s-1} A_s^\top + NI.$$

For a proposed total-noise radius r, recursively put $u_1 = 0$ and, for $s \geq 2$,

$$b_s = r\sqrt{\operatorname{diag}(W_s G_{s-1} W_s^\top)} + \sum_{t=2}^{s-1} |W_s P_{s-1,t}| u_t, \tag{29}$$

$$u_s = \frac{2}{3\sqrt{3}}\, b_s^{\odot 2}. \tag{30}$$

Absolute values, the square root, and the square in these expressions are coordinatewise. Finally define

$$B(r, x) = r\sqrt{v^\top G_H v + 1} + \sum_{t=2}^{H} |v^\top P_{H,t}| u_t. \tag{31}$$

Proposition D.1 (Finite-radius transport certificate)**.** *Every simultaneous postactivation and readout disturbance satisfying $\sum_{s=1}^{H} \|\eta_s\|^2/N + \eta_{\text{out}}^2 \leq r^2$ changes the output at x by at most $B(r, x)$. Consequently, if $B(r, x) < |z(x)|$ on the full input cube, the network preserves its own clean signs throughout this ball.*

Proof. Let Δh_s be the hidden displacement and write its exact recursion as

$$\Delta h_s = A_s \Delta h_{s-1} + \eta_s + R_s, \qquad \Delta h_s = \sum_{t=1}^{s} P_{s,t}\eta_t + \sum_{t=2}^{s} P_{s,t} R_t.$$

There is no input perturbation, so $R_1 = 0$. The global scalar bound $|\tanh''| \leq 4/(3\sqrt{3})$ implies $|R_s| \leq [2/(3\sqrt{3})]|W_s \Delta h_{s-1}|^{\odot 2}$. Assuming $|R_t| \leq u_t$ at preceding layers, Cauchy–Schwarz applied to the linear disturbance map, followed by the coordinatewise triangle inequality for the remainder terms, gives $|W_s \Delta h_{s-1}| \leq b_s$. This proves $|R_s| \leq u_s$ inductively. At the readout, Cauchy–Schwarz including scalar output noise gives the first term of (31); the remainder sum gives the second. No statistical independence or small-noise approximation is used. □

The test can be sharpened by replacing the global curvature bound in (30) with the supremum of $|\tanh''|$ on $[a_{s,j} - b_{s,j}, a_{s,j} + b_{s,j}]$. An alternative exact scalar remainder envelope evaluates $\tanh(a+q) - \tanh(a) - \operatorname{sech}^2(a) q$ at $q = \pm b$ and at $q = -2a$ when $|2a| \leq b$: these and $q = 0$ exhaust its extrema on the interval. The global-curvature version above already suffices for the numerical improvement below.

The numerical condition in Section 14 is certified at an input whenever $B(r_{\text{num}}, x) \leq 1/16$. Floating-point evaluations of this sufficient test are distinguished from interval-verified statements in the experimental section.

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